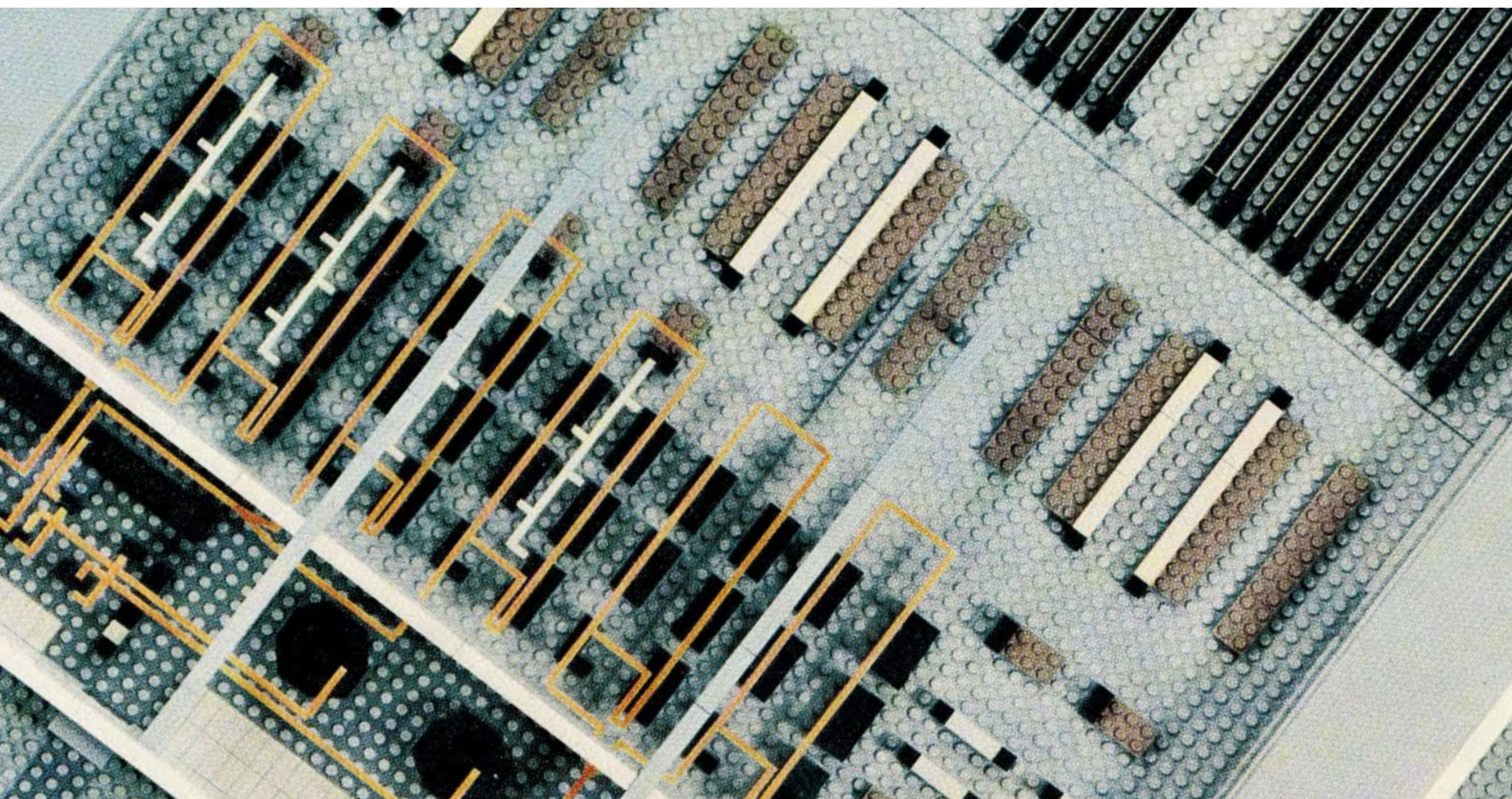




Rieter Spinning Documentation

Reference for planning a short-staple fiber mill including all four end spinning systems



Rieter Spinning Documentation

Documentation for designing a short-staple fiber spinning mill

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Introduction

Dear spinning specialist,

This is the latest edition of the Rieter Spinning Documentation. This documentation contains present and generally recognized fundamental principles and key values of short-staple spinning which have been collected in mill operations.

In order to achieve this objective, prominent experts from the academic, institutional and industrial fields have been consulted to contribute their wide-ranging know-how and experience. In particular, we wish to thank the management of the Bremen Cotton Exchange for providing extensive data and all personnel at Rieter who have placed their expertise and experience at our disposal and therefore made this publication possible.

This documentation is intended to serve as a tool for practical spinning mill planning for:

- spinning managers and technologists, as a work of reference and an aid for daily planning work,
- interested spinning specialists, to whom this collection of practical and basic data can provide support in planning decisions and
- students at technical colleges and universities, to whom a systematic collection of basic spinning planning material should be communicated.

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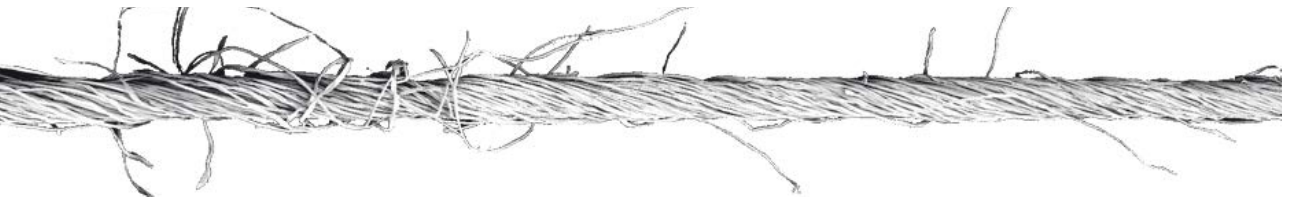
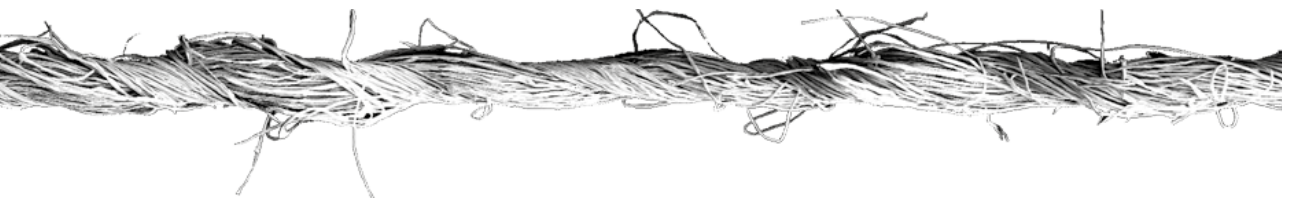
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Com4ring**Com4**compact**Com4**rotor**Com4**jet

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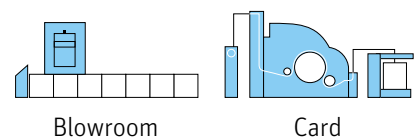
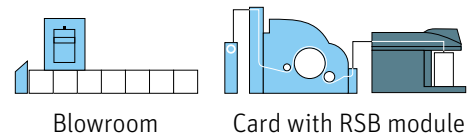
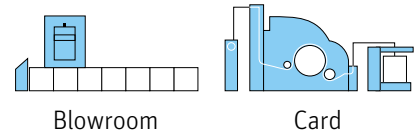
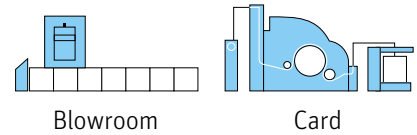
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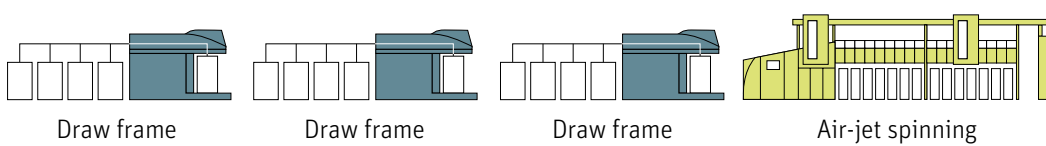
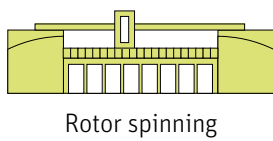
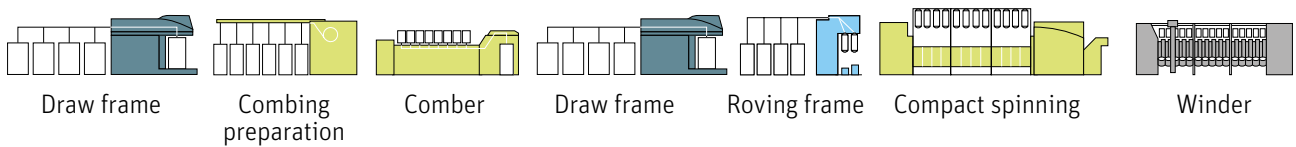
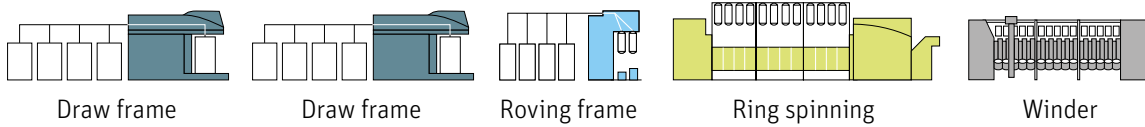
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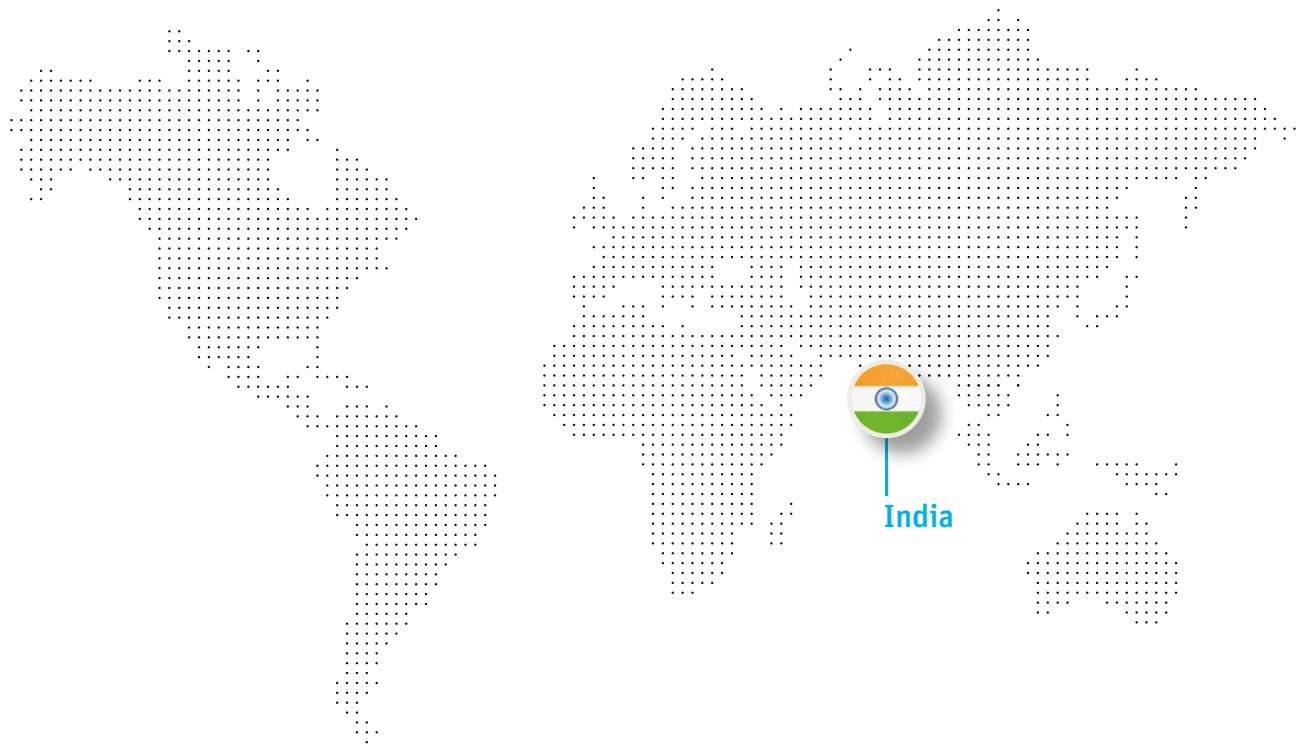
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In this latest edition of the spinning documentation, the main aim is to provide all relevant data and information as part of Rieter knowledge to the reader. However, to keep the documentation in printable dimensions, some restrictions needed to be made. Rieter sells its machines and systems to customers from all over the world and therefore can help with calculations and data for most countries. In this document, the decision was made not only to transfer information about all four spinning processes but also about four main markets of the spinning industry – China, India, Turkey and USA.





1.1. Ring spinning



Basis of calculation			
Wage for operating personnel per h	INR		42.95
Wage for skilled maintenance personnel per h	INR		62.42
Wage for unskilled maintenance personnel per h	INR		36.24
Personnel in reserve	%		10
Cost of electric power per kWh	INR		6.174
Building costs per m ²	INR		11 007
Yearly building maintenance	%		1.5
Number of shifts for operation per day			3
Number of shifts for maintenance per day			1
Operating hours per year	h		8 700
Customs, sales tax etc. in of machine price	%		0
Capital interest	%		12.1
Depreciation for machinery	Years		10
Depreciation for accessories	Years		5
Depreciation for building	Years		30
Exchange rate for offer	INR		0.014 9
Exchange rate for manufacturing cost	INR		0.014 9

1.1.1. Ring spinning process for cotton/polyester blend



Cotton



Polyester

Market

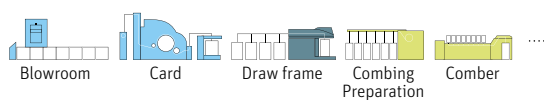
Ring spinning is presented in the Indian market. This is relevant, as for the calculation of extended spinning and production schedules, the basis of calculation must always be precised.

The spinning schedule is made for a spinning mill with 20 machines with 1 824 spindles each. This represents the average capacity of a modern ring spinning mill. The produced yarn is a 50% cotton/50% polyester blend with a yarn count of Ne 40 and an alpha e 3.4 for knitting applications.

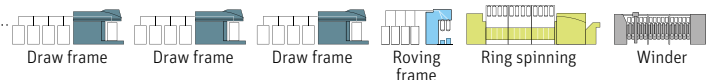
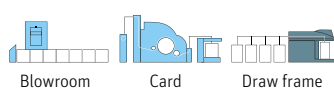
Process

The exact spinning process for a cotton/polyester blend is symbolized here:

Cotton



Polyester



Yarn

The best known advantages of ring yarn are the pleasant soft touch, the very good opacity of the fabric and the widest range of fabric designs.



1.1.2. Ring spinning mill for cotton/polyester blend



- 1 Blowroom
- 2 Cards
- 3 Combing
- 4 Draw frames
- 5 Roving frames
- 6 Ring spinning machines
- 7 Winders



1.1.3. Data for yarn and fiber

Raw material blend 50/50	Combed cotton	Polyester	Yarn	1
Staple length	1 3/16"	38 mm	Yarn count	Ne 40
Fiber specification	3.5 mic	1.3 dtex	Ring	28 mm
Example for a spinning mill in:	India			
End spinning capacity	20 machines à 1 824 spindles			
Utilisation	Knitting			

1.1.4. Spinning and production schedule

Machines				Spinning schedule			
Machine description	Number of machines	Number of units per machine	Degree of utilisation in %	Count delivered in Ne or g/m	Doubling	Draft	Twist
							alpha e
Blend							
Link	20	36	87.2	40			
G 36	20	1 824	100	40	1	40	3.44
SERVOftrail-F-A	1						
F 38 – 110	4	208	86.7	1	1	8.3	0.91
RSB-D 50	4		99.7	0.12	6	6	
SB-D 50	4		98	0.12	6	6	
SB-D 50	4		98.6	0.12	6	6	
Cotton							
E 86R	6		94.8	0.12	8		
E 26	1						
E 36	1		69.1	76	24	1.6	
SB-D 22	1	2	99.9	0.12	5	6	
C 70	6		97.4	0.1		126.6	
A 21	1		41.7				
Vision Shield	1		34.7				
B 76	1		41.7				
B 12	1		34.7				
A 22	1		34.7				
B 25	1		26.2				
A 12.2	1		25				
Blowroom total							
Raw material total							
Polyester							
SB-D 50	2		84.9	0.12	5	7.5	
C 70	5		90.5	0.08		95.3	
A 21	1		37.6				
Vision Shield	1		28.2				
B 76	1		37.6				
Blowroom total							
Raw material total							

	Production schedule						
alpha m	Recommended processing speed in rpm or m/min	Calculated production per unit in g/h or kg/h	Efficiency in %	Practical production per unit in g/h or kg/h	Waste total in %	Total production required in kg/h	Number of calculated production units
	1 300	1 151.6	88.9	1 023.8	1	642.9	628
21.74	17 900	18.5	94.6	17.5	2.3	649.3	36 477
0.91	1 260	1 240	74.4	922.6	1	664.9	721
	650	191.9	87.7	168.3	0.6	671.5	4
	650	191.9	89.8	172.3	0.6	675.5	3.9
	650	191.9	89.8	172.3	0.6	679.6	3.9
	500	65.5	91.8	60.1	17	341.8	5.7
	230	1 048.8	56.8	595.7	0.5	411.9	0.7
	850	251	81.6	207.3	0.6	414	2
	212	75	95	71.3	4.3	416.5	5.8
		1 100	95	1 045			0.4
		1 320	95	1 254			0.3
		1 100	95	1 045			0.4
		1 320	95	1 254			0.3
		1 320	95	1 254			0.3
		70	95	66.5			0.3
		1 760	95	1 672			0.2
					1.5	435.2	
						441.7	
	800	236.2	85.2	201.2	0.6	341.8	1.7
	181	80	95	76	0.6	343.9	4.5
		990	93	920.7			0.4
		990	93	920.7			0.4
					0.3	346	
						347	

1.1.5. Expenditure and manufacturing costs

Machines			Expenditure			
Machine description	Number of machines	Number of units per machine	Personnel		Installed power	
			Operators per shift	Maintenance per day	Machine kW	Illumination kW
Blend						
Link	20	36	4	6.07	520	8.5
G 36	20	1 824	11.8	8.66	2 187.00	31.1
SERVOfail-F-A	1				5	
F 38 – 110	4	208	2.1	0.7	174	6
RSB-D 50	4		0.7	0.33	44.6	2.6
SB-D 50	4		0.5	0.33	34	2.9
SB-D 50	4		0.6	0.33	34	2.6
Cotton						
E 86R	6		0.5	0.41	41.7	1.4
E 26	1				2.3	
E 36	1		0.2	0.06	14.5	1
SB-D 22	1	2	0.4	0.12	13	0.7
C 70	6		0.3	0.57	166.8	1.5
A 21	1				3.8	0.1
Vision Shield	1					
B 76	1				19	0.3
B 12	1				15.3	0.1
A 22	1				11.1	0.1
B 25	1				3.2	0.3
A 12.2	1				15	1.6
Blowroom total			0.9	0.1	67.40	2.5
Polyester						
SB-D 50	2		0.5	0.17	17	1.4
C 70	5		0.3	0.48	139	1.2
A 21	1				3.8	0.1
Vision Shield	1					
B76	1				19	0.3
MT	1				37.2	
AT	1					
UNIcontrol	1					
SPIDERweb	1					
Blowroom total				0.03	71	0.4
Total			22.8	18.36	3 531.70	63.8

		Conversion costs in INR/kg Yarn
Air condition kW	Space required m ²	Total costs
81.4	1 000	20.5169
336.7	3 660	39.181
	20	0.9865
28.5	1 028	5.5101
8.8	452	1.2348
8	504	1.1073
7.6	452	1.0738
6.7	234	10.8388
		0.4027
3.1	173	0.7248
2.5	123	0.4295
18.2	258	7.6174
0.5	24	
	8	
2	70	
1.4	31	
1.1	24	
0.6	60	
3.2	340	
8.8	587	3.3756
3.9	236	0.5435
15.2	215	3.2685
0.5	24	
	8	
2	70	
4		
6.5	102	1.6306
535.9	9 014	98.9586

The total conversion costs of the produced yarn of this mill are calculated respecting:

1. The waste costs
2. The labor costs
3. The energy costs
4. The running costs
5. And the capital costs

In case you would like to have detailed information for your own mill project, please contact a Rieter sales agent.

1.1.6. Machine data

				Specification	
Model	Number of machines	Units per machine	Name of machine	Indication	Size
Blend					
Link	20	36	Package winder	Package diameter	260 mm
				Package traverse	150 mm
G 36	20	1 824	Ring spinning with doffer	Gauge	70 mm
				Ring	38 mm
				Tube	200 mm
SERVotrail F-A	1		Bobbin transport		
F 38 – 110	4	208	Roving frame	Gauge	110 mm
				Bobbin diameter	152 mm
				Bobbin traverse	405 mm
RSB-D 50	4		Draw frame 3. passage	Can diameter	500 mm
				Can height	1 200 mm
SB-D 50	4		Draw frame 2. passage	Can diameter	1 000 mm
				Can height	1 200 mm
SB-D 50	4		Draw frame 1. passage	Can diameter	1 000 mm
				Can height	1 200 mm
Cotton					
E 86R	6		Comber	Can diameter	600 mm
				Can height	1 200 mm
				Feed amount	4.7 mm
E 26	1		SERVolap		
E 36	1		OMEGAlap	Lap width	300 mm
SB-D 22	1	2	Draw frame for combing	Can diameter	1 000 mm
				Can height	1 200 mm
C 70	6		Card	Can diameter	1 000 mm
				Can height	1 200 mm
Blowroom					
A 21	1		Condenser		
Vision Shield	1		Vision Shield		
A 79	1		UNIstore		
B 12	1		UNiclean		
A 22	1		Condenser		
B 25	1		Waste opener	Feed apron	3 m
A 12	1		6UNIfloc	Length of channel	30 m
Polyester					
SB-D 50	2		Pre-draw frame	Can diameter	600 mm
				Can height	1 200 mm
C 70	5		Card	Can diameter	1 000 mm
				Can height	1 200 mm
Blowroom					
A 21	1		Condenser		
Vision Shield	1		Vision Shield		
B 76	1		UNImix		
UNIconrol	1		Control of Blowroom		

1.2. Compact spinning



Basis of calculation

Wage for operating personnel per h	EUR	3.36
Wage for skilled maintenance personnel per h	EUR	5.27
Wage for unskilled maintenance personnel per h	EUR	2.96
Personnel in reserve	%	10
Cost of electric power per kWh	EUR	0.068
Building costs per m ²	EUR	202
Yearly building maintenance	%	1.5
Number of shifts for operation per day		3
Number of shifts for maintenance per day		1
Operating hours per year	h	8520
Customs, sales tax etc. in of machine price	%	0
Capital interest	%	10.5
Depreciation for machinery	Years	10
Depreciation for accessories	Years	5
Depreciation for building	Years	30
Exchange rate for offer	EUR	1.14
Exchange rate for manufacturing cost	EUR	1.14

1.2.1. Compact spinning process for 100% cotton



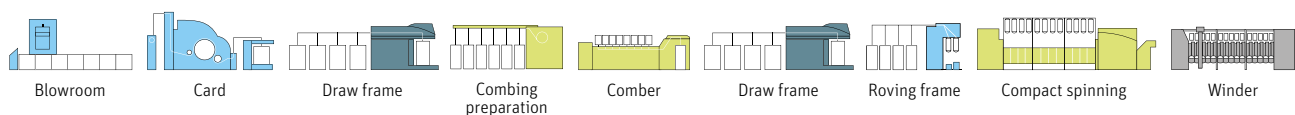
Cotton

Market

Compact spinning is represented in the main market Turkey. The basis of calculation is shown on page 20. The represented mill is running 20 compact spinning machines with 1 824 spindles each. The raw material is 100% cotton and the endproduct is a Ne 60 compact yarn with an alpha e 3.9 for warp application. The total production is 339.5 kg yarn per hour.

Process

In the following machine line, the exact process of the transformation from cotton fibers to compact yarn is symbolized by pictograms.

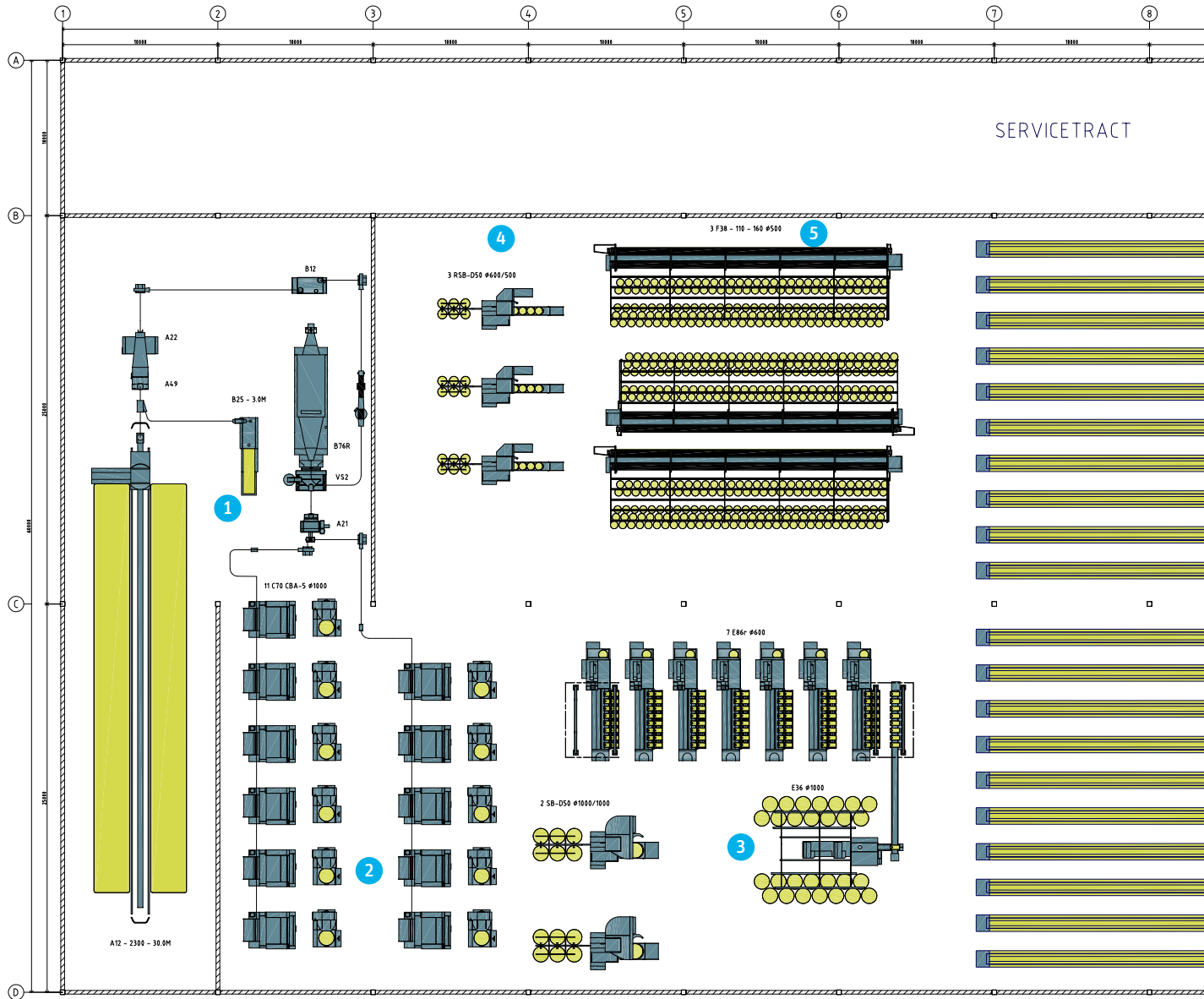


Yarn

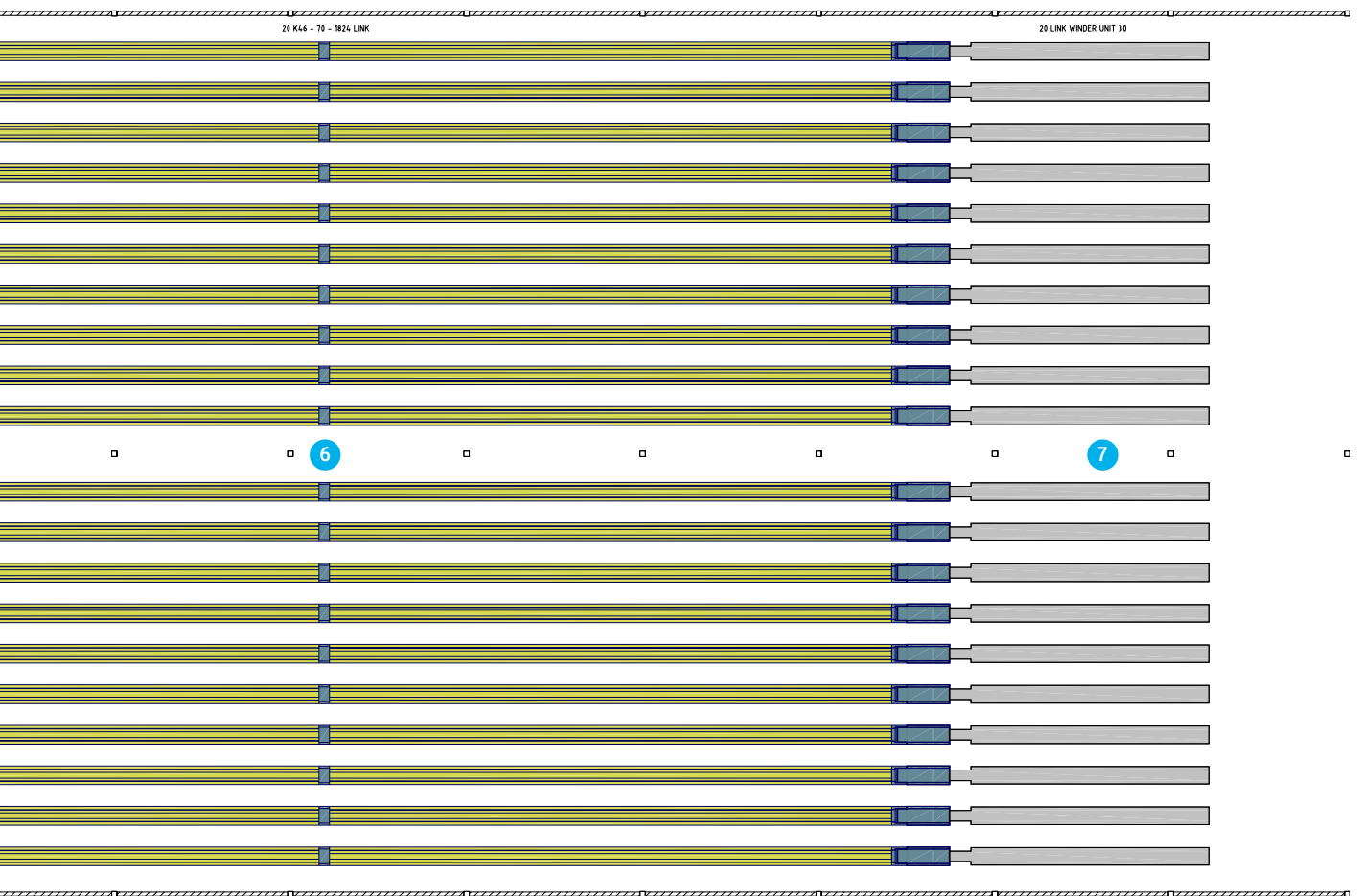
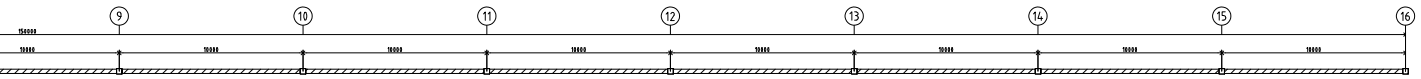
Compact yarn is known for its highest strength of all yarns from the four different end spinning technologies, as well as its optimal processing characteristics, the high wearing comfort and the finest luster.



1.2.2. Compact spinning mill for 100% cotton



- 1 Blowroom
- 2 Cards
- 3 Combing section
- 4 Draw frames
- 5 Roving frames
- 6 Ring spinning machines
- 7 Winders



Raw material	100% combed cotton
Staple length	1 1/4"
Fiber specification	3.5 mic
Example for a spinning mill in:	Turkey
Endspinning capacity	20 machines à 1 824 spindles
Utilisation	Warp

Yarn	1
Yarn count	Ne 60
Ring	38 mm

Machines				Spinning schedule			
Machine description	Number of machines	Number of units per machine	Degree of utilisation in %	Count delivered in Ne or g/m	Doubling	Draft	Twist
							alpha e
Link	20	30	88.3	60			
K 46	20	1824	100	60	1	60	3.9
SERVOftrail F-A	1						
F 38 – 110	3	160	88.3	1	1	8.3	1.09
RSB-D 50	3		71.9	0.12	6	6	
E 86R	7		97.2	0.12	8		
E 26	1						
E 36	1		74	74	24	1.6	
SB-D 50	2		99.1	0.12	5	6	
C 70	11		92.9	0.1		126.6	
A 21	1		43.8				
Vision Shield	1		36.5				
A 79	1		43.8				
B 76	1		43.8				
B 12	1		36.5				
A 22	1		36.5				
B 25	1		27.5				
A 12	1		21				
Blowroom total							
Raw material							

	Production schedule						
alpha m	Recommended processing speed in rpm or m/min	Calculated production per unit in g/h or kg/h	Efficiency in %	Practical production per unit in g/h or kg/h	Waste total in %	Total production required in kg/h	Number of calculated production units
	1200	708.6	90.4	640.6	1	339.5	530
30.2	20000	9.9	94.5	9.4	3	342.9	36478
1.09	1300	1071.1	78	835.5	1	353.5	424
	650	191.9	86.3	165.6	0.6	357	2.2
	450	57.4	92	52.8	17	359.1	6.8
	230	1021.2	57.3	585.1	0.5	432.7	0.7
	850	251	87.4	219.4	0.6	434.9	2
	127	45	95	42.8	4.3	437.5	10.2
		1100	95	1045			0.4
		1320	95	1254			0.4
		1100	95	1045			0.4
		1100	95	1045			0.4
		1320	95	1254			0.4
		1320	95	1254			0.4
		70	95	66.5			0.3
		2200	95	2090			0.2
					1.5	457.2	
						464.1	

1.2.5. Expenditure and manufacturing costs

Machines			Expenditure					
Machine description	Number of machines	Number of units per machine	Personnel		Installed power			Space required m²
			Operators per shift	Maintenance per day	Machine kW	Illumination kW	Air condition kW	
Link	20	30	2.5	4.56	400	9.1	64.1	900
K 46	20	1 824	9.6	12.13	2 317	37.8	356.3	3 740
SERVOntrail F-A	1	160			6			20
F 38 – 110	3		1.1	0.37	101.7	3.7	16.9	633
RSB-D 50	3		0.5	0.22	33.5	1.8	6.3	303
E 86R	7		0.5	0.43	48.7	1.6	7.8	273
E 26	1				2.3			
E 36	1		0.2	0.06	14	1	3	173
SB-D 50	2		0.3	0.14	17	1.5	4	252
C 70	11		0.5	0.85	305.8	2.7	33.4	473
A 21	1				3.8	0.1	0.5	24
Vision Shield	1							8
A 79	1				12.6	0.1	1.2	30
B 76	1				19	0.3	2	70
B 12	1				15.3	0.1	1.4	31
A 22	1				11.1	0.1	1.1	24
B 25	1				3.2	0.3	0.6	60
A 12	1				15	1.6	3.2	340
MT	1				42.7		3.5	
AT	1							
UNIconrol	1							
SPIDERweb	1							
Blowroom total			0.8	0.1	122.7	2.6	13.5	587
Total			16	18.86	3 368.70	61.8	505.3	7 354

Conversion costs in EUR/kg yarn**Total
costs**

0.3921

1.0465

0.0233

0.0728

0.0234

0.2847

0.0095

0.0181

0.016

0.2376

0.1048

2.2288

The total conversion costs of the produced yarn of this mill are calculated respecting:

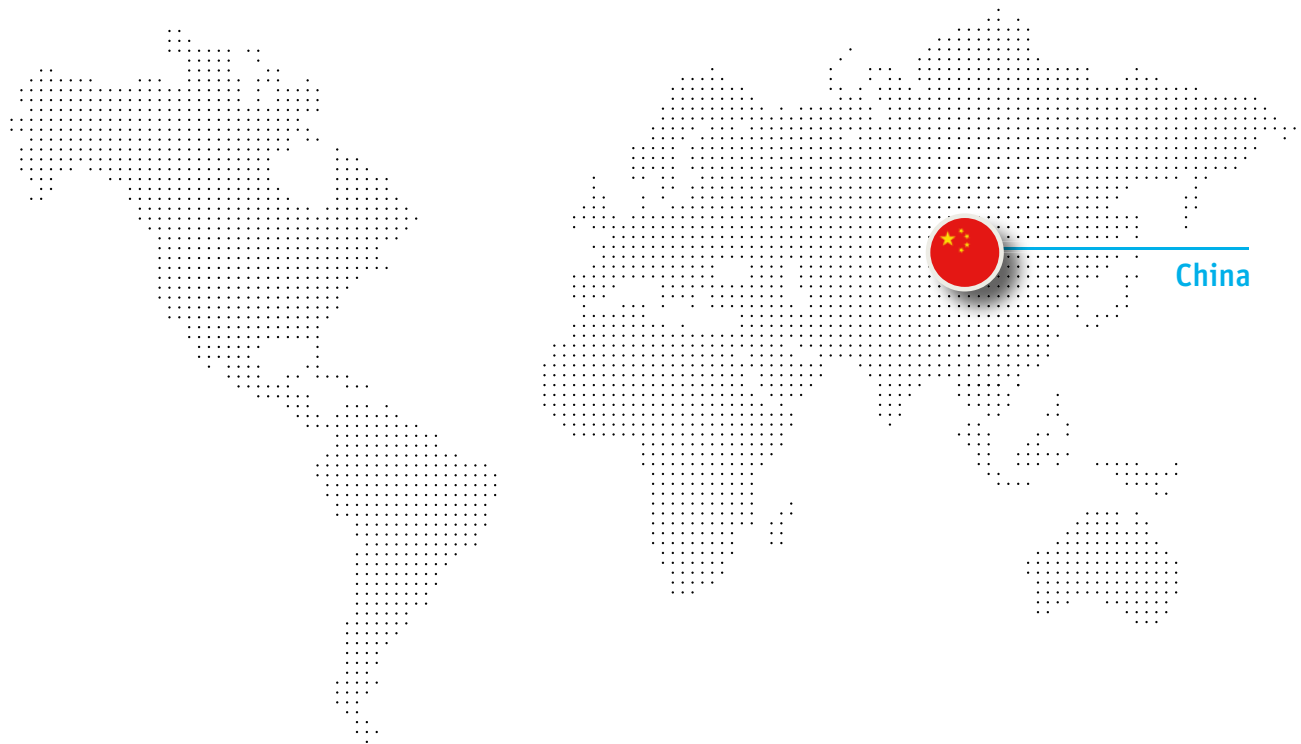
1. The waste costs
2. The labor costs
3. The energy costs
4. The running costs
5. And the capital costs

In case you would like to have detailed information for your own mill project, please contact a Rieter sales agent.

1.2.6. Machine data

				Specification	
Model	Number of machines	Units per machine	Name of machine	Indication	Size
Link	20	30	Package Winder	Package diameter	220 mm
				Package traverse	150 mm
K 46	20	1 824	Compact spinning with doffer	Gauge	70 mm
				Ring	38 mm
				Tube	190 mm
SERVOTrail F-A	1		Bobbin transport		
F 38 – 110	3	160	Roving frame	Gauge	110 mm
				Bobbin diameter	152 mm
				Bobbin traverse	405 mm
RSB-D 50	3		Draw frame 1. passage	Can diameter	500 mm
				Can height	1 200mm
E 86R	7		Comber	Can diameter	600 mm
				Can height	1 200mm
				Feed amount	4.7 mm
E 26	1		SERVOLap		
E 36	1		OMEGALap	Lap width	300 mm
SB-D 50	3		Draw frame for combing	Can diameter	1 000mm
				Can height	1 200 mm
C 70	11		Card	Can diameter	1 000mm
				Can height	1 200 mm
Blowroom					
A 21	1		Condenser		
Vision Shield 2	1		Vision Shield		
A 79	1		UNISTore		
B 76	1		UNImix		
B 12	1		UNIClean		
A 22	1		Condenser		
B 25	1		Waste Opener	Feed Apron	3 m
A 12	1		UNIfloc	Length of Channel	30 m
UNIcontrol	1		Control of Blowroom		

1.3. Semi-automated rotor spinning



Basis of calculation		
Wage for operating personnel per h	CNY	16.94
Wage for skilled maintenance personnel per h	CNY	19.93
Wage for unskilled maintenance personnel per h	CNY	13.95
Personnel in reserve	%	10
Cost of electric power per kWh	CNY	0.415
Building costs per m ²	CNY	1796
Yearly building maintenance	%	1.5
Number of shifts for operation per day		3
Number of shifts for maintenance per day		1
Operating hours per year	h	7 500
Customs, sales tax etc. in of machine price	%	6
Capital interest	%	6
Depreciation for machinery	Years	10
Depreciation for accessories	Years	5
Depreciation for building	Years	30
Exchange rate for offer	CNY	0.147
Exchange rate for manufacturing cost	CNY	0.147

1.3.1. Semi-automated rotor spinning process for regenerated fibers/polyester blends



Regenerated fibers



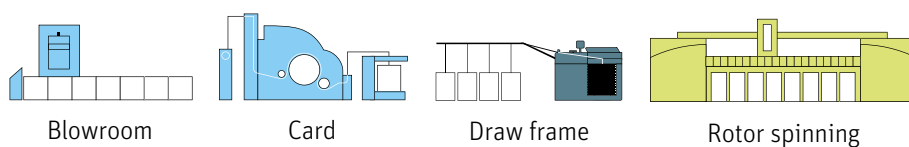
Polyester

Market

The rotor process is represented in the market China. The trend of recycling or regenerated fibers have shown an important increase in the last decade. Here shown is a rotor spinning mill for yarns out of regenerated fibers. Regenerated fibers are, in the terms of Rieter, fibers that have been reused out of fabrics and end product which were torn apart. The 10% polyester fibers in the blend are the so called carrier material. It helps to increase the strength and build a good base for a recycled fiber yarn. The represented spinning mill produces rotor yarns out of the blend in yarn count Ne 8 and an alpha e of 4.9 for warp application. The spinning mill counts four rotor spinning machines and 600 rotors per machine and is running with a total production of 1 076.1 kg yarn per hour. The blend is done in the blowroom.

Process

These pictograms symbolize the exact process line.

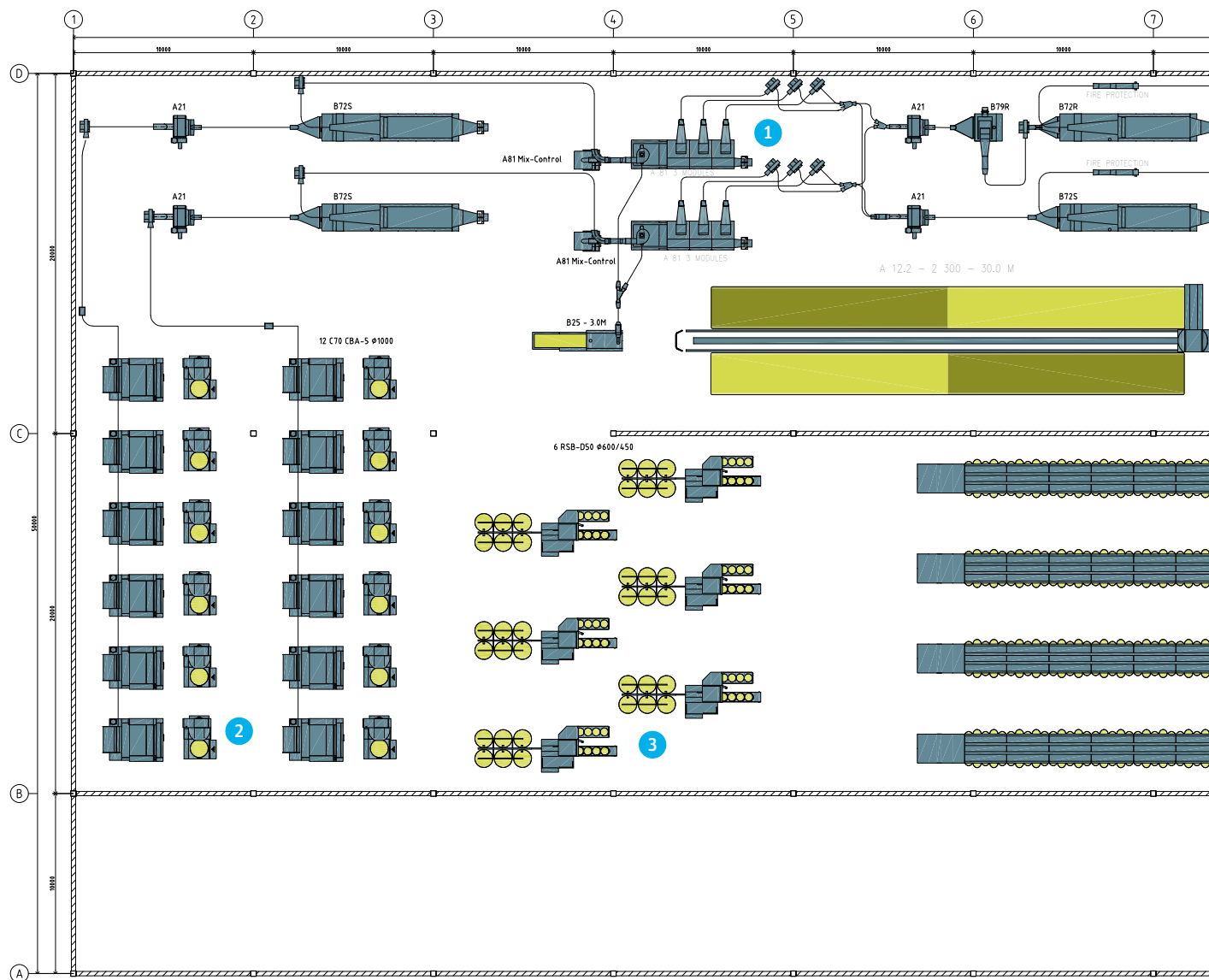


Yarn

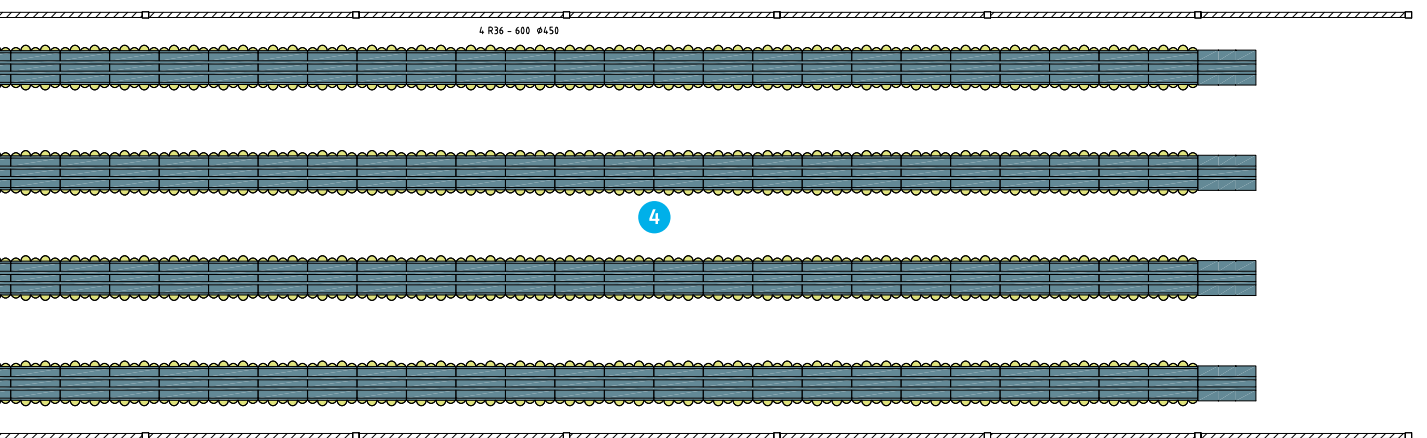
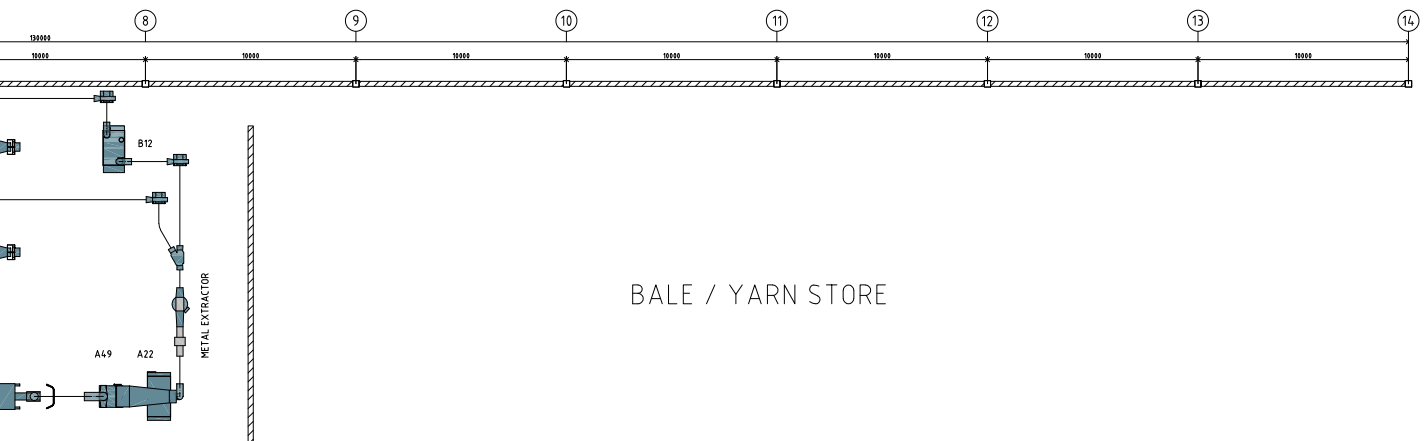
Rotor yarns are typical for their very uniform appearance in fabrics, their high abrasion resistance and the good appearance after razing.



1.3.2. Semi-automated rotor spinning mill for regenerated fibers/polyester blends



- 1 Blowroom
- 2 Cards
- 3 Draw frames
- 4 Semi-automated rotor spinning machine



1.3.3. Data for yarn and fiber

Raw material blend 90/10	Regenerated fibers (90%)	Polyester (10%)
Staple length		38 mm
Fiber specification		1.5 dtex

Example for a spinning mill in:	China
Endspinning capacity	5 machines x 600 units
Utilisation	Warp

Yarn	1
Yarn count	Ne 8
Rotor diameter	41
Rotor speed	60 000 rpm

1.3.4. Spinning and production schedule

Machines				Spinning schedule			
Machine description	Number of machines	Number of units per machine	Degree of utilisation in %	Count delivered in Ne or g/m	Doubling	Draft	Twist
							alpha e
Blend							
R 36	4	600	100	8			
Yarn 1			100	8	1	80	4.99
RSB-D 50	6		99.9	0.1	6	6.7	
C 70	12		97.7	0.09		113.9	
A 21	2		59.5				
B 72	2		68.4				
A 81	2		65.5				
Regenerated fibers							
B 76	1		98.5				
B 25	1		61.9				
A 12	1		78.8				
Blowroom total							
Raw material							
Polyester							
B 76	1		12.2				
Blowroom total							
Raw material							

	Production schedule						
alpha m	Recommended processing speed in r.p.m or m/min	Calculated production per unit in g/h or kg/h	Efficiency in %	Practical production per unit in g/h or kg/h	Waste total in %	Total production required in kg/h	Number of calculated production units
						1076.1	
14.12	60 000	478	93.8	448.4	0.7	1076.1	2 399
	600	212.6	85	180.7	0.6	1083.6	6
	254	100	93	93	2.6	1090.1	11.7
		990	95	940.5			1.2
		880	93	818.4			1.4
		900	95	855			1.3
		1100	93	1023			1
		70	93	65.1			0.6
		1320	93	1 227.6			0.8
					1.4	1007.6	
						1021.7	
		990	93	920.7			0.1
					0.3	112	
						112.3	

1.3.5. Expenditure and manufacturing costs

Machines			Expenditure					
Machine description	Number of machines	Number of units per machine	Personnel		Installed power			Space required m ²
			Operators per shift	Maintenance per day	Machine kW	Illumination kW	Air condition kW	
Blend								
R36	4	600	7	1.64	668.70	10.5	106.2	1 464
Yarn 1			7					
RSB-D 50	6		0.7	0.39	66.9	3.9	13.2	678
C 70	12		0.7	0.95	333.6	3	36.5	516
A 21	2				7.5	0.2	0.9	48
B 72	2				34	0.6	3.6	126
A 81	2				26.4	0.5	2.8	108
Regenerated fibers								
B 76	1				19	0.3	2	70
B 25	1				3.2	0.3	0.6	60
A 12.2	1				15	1.6	3.2	340
Blowroom total			1.2	0.22	105.1	3.5	13.1	752
Polyester								
B 76	1				19	0.3	2	70
MT					71.5		5.9	
UNicontrol								
SPIDERweb								
Blowroom total	1			0.02	90.5	0.3	7.9	70
Total			9.6	3.22	1 264.80	21.2	176.9	3 480
Yarn 1								

Conversion costs in CNY/kg yarn	
Totals	costs
	0.6755
	0.0884
	0.5488
	0.2433
	0.0511
	1.6071

The total conversion costs of the produced yarn of this mill are calculated respecting:

1. The waste costs
2. The labor costs
3. The energy costs
4. The running costs
5. And the capital costs

In case you would like to have detailed information for your own mill project, please contact a Rieter sales agent.

1.3.6. Machine data

				Specification	
Model	Number of machines	Units per machine	Name of machine	Indication	Size
Blend					
R 36	4	600	Rotor spinning machine		
	4		Yarn 1	Rotor diameter	41.0 mm
				Package diameter	300.0 mm
				Package traverse	150.0 mm
RSB-D 50	6		Draw frame 1. passage	Can diameter	450.0 mm
				Can height	1070.0 mm
C 70	12		Card	Can diameter	1000.0 mm
				Can height	1200.0 mm
A 21	2		Condenser		
B 72	2		UNImix		
A 81	2		UNIBlend	Moduls	3
Regenerated fibers					
B 76	1		UNImix		
B 25	1		Waste opener	Feed apron	3.0 m
A 12	1		UNIfloc	Length of channel	30.0 m
Polyester					
B 76	1		UNImix		
UNicontrol	1		Control of blowroom		

1.4. Fully automated rotor spinning



Basis of calculation		
Wage for operating personnel per h	USD	16.74
Wage for skilled maintenance personnel per h	USD	25.13
Wage for unskilled maintenance personnel per h	USD	13.96
Personnel in reserve	%	10
Cost of electric power per kWh	USD	0.06
Building costs per m²	USD	598
Yearly building maintenance	%	1.5
Number of shifts for operation per day		3
Number of shifts for maintenance per day		1
Operating hours per year	h	8400
Customs, sales tax etc. in of machine price	%	3
Capital interest	%	4
Depreciation for machinery	Years	10
Depreciation for accessories	Years	5
Depreciation for building	Years	30
Exchange rate for offer	USD	0.99
Exchange rate for manufacturing cost	USD	0.99

1.4.1. Fully automated rotor spinning process for cotton/polyester blends



Cotton



Polyester

Market

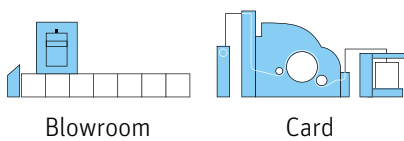
The automated rotor process is shown in the market USA. The represented spinning mill produces two rotor yarns out of a 60/40% cotton/polyester in yarn counts Ne 20 and Ne 30, with an alpha e of approximately 3.9 for knitting application.

The size of the spinning mill contains five Rotor machines with 600 rotors per machine. The total production is 774.2 kg yarn per hour. As Rotor spinning is an open end process and works with short fibers, cotton is always carded.

Process

The spinning process is symbolized in pictograms. It describes the exact process line for the blend:

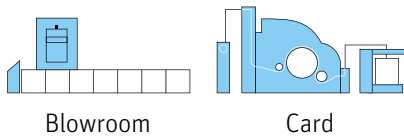
Cotton



Blowroom

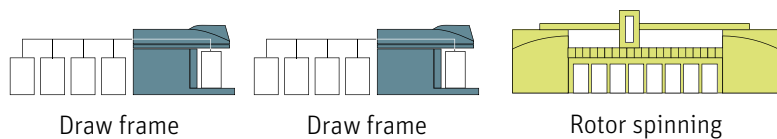
Card

Polyester



Blowroom

Card



Draw frame

Draw frame

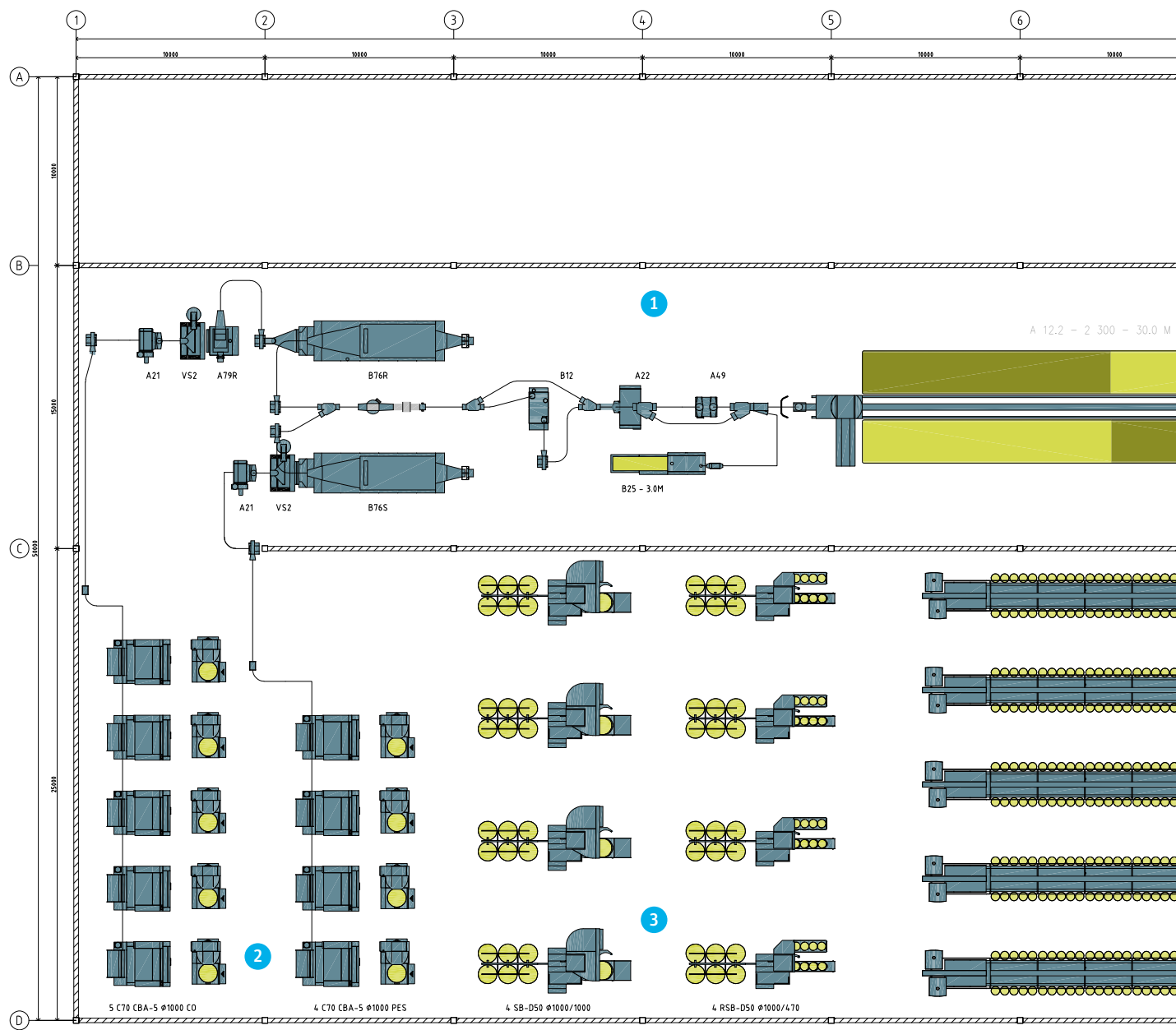
Rotor spinning

Yarn

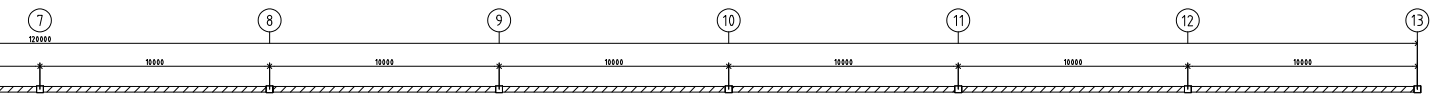
Rotor yarns are typical for their very uniform appearance in fabrics, their high abrasion resistance and the good appearance after razing.



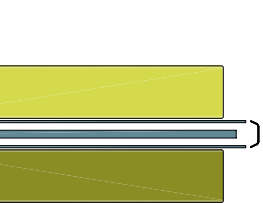
1.4.2. Fully automated rotor spinning mill for cotton/polyester blends



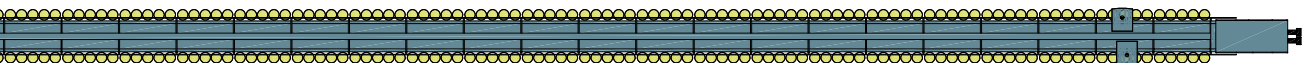
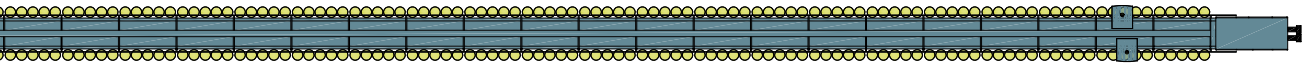
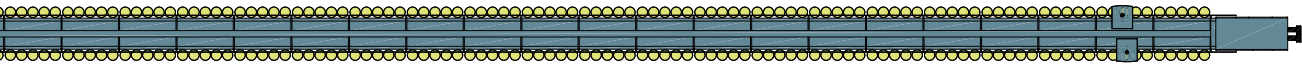
- 1 Blowroom
- 2 Cards
- 3 Draw frames
- 4 Automated rotor spinning machines



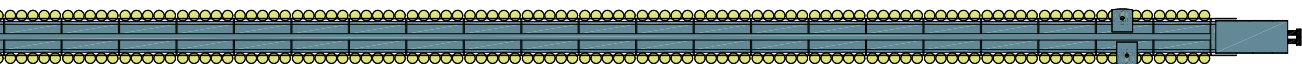
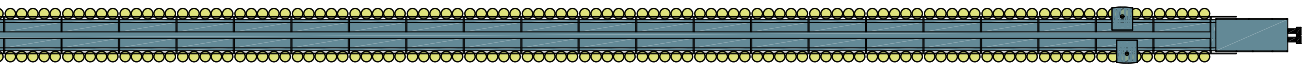
SERVICETRACK



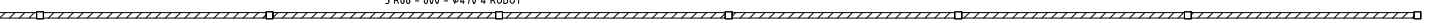
BALE / YARN STORE



4



5 R66 - 600 - Ø470 4 ROBOT



1.4.3. Data for yarn and fiber

Raw material blend 60/40	Cotton (60%)	PES (40%)
Staple length	1 1/16 "	32 mm
Fiber specification	4 mic	1.3 dtex
Example for a spinning mill in:	USA	
End spinning capacity	5 machines x 600 units	
Utilisation	Knitting	

Yarn	1	2
Yarn count	Ne 20	Ne 30
Rotor diameter	31	31
Rotor speed	125 000	125 000

1.4.4. Spinning and production schedule

Machines				Spinning schedule			
Machine description	Number of machines	Number of units per machine	Degree of utilisation in %	Count delivered in Ne or g/m	Doubling	Draft	Twist
							alpha e
Blend							
SL	5						
R 66 (4 robots)	5	600	100	22.635			
Yarn 1	3		100	20	1	166.7	3.86
Yarn 2	2		100	30	1	250	3.93
RSB-D 50	4		97.3	0.12	6	6	
SB-D 50	4		94.7	0.12	6	8	
Cotton							
C 70	5		94.9	0.1		135.2	
A 21	1		47.7				
Vision Shield	1		39.8				
A 79	1		47.7				
B 76	1		47.7				
B 12	1		39.8				
A 22	1		39.8				
B 25	1		29.9				
A 12	1		28.6				
Blowroom total							
Raw material							
Polyester							
C 70	4		80.7	0.075		89.3	
A 21	1		34.5				
Vision Shield	1		25.9				
B 76	1		34.5				
Blowroom total							
Raw material							

	Production schedule						
	Recommended processing speed in r.p.m or m/min	Calculated production per unit in g/h or kg/h	Efficiency in %	Practical production per unit in g/h or kg/h	Waste total in %	Total production required in kg/h	Number of calculated production units
alpha m							
						774.2	
17.27	125 000	325.7	97.4	317.2	0.7	571	1 800
21.51	125 000	174.3	97.1	169.2	0.7	203.2	1 200
	800	236.2	84.8	200.3	0.6	779.6	3.9
	800	236.2	87.7	207.1	0.6	784.3	3.8
	296	105	95	99.8	5	473.4	4.7
		1 100	95	1 045			0.5
		1 320	95	1 254			0.4
		1 100	95	1 045			0.5
		1 100	95	1 045			0.5
		1 320	95	1 254			0.4
		1 320	95	1 254			0.4
		70	95	66.5			0.3
		1 760	95	1 672			0.3
					2.5	498.5	
						511.5	
	222	105	93	97.7	0.6	315.6	3.2
		990	93	920.7			0.3
		1 320	93	1 227.6			0.3
		990	93	920.7			0.3
					0.3	317.5	
						318.5	

1.4.5. Expenditure and manufacturing costs

Machines			Expenditure			
Machine description	Number of machines	Number of units per machine	Personnel		Installed power	
			Operators per shift	Maintenance per day	Machine kW	Illumination kW
Blend						
SL	5					
R 66 (4 robots)	5	600	2.7	1.52	1 536.90	17.7
Yarn 1	3		1.8			
Yarn 2	2		0.9			
RSB-D 50	4		0.4	0.23	44.6	2.6
SB-D 50	4		0.5	0.23	34	2.9
Cotton						
C 70	5		0.3	0.38	139	1.2
A 21	1				3.8	0.1
Vision Shield	1					
A 79	1				12.6	0.1
B 76	1				19	0.3
B 12	1				15.3	0.1
A 22	1				11.1	0.1
B 25	1				3.2	0.3
A 12	1				15	1.6
Blowroom total			0.7	0.08	80	2.6
Polyester						
C 70	4		0.2	0.31	111.2	1
A 21	1				3.8	0.1
Vision Shield	1					
B 76	1				19	0.3
MT	1				53.7	
UNIcontrol	1					
SPIDERweb	1					
Blowroom total				0.02	76.5	0.4
Total			4.8	2.77	2 022.20	28.4
Yarn 1						
Yarn 2						

		Conversion costs in USD/lb yarn
Air condition kW	Space required m ²	Totals costs
231.7	2 085	
		0.1087
		0.1902
8.8	452	0.0105
8	504	0.011
15.2	215	0.037
0.5	24	
	8	
1.2	30	
2	70	
1.4	31	
1.1	24	
0.6	60	
3.2	340	
10	587	0.0259
12.2	172	0.0147
0.5	24	
	8	
2	70	
4.5		
7	102	0.0051
292.9	4 117	
		0.2129
		0.2944

The total conversion costs of the produced yarn of this mill are calculated respecting:

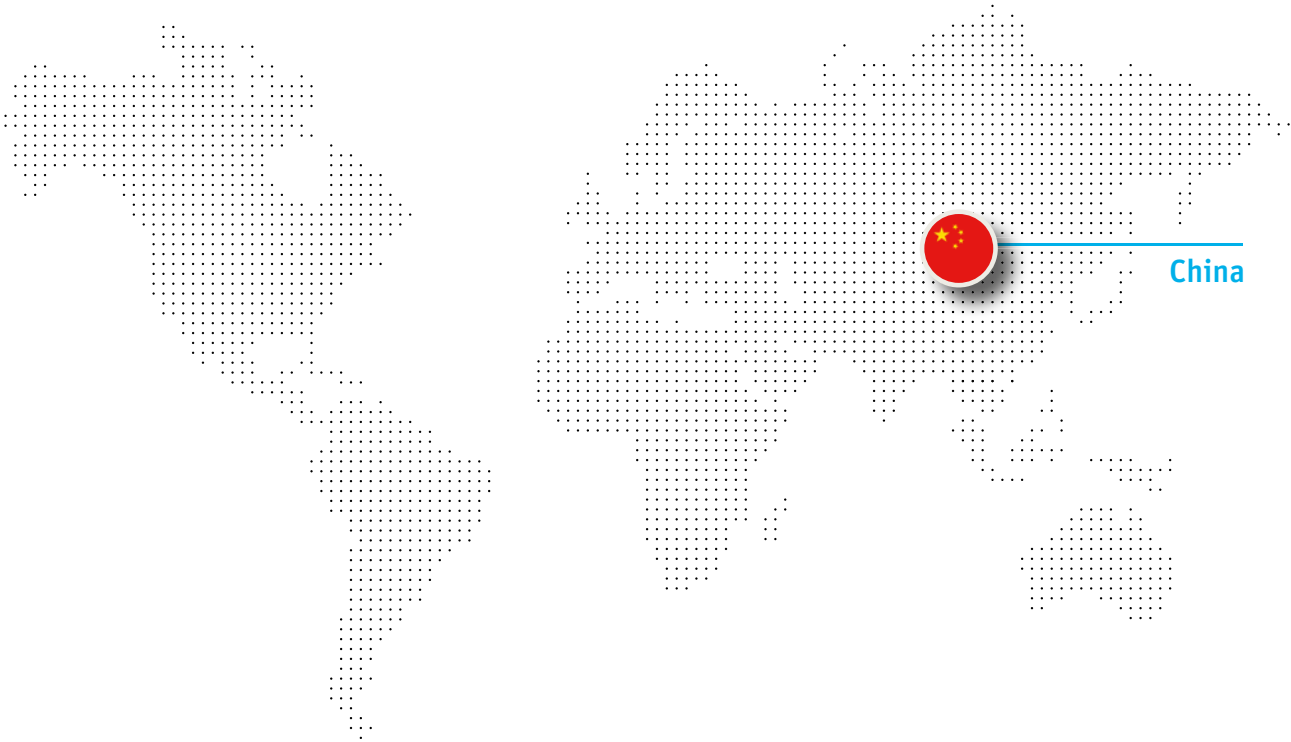
1. The waste costs
2. The labor costs
3. The energy costs
4. The running costs
5. And the capital costs

In case you would like to have detailed information for your own mill project, please contact a Rieter sales agent.

1.4.6. Machine data

				Specification	
Machines	Number of machines	Units per machine	Name of machine	Indication	Size
Model					
Blend					
SL	5		Bobbin lift system		
R 66	5	600	Rotor spinning machine		
	3		Yarn 1	Rotor diameter	31 mm
				Package diameter	300 mm
				Package traverse	150 mm
	2		Yarn 2	Rotor diameter	31 mm
				Package diameter	300 mm
				Package traverse	150 mm
Robots per machine	4		Automation		
RSB-D 50	4		Draw frame 2. Passage	Can diameter	470 mm
				Can height	1 070 mm
SB-D 50	4		Draw frame 1. Passage	Can diameter	1 000 mm
				Can height	1 200 mm
Cotton					
C 70	5		Card	Can diameter	1 000 mm
				Can height	1 200 mm
A 21	1		Condenser		
Vision Shield	1		Vision Shield		
A 79	1		UNIstore		
B 76	1		UNImix		
B 12	1		UNIclean		
A 22	1		Condenser		
B 25	1		Waste opener	Feed apron	3 m
A 12	1		Unifloc	Length of channel	30 m
Polyester					
C 70	4		Card	Can diameter	1 000 mm
				Can height	1 200 mm
A 21	1		Condenser		
Vision Shield	1		Vision Shield		
B 76	1		UNImix		
UNIcontrol	1		Control of blowroom		

1.5. Air-jet spinning



Basis of calculation		
Wage for operating personnel per h	CNY	16.94
Wage for skilled maintenance personnel per h	CNY	19.93
Wage for unskilled maintenance personnel per h	CNY	13.95
Personnel in reserve	%	10
Cost of electric power per kWh	CNY	0.415
Building costs per m²	CNY	1796
Yearly building maintenance	%	1.5
Number of shifts for operation per day		3
Number of shifts for maintenance per day		1
Operating hours per year	h	7 500
Customs, sales tax etc. in of machine price	%	6
Capital interest	%	6
Depreciation for machinery	Years	10
Depreciation for accessories	Years	5
Depreciation for building	Years	30
Exchange rate for offer	CNY	0.147
Exchange rate for manufacturing cost	CNY	0.147

1.5.1. Air-jet spinning process for 100% cellulosic fibers



Cellulosic fibers

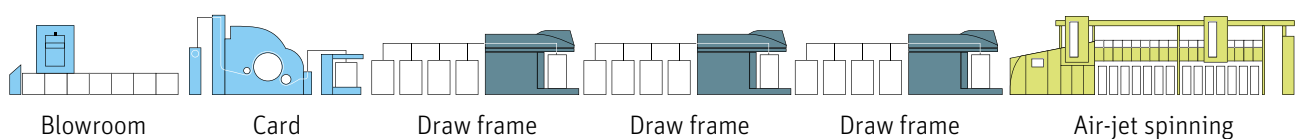
Market

The air-jet spinning mills will be represented in the market China. As raw material, 100% carded cellulosic fibers is processed to medium yarns counts. The three yarns which are produced in the air-jet spinning mill have yarn counts from Ne 24 to Ne 40 and production speed of 440 m/min. The yarn will further be transformed into knitting fabrics.

The capacity of the spinning mill contains six air-jet machines with 200 spinning units each. The total production is 594.4 kg yarn per hour.

Process

The process is the following:

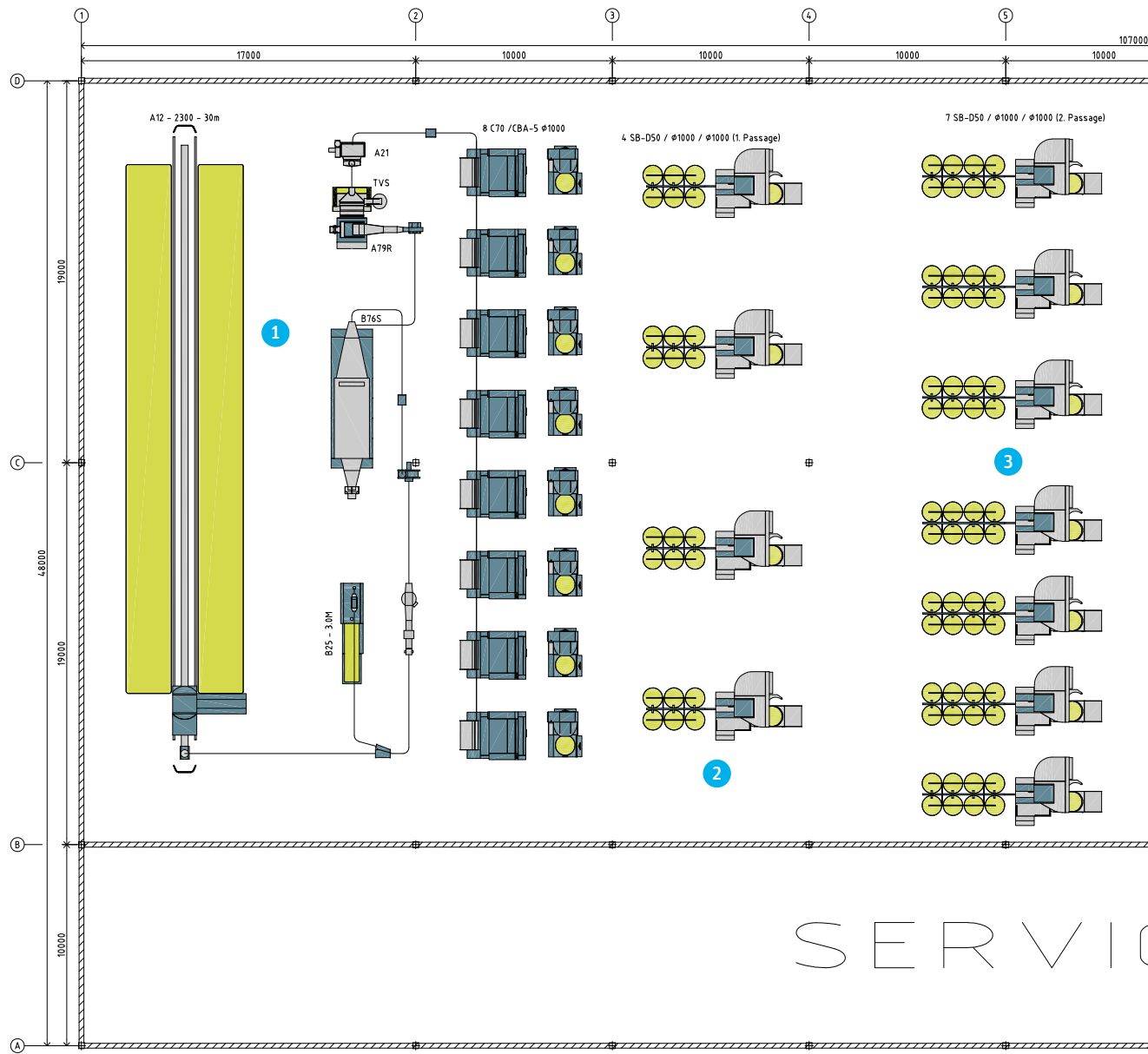


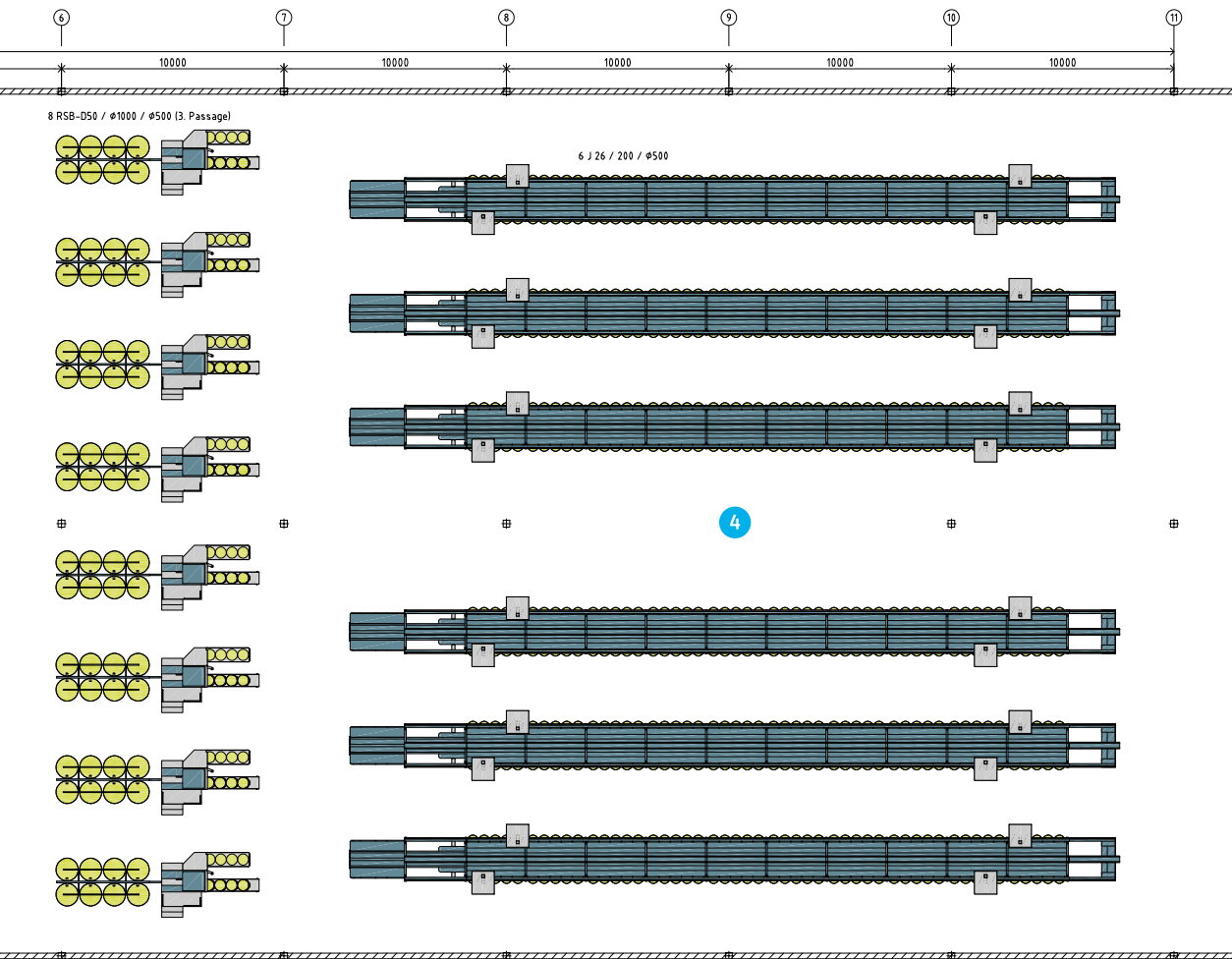
Yarn

Air-jet yarn is distinguished by a very low pilling tendency, uniform fabric appearance and high washing resistance.



1.5.1.1. Air-jet spinning mill for 100% cellulosic fibers





CE AREA

1.5.1.2. Data for yarn and fiber

Raw material 100% carded	Viscose
Staple length	38 mm
Fiber specification	1.3 dtex
Example for a spinning mill in:	China
Endspinning capacity	6 machines x 200 units
Utilisation	Knitting

Yarn	1	2	3
Yarn count	Ne 24	Ne 30	Ne 40
Speed (m/min)	440	440	440

1.5.1.3. Spinning and production schedule

Machines				Spinning schedule			
Machine description	Number of machines	Number of units per machine	Degree of utilisation in%	Count delivered in Ne or g/m	Doubling	Draft	Twist
							alpha e
J 26 (4 robots)	6	200		30			
Yarn 1	2		100	24	1	141.2	
Yarn 2	2		100	30	1	176.5	
Yarn 3	2		100	40	1	235.3	
RSB-D 50	8		80.8	0.17	6	6.4	
SB-D 50	7		86.1	0.16	6	7.4	
SB-D 50	4		96.1	0.13	6	6.5	
C 70	8		94.7	0.12		142.9	
A 21	1		60.8				
A 79	1		67.5				
B 76	1		60.8				
B 25	1		38.3				
A 12	1		29.2				
Blowroom total							
Raw material							

	Production schedule						
alpha m	Recommended processing speed in r.p.m or m/min	Calculated production per unit in g/h or kg/h	Efficiency in %	Practical production per unit in g/h or kg/h	Waste total in %	Total production required in kg/h	Number of calculated production units
						594.7	
	440	649.6	95	617.1	0.6	246.9	400
	440	519.7	96	498.9	0.6	199.6	400
	440	389.8	95	370.3	0.7	148.2	400
	500	104.2	89.3	93.1	0.6	601.8	6.5
	500	110.7	90.7	100.4	0.6	605.4	6
	650	177.1	88.9	157.4	0.6	609	3.9
	288	85	95	80.8	1.5	612.7	7.6
		1 100	93	1 023			0.6
		990	93	920.7			0.7
		1 100	93	1 023			0.6
		70	93	65.1			0.4
		2 200	93	2 046			0.3
					0.3	621.9	
						623.8	

1.5.1.4. Expenditure and manufacturing costs

Machines			Expenditure		
Machine description	Number of machines	Number of units per machine	Personnel		Installed power
			Operators per shift	Maintenance per day	Machine kW
J 26 (4 robots)	6	200	4.1	1.78	543
Yarn 1	2		1.4		
Yarn 2	2		1.4		
Yarn 3	2		1.3		
RSB-D 50	8		0.6	0.52	89.2
SB-D 50	7		0.5	0.45	59.6
SB-D 50	4		0.4	0.26	34
C 70	8		0.5	0.63	222.4
A 21	1				3.8
A 79	1				12.6
B 76	1				6.4
B 25	1				3.2
A 12	1				15
MT					27.5
UNIcontrol					
SPIDERweb					
Blowroom total			0.7	0.08	68.5
Total			6.8	3.72	1 016.70
Yarn 1					
Yarn 2					
Yarn 3					

		Conversion costs in CNY/kg yarn
		Total costs
Illumination kW	Air condition kW	
11.7	88.4	
		1.8564
		2.289
		3.0349
5.2	17.7	0.1966
5.1	14.1	0.1557
2.9	8	0.1007
2	24.3	0.6251
0.1	0.5	
0.1	1.2	
0.3	0.9	
0.3	0.6	
1.6	3.2	
	2.3	
2.4	8.7	0.2242
29.3	161.2	
		3.1587
		3.5913
		4.3372

The total conversion costs of the produced yarn of this mill are calculated respecting:

1. The waste costs
2. The labor costs
3. The energy costs
4. The running costs
5. And the capital costs

In case you would like to have detailed information for your own mill project, please contact a Rieter sales agent.

1.5.1.5. Machine data

				Specification	
Model	Number of machines	Units per machine	Name of machine	Indication	Size
J 26	6	200	Air-Jet		
	2		Yarn 1	Package diameter	300 mm
				Package traverse	150 mm
	2		Yarn 2	Package diameter	300 mm
				Package traverse	150 mm
	2		Yarn 3	Package diameter	300 mm
				Package traverse	150 mm
Robots p/m	4		Automation		
RSB-D 50	8		Draw frame 3. passage	Can diameter	500 mm
				Can height	1070 mm
SB-D 50	7		Draw frame 2. passage	Can diameter	1000 mm
				Can height	1200 mm
SB-D 50	4		Draw frame 1. passage	Can diameter	1000 mm
				Can height	1200 mm
C 70	8		Card	Can diameter	1000 mm
				Can height	1200 mm
Blowroom					
A 21	1		Condenser		
A 79	1		UNIstore		
B 76	1		UNImix		
B 25	1		Waste Opener	Feed Apron	3 m
A 12	1		UNIfloc	Lenght of Channel	30 m
UNIcontrol	1		Control of Blowroom		

1.5.2. Air-jet spinning process for cotton/polyester blends



Cotton



Polyester

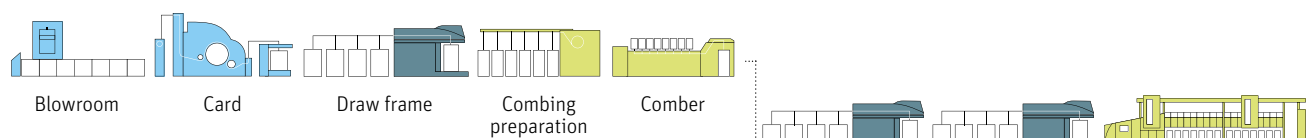
Market

The last spinning mill example shows an air-jet process for a 50% cotton/50% polyester blend. The combed cotton and the carded polyester are blended on the draw frame. The three produced yarns are in a yarn count range from Ne 24 to Ne 40 for knitting application. The spinning mill runs 10 air-jet spinning machines with 120 spinning units each and has running speed from 420 m/min and 430m/min. The total production is 565.1 kg yarn per hour.

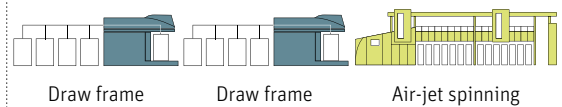
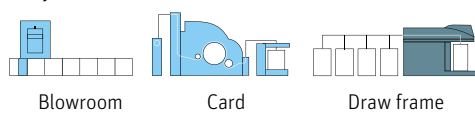
Process

The process is symbolized as follows:

Cotton



Polyester

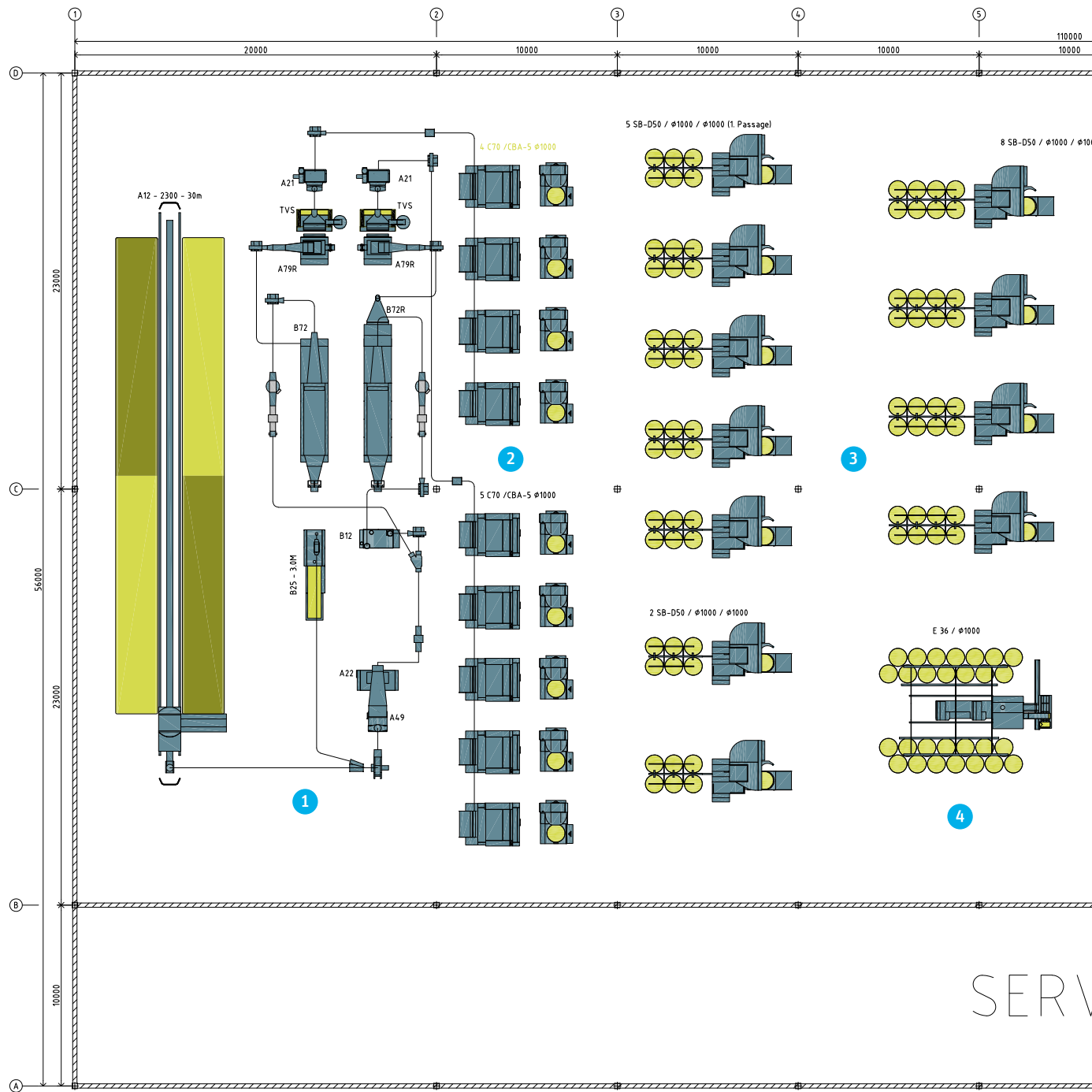


Yarn

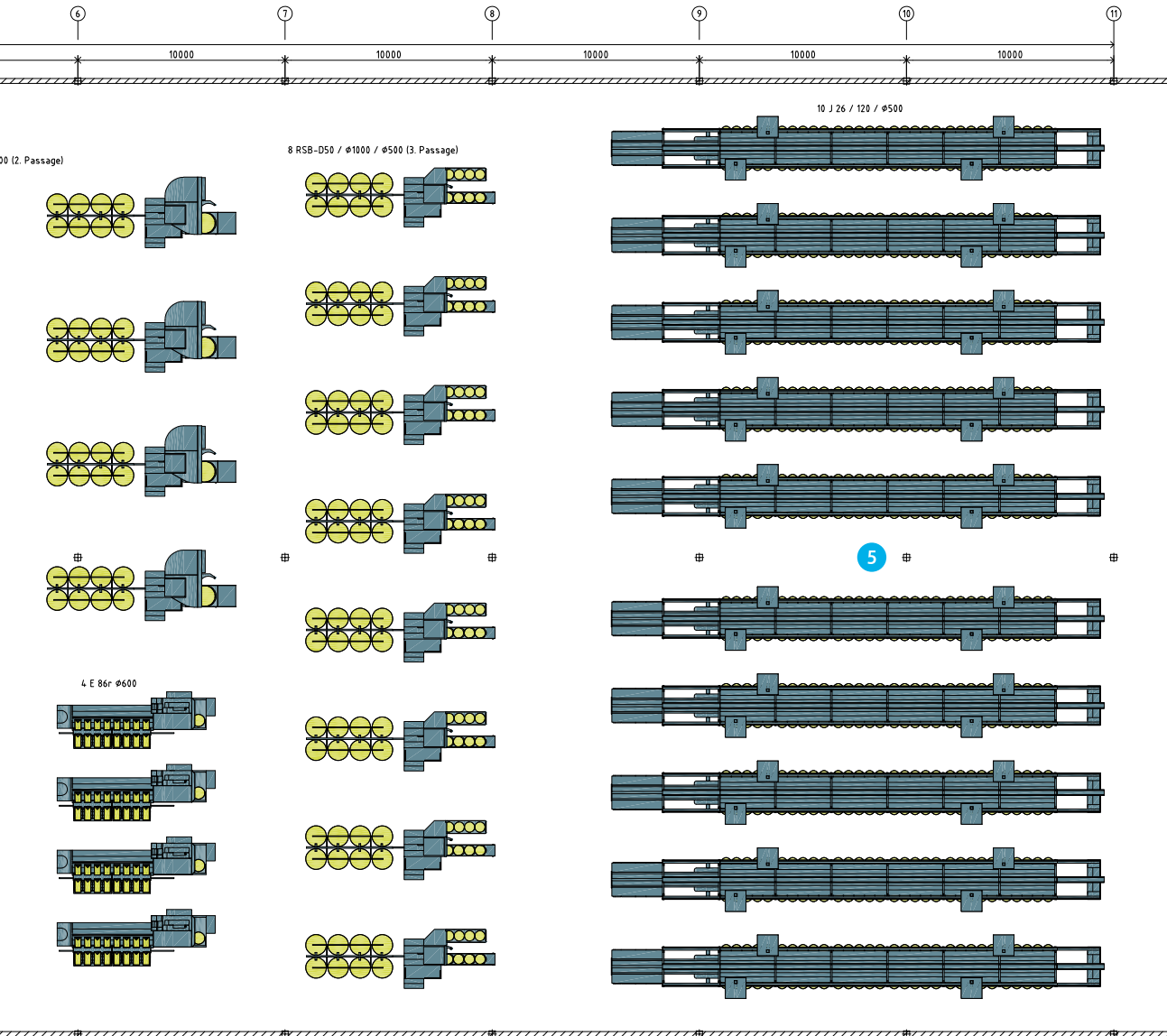
Air-jet yarn is distinguished by a very low pilling tendency, uniform fabric appearance and high washing resistance.



1.5.2.1. Air-jet spinning mill for cotton/polyester blends



- 1 Blowroom
- 2 Cards
- 3 Draw frames
- 4 Combing section
- 5 Air-jet spinning machines



VICE AREA

1.5.2.2. Data for yarn and fiber

Raw material blend 50/50	Combed cotton	Polyester	Yarn	1	2	3
Staple length	1 1/16 "	38 mm	Yarn count	Ne 24	Ne 30	Ne 40
Fiber specification	4.5 mic	1.3 dtex	Speed (m/min)	430	430	420
Example for a spinning mill in:			China			
Endspinning capacity	10 machines x 120 Units					
Utilisation	Knitting					

1.5.2.3. Spinning and production schedule

Machines				Spinning schedule			
Machine description	Number of machines	Number of units per machine	Degree of utilisation in%	Count delivered in Ne or g/m	Doubling	Draft	Twist alpha e
Blend							
J 26 (4 robots)	10	120	100	29.994			
Yarn 1	3		100	24	1	141.2	
Yarn 2	3		100	30	1	176.5	
Yarn 3	3		100	40	1	235.3	
RSB-D 50	8		96.4	0.17	6	6.4	
SB-D 50	8		90.1	0.16	6	7.4	
SB-D 50	5		96.7	0.13	6	6.5	
Cotton							
E 86R	4		99.8	0.12	8		
E26	1						
E36	1		58.9	80	24	1.5	
SB-D 50	2		86.5	0.12	5	6	
C 70	5		89.2	0.1		126.6	
A21	1		36.3				
Vision Shield	1		30.3				
A 79	1		36.3				
B 76	1		36.3				
B 12	1		30.3				
A 22	1		30.3				
B 25	1		22.9				
A 12	1		17.4				
Blowroom total							
Raw material							
Polyester							
C 70	4		91.5	0.12		142.9	
A 21	1		32.3				
Vision Shield	1		24.3				
A 79	1		36.4				
B 76	1		32.3				
Blowroom total							
Raw material							

Production schedule							
alpha m	Recommended processing speed in r.p.m or m/min	Calculated production per unit in g/h or kg/h	Efficiency in %	Practical production per unit in g/h or kg/h	Waste total in %	Total production required in kg/h	Number of calculated production units
						565.5	
	430	634.8	92	584	2.7	210.4	360
	430	507.9	94	477.4	2.7	229.2	480
	420	372	94	349.7	2.7	125.9	360
	400	83.4	90.4	75.4	0.6	581.3	7.7
	400	88.6	91.5	81.1	0.6	584.8	7.2
	500	136.3	89.3	121.7	0.6	588.3	4.8
	525	80.1	92.5	74.1	17	295.9	4
	230	1104	54.8	605	0.5	356.6	0.6
	800	236.2	87.7	207.1	0.6	358.4	1.7
	240	85	95	80.8	5	360.6	4.5
		1100	95	1045			0.4
		1320	95	1254			0.3
		1100	95	1045			0.4
		1100	95	1045			0.4
		1320	95	1254			0.3
		1320	95	1254			0.3
		70	95	66.5			0.2
		1500	95	1425			0.2
					2.5	379.7	
						389.6	
	288	85	95	80.8	0.6	295.9	3.7
		990	93	920.7			0.3
		880	93	818.4			0.4
		990	93	920.7			0.3
					0.3	297.7	
						298.6	

1.5.2.4. Expenditure and manufacturing costs

Machines			Expenditure		
Machine description	Number of machines	Number of units per machine	Personnel		Installed power
			Operators per shift	Maintenance per day	Machine kW
Blend					
J 26 (4 robots)	10	120	5.2	2.07	563
Yarn 1	3		1.6		
Yarn 2	4		2.1		
Yarn 3	3		1.5		
RSB-D 50	8		0.5	0.52	89.2
SB-D 50	8		0.5	0.52	68.1
SB-D 50	5		0.5	0.33	42.6
Cotton					
E 86R	4		0.3	0.21	27.8
E 26	1			0.04	2.3
E 36	1		0.2	0.13	14
SB-D 50	2		0.3	0.4	17
C 70	5		0.3		139
A 21	1				3.8
Vision Shield	1				
A 79	1				12.6
B 76	1				19
B 12	1				15.3
A 22	1				11.1
B 25	1				3.2
A12	1				15
Blowroom total			0.7	0.08	80
Polyester					
C 70	4		0.4	0.32	111.2
A 21	1				3.8
Vision Shield	1				
A 79	1				12.6
B 76	1				19
MT	1				59.2
AT	1				
UNIcontrol	1				
SPIDERweb					
Blowroom total				0.03	94.6
Total			8.9	4.65	1 249.20
Yarn 1					
Yarn2					
Yarn3					

		Conversion costs in RMB/kg yarn
		Total costs
Illumination kW	Air condition kW	
14.1	94	
		2.1278
		2.5937
		3.4795
5.2	17.7	0.1945
5.8	16.1	0.1775
3.3	9.5	0.1177
0.9	4.5	0.4791
		0.0361
1	3	0.0674
1.5	4	0.0565
1.2	15.2	0.5974
0.1	0.5	
0.1	1.2	
0.3	2	
0.1	1.4	
0.1	1.1	
0.3	0.6	
1.6	3.2	
2.6	10	0.3602
0.9	12	0.2653
0.1	0.5	
0.1	1.2	
0.3	2	
	3.5	
0.5	8.6	0.1566
37	194.7	
		4.6361
		5.102
		5.9878

The total conversion costs of the produced yarn of this mill are calculated respecting:

1. The waste costs
2. The labor costs
3. The energy costs
4. The running costs
5. And the capital costs

In case you would like to have detailed information for your own mill project, please contact a Rieter sales agent.

1.5.2.5. Machine data

Model	Number of machines	Units per machine	Name of machine	Specification	
				Indication	Size
Blend					
J 26	10	120	Air-jet		
	3		Yarn 1	Package diameter	300 mm
				Package traverse	150 mm
	4		Yarn 2	Package diameter	300 mm
				Package traverse	150 mm
	3		Yarn 3	Package diameter	300 mm
				Package traverse	150 mm
Robots p/m	4		Automation		
RSB-D 50	8		Draw frame 3. passage	Can diameter	500 mm
				Can height	1 070 mm
SB-D 50	8		Draw frame 2. passage	Can diameter	500 mm
				Can height	1 070 mm
SB-D 50	5		Draw frame 1. passage	Can diameter	500 mm
				Can height	1 070 mm
Cotton					
E 86R	4		Comber	Can diameter	600mm
				Can height	1 200 mm
				Feed amount	5.2 mm
E 26	1		SERVOLap		
E 36	1		OMEGALap	Lap width	300 mm
SB-D 50	2		Draw frame for combing	Can diameter	1 000 mm
				Can height	1 200 mm
C 70	5		Card	Can diameter	1 000 mm
				Can height	1 200 mm
A 21	1		Condenser		
Vision Shield	1		Vision Shield		
A 79	1		UNIstore		
B 76	1		UNImix		
B 12	1		UNiclean		
B 22	1		Condenser		
B 25	1		Waste opener	Feed upon	3 m
A 12	1		UNIfloc	Length of channel	30 m
Polyester					
C 70	4		Card	Can diameter	600 mm
				Can height	1 200 mm
A 21	1		Condenser		
Vision Shield	1		Vision Shield		
A 79	1		UNIstore		
B 76	1		UnImix		
UNIcontrol	1		Contrl of blowroom		

2. Data for Spinning Schedules

2.1. Proposed spinning schedules in Ne

2.1.1. Ring and compact spinning schedules

Ring carded cotton	Proposed spinning schedule in Ne/g/m								
Ring spinning machine	4.5	8	12	16	20	24	30	36	40
Roving frame	0.5	0.6	0.6	0.7	0.8	0.8	0.8	1	1
Draw frame RSB-D 50	0.1	0.1	0.1	0.11	0.11	0.11	0.11	0.12	0.12
Draw frame SB-D 50	0.1	0.1	0.1	0.11	0.11	0.11	0.11	0.12	0.12
Card C 70	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.08	0.08

Ring carded MMF	Proposed spinning schedule in Ne/g/m												
Ring spinning machine	4.5	8	12	16	20	24	30	36	40	50	60	70	80
Roving frame	0.7	0.7	0.7	0.7	0.8	0.8	0.8	1	1	1.1	1.2	1.3	1.8
Draw frame RSB-D 50	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.12	0.12	0.13	0.13	0.14	0.14
Draw frame SB-D 50	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.12	0.12	0.13	0.13	0.14	0.14
Card C 70	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.7	0.7	0.08	0.08	0.08	0.08

Ring combed cotton	Proposed spinning schedule in Ne/g/m											
Ring spinning machine	20	24	30	36	40	50	60	80	100	120	140	160
Roving frame	0.6	0.7	0.7	0.8	0.8	0.9	1	1.2	1.4	2	2	2.6
Draw frame RSB-D 50	0.1	0.11	0.11	0.11	0.11	0.12	0.12	0.13	0.14	0.16	0.16	0.17
Comber E 86	0.1	0.11	0.11	0.11	0.11	0.12	0.12	0.13	0.14	0.16	0.16	0.16
Combing preparation E 36	76	76	76	76	76	80	80	74	74	74	74	70
Draw frame SB-D 50	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12
Card C 70	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1

Ring combed CO/MMF blend (MMF < 50%)	Proposed spinning schedule in Ne/g/m			
Ring spinning machine	4.5	8	12	16
Roving frame	0.7	0.7	0.7	0.7
Draw frame RSB-D 50	0.11	0.11	0.11	0.11
Draw frame SB-D 50	0.11	0.11	0.11	0.11
Draw frame SB-D 50	0.11	0.11	0.11	0.11
Cotton				
Comber E 86	0.12	0.12	0.12	0.12
Combing preparation E 36	80	80	80	80
Draw frame SB-D 50	0.12	0.12	0.12	0.12
Card C 70	0.1	0.1	0.1	0.1
Polyester				
Card C 70	0.92	0.92	0.92	0.92

The spinning schedule tables are based on some of Rieter's newest process recommendations and technological essays. Anyway, they can serve as a reference without any assurance. Please contact your Rieter agent when in need of detailed spinning schedules for your mill.

Rotor carded cotton waste noils direct process	Proposed spinning schedule in Ne/g/m			
Rotor spinning machine	3	6	8	10
Draw frame RSB modul	0.1	0.1	0.1	0.1
Card	0.05	0.05	0.05	0.05

Rotor carded cotton waste noils	Proposed spinning schedule in Ne/g/m			
Rotor spinning machine	11	12	14	16
Draw frame RSB-D 50	0.11	0.11	0.11	0.11
Draw frame SB-D 50	0.11	0.11	0.11	0.11
Card	0.09	0.09	0.09	0.09

Rotor carded regenerated fibers	Proposed spinning schedule in Ne/g/m			
Rotor spinning machine	11	12	14	16
Draw frame RSB-D 50	0.11	0.11	0.11	0.11
Card	0.09	0.09	0.09	0.09

Rotor carded regenerated fibers direct process	Proposed spinning schedule in Ne/g/m			
Rotor spinning machine	3	6	8	10
Draw frame RSB modul	0.1	0.1	0.1	0.1
Card	0.06	0.06	0.06	0.06

The spinning schedule tables are based on some of Rieter's process recommendations and technological essays. Anyway, they can serve as a reference without any assurance. Please contact your Rieter agent when in need of detailed spinning schedules for your mill.

2.1.3. Air-jet spinning schedules

Cotton		PES
Length:	1 1/16 inch	38 mm
Fineness:	3.5 – 4.5 mic	1.3 dtex

Air-jet carded viscose 1 dtex	Proposed spinning schedule in Ne/g/m		
Air-jet spinning machine	50	60	70
Draw frame RSB-D 50	0.28	0.28	0.31
Draw frame SB-D 50	0.22	0.22	0.24
Draw frame SB-D 50	0.16	0.16	0.17
Card	0.13	0.13	0.14

Air-jet carded viscose > 1 dtex	Proposed spinning schedule in Ne/g/m		
Air-jet spinning machine	24	30	40
Draw frame RSB-D 50	0.17	0.17	0.23
Draw frame SB-D 50	0.16	0.16	0.19
Draw frame SB-D 50	0.13	0.13	0.15
Card	0.12	0.12	0.13

Air-jet carded cotton	Proposed spinning schedule in Ne/g/m		
Air-jet spinning machine J 26	20	30	40
Draw frame RSB-D 50	0.14	0.17	0.23
Draw frame SB-D 50	0.14	0.16	0.19
Draw frame SB-D 50	0.12	0.13	0.15
Card C 70	0.12	0.12	0.13

Air-jet blend PES/combed CO	Proposed spinning schedule in Ne/g/m		
Air-jet spinning machine J 26	30	40	
Draw frame RSB-D 50	0.17	0.23	
Draw frame SB-D 50	0.16	0.19	
Draw frame SB-D 50	0.13	0.15	
Cotton			
Comber E 86	0.12	0.13	
Combing preparation OMEGAlap	80	80	
Draw frame SB-D 50	0.12	0.12	
Card C 70	0.1	0.1	
Polyester			
Card C 70	0.12	0.13	

The spinning schedule tables are based on some of Rieter's process recommendations and technological essays. Anyway, they can serve as a reference without any assurance. Please contact your Rieter agent when in need of detailed spinning schedules for your mill.

2.2. Production data

2.2.1. Ring and compact spinning

2.2.1.1. Machine data ring spinning

Yarn count			Machine data		
Ne	Nm	tex	Gauge in mm	Ring diameter in mm	Tube lenght in mm
4.5 – 15	7.6 – 26	131 – 38	75	51	250
16 – 19	27 – 32	37 – 31	75	48	240
20 – 23	33 – 39	30 – 26	70	45	230
24 – 29	40 – 49	25 – 21	70	42	210
30 – 49	50 – 83	20 – 13	70	40	200
50 – 99	84 – 168	12 – 7	70	38	190
100 – 250	169 – 424	6 – 2.3	70	36	180

2.2.1.2. Raw material

Material	Staple lenth	
	Inch	mm
CO	>= 1 1/16	27
PES	<= 2 1/2	60
Blends	<= 2	51

2.2.1.3. Yarn count range

Raw material	Yarn range		
	Ne	Nm	Tex
100% cotton	4.5 – 250	7.5 – 420	132 – 2.3
MMF, blends	4.5 – 160	7.5 – 270	132 – 3.6

2.2.1.4. Machine data compact spinning

Yarn count			Machine data		
Ne	Nm	tex	Gauge in mm	Ring diameter in mm	Tube lenght in mm
10 – 15	17 – 26	59 – 38	75	51	250
16 – 19	27 – 32	37 – 31	75	48	240
20 – 23	33 – 39	30 – 26	70	45	230
24 – 29	40 – 49	25 – 21	70	42	210
30 – 49	50 – 83	20 – 13	70	40	200
50 – 99	84 – 168	12 – 7	70	38	190
100 – 250	169 – 424	6 – 2.3	70	36	180

2.2.1.5. Raw material

Material	Staple lenth	
	Inch	mm
CO	>= 1 1/16	27
PES	<= 2 1/2	51
Blends	<= 2 1/2	51

2.2.1.6. Yarn twist

200 – 3 000 T/m
5.1 – 76.1 T/°

2.2.1.7. Delivery speed

4 – 35 m /min

2.2.1.8. Spindle speed (ring T-flange) on ring spinning in rpm

		For denim yarn							
Max. spindle speed rpm	Alpha e	3.4 – 3.59	3.6 – 4.2	3.4 – 3.59	3.6 – 4.2	3.0 – 3.49	3.5 – 4.0	3.0 – 3.49	3.5 – 4.0
Yarn count in Ne	Ring diameter in mm	Cotton 100%		Blends > 65% CO		Blends =< 65% CO		Man-made fibers	
30	36	17 900	18 800	17 900					
36		18 700	19 600	18 700	19 600	18 500	19 900	18 000	19 300
40		19 000	20 000	19 000	20 000	18 800	20 300	18 400	19 700
50		19 000	20 000	19 000	20 000	18 800	20 300	18 400	19 700
60		18 700	19 800	18 700	19 800	18 500	20 000	18 000	19 400
70		18 000	19 300	18 000	19 300	18 000	19 500	17 600	18 900
80		17 400	18 800	17 400	18 800	17 600	19 100	17 000	18 500
90		16 800	18 100	16 800	18 100	17 100	18 600	16 600	18 000
100		16 000	17 500		17 500				
120		15 200	16 700		16 700				
140		14 700	15 400		15 400				
150		12 600	14 000		14 000				
160		11 600	13 000		13 000				
24	38	16 400	17 300	16 400	17 300				
30		17 400	18 300	17 400	18 300	17 200	18 600	16 600	18 000
36		18 200	19 100	18 200	19 100	18 000	19 300	17 600	18 800
40		18 500	19 500	18 500	19 500	18 300	19 800	17 900	19 200
50		18 500	19 500	18 500	19 500	18 300	19 800	17 900	19 200
60		18 200	19 300	18 200	19 300	18 000	19 400	16 000	18 900
70		17 500	18 800	17 500	18 800	17 600	19 000	17 100	18 400
80		16 900	18 300	16 900	18 300	17 100	18 600	16 600	18 000
90		18 300	17 600	16 300	17 600	16 700	18 100	16 100	17 600
100		15 600	17 000		17 000				
120		14 800	16 200		16 200				
140		13 600	15 000		15 000				
20	40	15 000	16 000	15 000	16 000				
24		16 000	16 900	16 000	16 900	15 400	16 800	14 800	16 200
30		17 000	17 800	17 000	17 800	16 800	18 100	16 200	17 500
36		17 700	18 600	17 700	18 600	17 500	18 800	17 100	18 300
40		18 000	19 000	18 000	19 000	17 900	19 300	17 400	18 700
50		18 000	19 000	18 000	19 000	17 900	19 300	17 400	18 700
60		17 700	18 800	17 700	18 800	17 500	18 900	17 400	18 700
70		17 100	18 300	17 100	18 300	17 100	18 500	17 100	18 400
80		16 500	17 800	16 500	17 800	16 700	18 100	16 700	18 000
90		15 900	17 200	15 900	17 200	16 300	17 700	16 200	17 500
100		15 200	16 600		16 600			15 700	17 100

		For denim yarn							
Max. spindle speed rpm	Alpha e	3.4 – 3.59	3.6 – 4.2	3.4 – 3.59	3.6 – 4.2	3.0 – 3.49	3.5 – 4.0	3.0 – 3.49	3.5 – 4.0
Yarn count in Ne	Ring diameter in mm	Cotton 100%		Blends > 65% CO		Blends =< 65% CO		Man-made fibers	
12	42	11 400	12 800	11 400	12 800				
14		12 300	13 700	12 300	13 700				
16		13 400	14 600	13 400	14 600				
20		14 600	15 600	14 600	15 600	13 800	15 000	13 100	14 300
24		15 600	16 500	15 600	16 500	15 000	16 400	14 400	15 800
30		16 600	17 400	16 600	17 400	16 400	17 600	15 800	17 100
36		17 300	18 200	17 300	18 200	17 100	18 400	16 700	17 900
40		17 600	18 500	17 600	18 500	17 400	18 800	17 000	18 300
50		17 600	18 500	17 600	18 500	17 400	18 800	17 000	18 300
60		17 300	18 300	17 300	18 300	17 100	18 500	16 700	18 000
70		16 700	17 900	16 700	17 900	16 700	18 100	16 300	17 500
10	45	9 900	11 200	9 900	11 200	9 600	10 500	9 100	10 000
11		10 400	11 700	10 400	11 700	10 000	10 900	9 400	10 300
12		11 400	12 400	11 100	12 400	10 500	11 300	9 700	10 700
14		11 900	13 200	11 900	13 200	11 300	12 100	10 600	11 700
16		12 900	14 100	12 900	14 100	12 200	13 100	11 500	12 700
20		14 100	15 100	14 100	15 100	13 300	14 500	12 600	13 800
24		15 100	15 900	15 100	15 900	14 500	15 800	13 900	15 200
30		16 000	16 800	16 000	16 800	15 800	17 000	15 200	16 500
36		16 700	17 500	16 700	17 500	16 500	17 800	16 100	17 300
40		17 000	17 900	17 000	17 900	16 800	18 200	16 400	17 700
6	48	7 500	8 700	7 500	8 700				
7		7 900	9 200	7 900	9 200	7 900	8 800	7 600	8 400
8		8 400	9 800	8 400	9 800	8 300	9 200	8 100	9 000
9		8 900	10 200	8 900	10 200	8 800	9 700	8 400	9 300
10		9 500	10 800	9 500	10 800	9 300	10 100	8 800	9 700
11		10 000	11 300	10 000	11 300	9 700	10 500	9 100	10 000
12		10 700	12 000	10 700	12 000	10 200	10 900	9 400	10 400
14		11 500	12 800	11 500	12 800	10 900	11 700	10 300	11 300
16		12 500	13 700	12 500	13 700	11 800	12 700	11 100	12 300
20		13 700	14 600	13 700	14 600	12 900	14 100	12 200	13 400
24		14 600	15 400	14 600	15 400	14 100	15 300	13 500	14 700
30		15 500	16 200	15 500	16 200	15 300	16 500	14 700	16 000

		For denim yarn							
Max. spindle speed rpm	Alpha e	3.4 – 3.59	3.6 – 4.2	3.4 – 3.59	3.6 – 4.2	3.0 – 3.49	3.5 – 4.0	3.0 – 3.49	3.5 – 4.0
Yarn count in Ne	Ring diameter in mm	Cotton 100%		Blends > 65% CO		Blends =< 65% CO		Man-made fibers	
4.5	51	6000	6900	6100	7300	6100	6600	5600	6200
5		6400	7400	6500	7800	6400	7000	6000	6600
5.5		6900	7900	6800	8400	6800	7500	6400	7100
6		7300	8400	7300	8900	7200	8000	6800	7600
7		7700	8900	7700	9500	7700	8500	7400	8100
8		8100	9500	8100	9900	8100	8900	7900	8700
9		8700	9900	8700	10500	8500	9400	8100	9000
10		9200	10500	9200	11000	9000	9800	8500	9400
11		9700	11000	9700	11600	9400	10100	8800	9700
12		10400	11600	10400	12400	9900	10600	9100	10000
14		11200	12400	11200	13300	10600	11400	9900	11000
16		12100	13300	12100	14200	11500	12300	10800	11900
20		13300	14200	13300	15000	12500	13600	11800	13000
24		14200	15000	14200	15000	13600	14900	13100	14300
4.5	54	5800	6700	5900	6700	5900	6400	5400	6000
5		6200	7200	6300	7100	6300	6800	5800	6400
5.5		6700	7700	6600	7600	6600	7300	6300	6900
6		7100	8200	7100	8200	7000	7700	6600	7400
7		7500	8700	7500	8700	7500	8300	7200	7900
8		7900	9200	7900	9200	7800	8700	7600	8500
9		8400	9600	8400	9600	8300	9100	7900	8700
10		9000	10200	9000	10200	8700	9500	8300	9100
11		9500	10700	9500	10700	9100	9900	8600	9400
12		10100	11300	10100	11300	9600	10300	8800	9800
14		10800	12000	10800	12000	10300	11100	9700	10700
16		11800	12900	11800	12900	11100	12000	10500	11600
20		12900	13800	12900	13800	12200	13300	11500	12600
24		13800	14500	13800	14500	13300	14500		

2.2.1.9. Spindle speed (ring S-flange) on ring spinning in rpm

Max. spindle speed rpm	Alpha e	3.4 – 3.59	3.6 – 4.2	3.0 – 3.49	3.5 – 4.0	3.0 – 3.49	3.5 – 4.0
Yarn count (Ne)	Ring diameter in mm	Cotton 100%		Blends =< 65% CO		Man-made fibers	
36	36	19 000	20 800	18 600	20 400	17 800	19 700
40		19 400	21 400	18 900	21 000	19 300	20 300
50		19 400	21 400	18 900	21 000	19 300	20 300
60		18 900	21 000	18 600	20 500	17 900	19 700
70		18 500	20 400	18 200	20 000	17 500	19 200
80		17 800	19 700	17 600	19 400	16 900	18 600
90		17 200	19 000	17 000	18 800	16 600	18 300
30	38	17 800	19 200	17 100	19 000	16 600	18 400
36		18 500	20 200	18 100	19 900	17 300	19 200
40		18 800	20 800	18 400	20 400	17 800	19 800
50		18 800	20 800	18 400	20 400	17 800	19 800
60		18 400	20 400	18 100	20 000	17 500	19 200
70		18 000	19 800	17 700	19 400	17 000	18 700
80		17 400	19 200	17 100	18 900	16 500	18 100
90	40	16 700	18 500	16 600	18 300	16 100	17 800
24		16 300	17 700	15 500	17 200	15 000	16 600
30		17 300	18 700	16 700	18 500	16 200	18 000
36		18 100	19 700	17 700	19 400	16 900	18 700
40		18 400	20 300	18 000	19 900	17 300	19 300
50		18 400	20 300	18 000	19 900	17 300	19 300
60		18 000	20 000	17 700	19 500	17 000	18 700
70	42	17 500	19 300	17 200	18 900	16 600	18 200
80		16 900	18 700	16 700	18 400	16 100	17 700
90		16 300	18 100	16 200	17 900	15 700	17 300
20		14 900	16 400	14 000	15 600	13 600	14 900
24		15 900	17 300	15 100	16 800	14 600	16 200
30		16 900	18 200	16 300	18 100	15 800	17 500
36		17 600	19 200	17 200	18 900	16 500	18 300
40	42	17 900	19 800	17 500	19 400	16 900	18 800
50		17 900	19 800	17 500	19 400	16 900	18 800
60		17 500	19 400	17 200	19 000	16 600	18 300
70		17 100	18 800	16 800	18 500	16 200	17 800

2.2.1.10. Yarn twist for ring spinning

Raw material	Yarn count			Warp				Knitting			
				Twist		Alpha of twist		Twist		Alpha of twist	
	<= Ne	<= Nm	>= Tex	Per inch	Per meter	Alpha e	Alpha m	Per inch	Per meter	Alpha e	Alpha m
Cotton 1" - 1 1/8"	8	13.55	73.82	11.31	445.35	4	121	10.1	397.51	3.57	108
	12	20.32	49.21	13.85	545.45	4	121	12.37	486.84	3.57	108
	16	27.09	36.91	16	629.83	4	121	14.28	532.16	3.57	108
	20	33.87	29.53	17.89	704.17	4	121	15.96	628.51	3.57	108
	24	40.64	24.61	19.95	771.38	4	121	17.49	688.5	3.57	108
Cotton > 1 1/8"	8	13.55	73.82	11.03	434.31	3.9	118	9.82	386.46	3.47	105
	16	27.09	36.91	15.6	614.21	3.9	118	13.88	546.54	3.47	105
	24	40.64	24.606	19.11	752.25	3.9	118	17	669.38	3.47	105
	30	50.8	19.69	21.36	841.04	3.9	118	19.01	748.38	3.47	105
	40	67.74	14.76	24.67	971.15	3.9	118	21.95	864.16	3.47	105
	60	101.6	9.84	30.21	1189.41	3.9	118	26.88	1058.38	3.47	105
	80	135.469	7.382	34.88	1373.42	3.9	118	31.04	1222.11	3.47	105
	100	169.336	5.905	39	1535.53	3.9	118	34.71	1366.36	3.47	105
Man made fibers	120	203.204	4.921	42.72	1682.08	3.9	118	38.02	1496.77	3.47	105
	8	13.55	73.82	10.47	412.23	3.7	112	9.63	379.1	3.40-3.41	103
	16	27.09	36.91	14.81	582.98	3.7	112	13.62	536.13	3.40-3.41	103
	24	40.64	24.606	18.14	714	3.7	112	16.68	656.63	3.40-3.41	103
	30	50.8	19.69	20.28	798.28	3.7	112	18.65	734.13	3.40-3.41	103
	40	67.74	14.76	23.41	921.77	3.7	112	21.53	847.7	3.40-3.41	103
	60	101.6	9.84	26.18	1030.57	3.7	112	24.07	947.76	3.40-3.41	103
	80	135.469	7.382	33.11	1303.58	3.7	112	30.45	1198.83	3.40-3.41	103
Regenerated fibers	100	169.336	5.905	37.02	1457.45	3.7	112	34.04	1340.33	3.40-3.41	103
	120	203.204	4.921	40.55	1596.55	3.7	112	37.29	1468.26	3.40-3.41	103
	8	13.55	73.82	14.72	579.33	5.21	157.4				
	16	27.09	36.91	22.4	881.76	5.6	169.4				
Waste	24	40.64	24.606	30.38	1195.95	6.2	187.6				
	30	50.8	19.69	33.96	1337.11	6.2	187.6				
	8	13.55	73.82	11.87	467.44	4.19	127	10.75	423.77	3.8	115
Linen	16	27.09	36.91	16	629.83	4	121	14.28	562.16	3.57	108
	24	40.64	24.606	19.59	771.38	4	121	17.49	688.5	3.57	108
	8	13.55	73.82	14.72	579.33	5.21	157.4				
	16	27.09	36.91	22.4	881.76	5.6	169.4				
	24	40.64	24.606	30.38	1195.95	6.2	187.6				
	30	50.8	19.69	33.96	1337.11	6.2	187.6				

2.2.1.11. Cops weight on ring spinning in gramm

Ring diameter in mm	Tube diameter inside in mm	Cops length in mm	Combed cotton in g	Carded cotton in g	PES/PAN in g	Cellulosic in g	Blends CO/MMF in g
36	18	180	37	36	41	40	39
	18	190	40	38	45	43	42
	18	200	44	43	50	48	46
	18	210	47	45	53	51	49
	20	220	42	41	48	46	45
	20						
38	18	180	45	43	50	48	47
	18	190	48	47	54	52	51
	18	200	53	52	60	58	56
	18	210	57	55	64	62	60
	20	220	53	51	60	58	56
	20	230	56	55	64	61	59
40	18	190	52	50	64	62	60
	18	200	58	56	71	68	66
	18	210	62	59	76	73	70
	20	220	59	57	72	69	67
	20	230	63	60	77	74	72
	20	240	66	64	81	78	76
42	18	200	66	64	81	78	76
	18	210	71	69	87	84	81
	20	220	69	67	85	82	79
	20	230	74	71	91	87	84
	20	240	78	75	96	92	89
	20	250	82	80	101	97	94
45	18	200	80	77	98	94	92
	18	210	86	83	106	101	99
	20	220	86	83	105	101	98
	20	230	91	88	112	107	104
	20	240	97	93	119	114	111
	20	250	102	99	125	120	117
48	18	200	85	82	108	103	101
	18	210	91	88	116	111	108
	20	220	92	89	117	113	109
	20	230	98	95	125	120	117
	20	240	104	101	133	127	124
	20	250	110	107	141	135	131
51	18	210	105	102	134	129	125
	20	220	108	104	137	132	128
	20	230	115	111	146	141	137
	20	240	122	118	156	149	145
	20	250	129	125	165	158	154
54	20	220	124	120	158	152	148
	20	230	133	128	169	162	158
	20	240	141	136	180	173	168
	20	250	150	145	191	183	178

2.2.1.12. Package weight on winder

	Lenght of H	Diameter D/ Ds	Cylindrical package	Conical package 1-51		Conical package 4-20	
			Netto package weight	Lenght of ds	Netto package weight	Diameter ds	Netto package weight
			in g	in mm	in g	in mm	in g
Combed cotton spec. weight	150	160	1 060	149	910	134	890
		180	1 380	169	1 220	154	1 180
		200	1 750	189	1 570	174	1 510
		220	2 150	209	1 950	194	1 880
		240	2 590	229	2 370	214	2 280
		260	3 070	249	2 820	234	2 730
		280	3 590	269	3 320	254	3 210
		300	4 140	289	3 850	274	3 720
Carded cotton spec. weight	150	320	4 740	309	4 420	294	4 280
		160	1 030	149	890	134	870
		180	1 350	169	1 190	154	1 150
		200	1 700	189	1 530	174	1 470
		220	2 100	209	1 900	194	1 830
		240	2 530	229	2 310	214	2 230
		260	2 990	249	2 750	234	2 660
		280	3 500	269	3 240	254	3 130
PES/PAN spec. weight	150	300	4 040	289	3 750	274	3 630
		320	4 620	309	4 310	294	4 170
		160	1 210	149	1 050	134	1 020
		180	1 580	169	1 400	154	1 350
		200	2 000	189	1 790	174	1 730
		220	2 460	209	2 230	194	2 150
		240	2 970	229	2 710	214	2 620
		260	3 520	249	3 240	234	3 120
Cellulosic viscose spec. weight	150	280	4 110	269	3 800	254	3 670
		300	4 750	289	4 410	274	4 270
		320	5 430	309	5 060	294	4 900
		160	1 160	149	1 000	134	970
		180	1 520	169	1 340	154	1 300
		200	1 920	189	1 720	174	1 660
		220	2 360	209	2 140	194	2 060
		240	2 840	229	2 600	214	2 510
Blends CO-MMF spec. weight	150	260	3 370	249	3 100	234	2 990
		280	3 940	269	3 640	254	3 520
		300	4 550	289	4 220	274	4 090
		320	5 200	309	4 850	294	4 690
		160	1 130	149	980	134	950
		180	1 480	169	1 310	154	1 270
		200	1 870	189	1 680	174	1 620
		220	2 310	209	2 090	194	2 020
	150	240	2 780	229	2 540	214	2 450
		260	3 290	249	3 030	234	2 930
		280	3 850	269	3 560	254	3 440
		300	4 450	289	4 130	274	4 000
		320	5 080	309	4 740	294	4 590

[illegible]

2.2.1.14. Efficiency compact spinning

[illegible]

2.2.2.1. Raw material

2.2.2.2. Yarn count range

2.2.2.3. Rotor speed R 66 (in 1 000 rpm)

Knitting yarn

Raw material	Rotor diameter in mm						
	26	28	29	30	31	33	34
CO > 1 1/16"		140		130	125	115 – 120	100 – 110
CO 1" – Noils- Waste; Blends: CO/Waste		130			105 – 115	100 – 110	85 – 105
CV			140		130	110 – 125	
100% PES; Blends: PES > 65%						80 – 105	80 – 105
Blends: PES < 65%		135		130	125	110 – 120	105 – 115
PAN						95 – 105	95 – 105
Blends:MMF/CO – MMF except PES		140		130	125	115 – 120	100 – 115
Regenerated fibers						110 – 85	110 – 85

36	37	40	41	46	47	56	57
90 – 100	90 – 100	75 – 90	75 – 90	55 – 70	55 – 70	45 – 65	45 – 65
75 – 95	75 – 95	72 – 85	72 – 85	60 – 70	60 – 70	55 – 60	55 – 60
90 – 100	90 – 100		75 – 90		55 – 65		45 – 55
80 – 95	75 – 80	65 – 80	65 – 80	60 – 65	60 – 65	50 – 55	50 – 55
85 – 105	85 – 105	80 – 90	80 – 90	75	75	60 – 65	60 – 65
		75 – 85	75 – 85	70 – 75	70 – 75	55 – 60	55 – 60
90 – 100	90 – 100	75 – 90	75 – 90	55 – 70	55 – 70	45 – 65	45 – 65
80 – 95	80 – 95	80	80	60 – 75	60 – 75	45 – 55	45 – 55

36	37	40	41	46	47	56	57
90 – 100	90 – 100	75 – 90	75 – 90	55 – 70	55 – 70	45 – 65	45 – 65
75 – 95	75 – 95	72 – 85	72 – 85	60 – 70	60 – 70	55 – 60	55 – 60
90 – 100		75 – 90		55 – 65		45 – 55	
80 – 95	75 – 95	65 – 80	65 – 80	55 – 65	55 – 65	50 – 55	50 – 55
90 – 95	90 – 95	75 – 85	75 – 85	55 – 65	55 – 65	50 – 55	50 – 55
80 – 95	80 – 95	75 – 85	75 – 85	70 – 75	70 – 75	55 – 60	55 – 60
90 – 100	90 – 100	75 – 90	75 – 90	55 – 70	55 – 70	45 – 65	45 – 65
80 – 95	80 – 95	80	80	60 – 75	60 – 75	45 – 55	45 – 55

2.2.2.4. Typical preferred rotor diameter depending on yarn count and material

- Take the influence of the rotor diameter into account when selecting the rotor.
- A smaller rotor diameter equals greater yarn strength, as well as greater elongation and lower power requirements at the same speed.
- A smaller rotor diameter enables a higher operating speed and increased production.
- A larger rotor diameter produces fewer bands and more voluminous (softer) yarn.
- A larger rotor diameter enables a coarser yarn, longer fibers and smaller alpha.
- Determine the rotor type based on the material and area of application.

	Rotor diameter in mm	Weaving yarn
CO 100% fiber length 1 1/16"	28	Ne 24 – 60
	31	
	33	Ne 10 – 24
	36	
	37	Ne 8 – 10
	40	
	41	Ne 6 – 8
	46	
CO 100% fiber length < 1 1/16" noil 100 % blends CO/waste	47	Ne 4 – 6
	57	Ne 3 – 4
	31	
	33	Ne 10 – 20
	34	
	36	
	37	Ne 8 – 10
	40	
PES 100% and blends with PES > 65 %	41	Ne 6 – 8
	46	
	47	Ne 4 – 6
	56	
	57	Ne 3 – 4
	34	Ne 30 – 40
	36	Ne 16 – 30
	37	Ne 8 – 16
CV 100%	40	
	41	Ne 6 – 8
	46	
	47	Ne 4 – 6
	57	Ne 3 – 4
	29	Ne 16 – 40
	33	Ne 10 – 16
	36	Ne 8 – 10
	40	
	41	Ne 6 – 8

Knitting yarn	Denim yarn
Ne 28 – 60	
Ne 14 – 28	
Ne 8 – 14	Ne 10 – 24
Ne 8 – 10	Ne 8 – 10
Ne 5 – 8	Ne 3 – 8
Ne 3 – 5	
Ne 10 – 20	
Ne 8 – 10	Ne 12 – 20
Ne 5 – 8	Ne 6 – 12
Ne 3 – 5	Ne 3 – 6
Ne 24 – 40	
Ne 8 – 24	Ne 8 – 40
Ne 6 – 8	Ne 6 – 8
Ne 4 – 6	Ne 4 – 6
Ne 3 – 4	Ne 3 – 4
	Ne 16 – 40
	Ne 10 – 16
	Ne 8 – 10
	Ne 6 – 8

	Rotor diameter in mm	Weaving yarn
Regenerated fibers 100%	36	
	37	Ne 10 – 24
	40	
	41	Ne 6 – 10
	46	
	47	Ne 4 – 6
	56	
	57	Ne 3 – 4
PAN 100%	37	Ne 18 – 30
	41	Ne 8 – 18
	47	Ne 4 – 8
	56	
	57	Ne 3 – 4
Blends (except for PES)	31	Ne 24 – 60
	33	Ne 10 – 24
	36	
	37	Ne 8 – 10
	40	
	41	Ne 6 – 8
	46	
	47	Ne 4 – 6
	57	Ne 3 – 4



Knitting yarn	Denim yarn
	Ne 10 – 24
	Ne 6 – 10
	Ne 4 – 6
	Ne 3 – 4
	Ne 18 – 30
	Ne 8 – 18
	Ne 4 – 8
	Ne 3 – 4
	Ne 24 – 60
	Ne 10 – 24
	Ne 8 – 10
	Ne 5 – 8
	Ne 3 – 5

Raw material	Yarn		
Name	Ne	Nm	tex
Cotton up to 1"	10.6	18	55
	20.1	34	29.4
	21.3	36	28
	22.4	38	26
	24.8	42	24
	28.4	48	21
	30.1	51	19.6
	34.8	59	17
Cotton up to 1 1/16"	40.2	68	14.7
	10.6	18	55
	20.1	34	29.4
	21.3	36	28
	22.4	38	26
	24.8	42	24
	28.4	48	21
	30.1	51	19.6
	34.8	59	17
	40.2	68	14.7
	45.7	77	13
Cotton up to 1 1/2"	59.1	100	10
	73.8	125	8
	10.6	18	55
	20.1	34	29.4
	21.3	36	28
	22.4	38	26
	24.8	42	24
	28.4	48	21
	30.1	51	19.6
	34.8	59	17
	40.2	68	14.7
Regenerated fibers	45.7	77	13
	59.1	100	10
	73.8	125	8
	10.6	18	55
	24.8	42	24
CV, PES up to 40 mm	31.1	53	19
	10.6	18	55.0
	24.8	42	24.0
	26.6	45	22.0
	31.1	53	18.7
	36.0	61	16.5
	10.6	18	55.0
	24.8	42	24.0
	41.9	71	14.0
	49.0	83	12.0
53.7	91	11.0	
60.8	103	9.7	

Twist warp		Twist knitting	
Alpha e	Alpha m	Alpha e	Alpha m
4.46	135	4.23	128
4.53	137	4.3	130
4.53	137	4.3	130
4.53	137	4.3	130
4.53	137	4.3	130
4.6	139	4.33	131
4.6	139	4.33	131
4.6	139	4.33	131
4.6	139	4.33	131
4.23	128	4.03	122
4.33	131	4.1	124
4.33	131	4.1	124
4.33	131	4.1	124
4.33	131	4.1	124
4.4	133	4.17	126
4.4	133	4.17	126
4.4	133	4.17	126
4.4	133	4.17	126
4.46	135	4.23	128
4.46	135	4.23	128
4.46	135	4.23	128
4.1	124	3.87	117
4.1	124	3.9	118
4.1	124	3.9	118
4.1	124	3.9	118
4.1	124	3.9	118
4.17	126	3.97	120
4.17	126	3.97	120
4.17	126	3.97	120
4.17	126	3.97	120
4.23	128	4.03	122
4.23	128	4.03	122
4.23	128	4.03	122
	147		140
	148		140
	149		141
3.77	114	3.57	108
3.84	116	3.64	110
3.9	118	3.7	112
3.9	118	3.7	112
3.9	118	3.7	112
3.64	110	3.44	104
3.7	112	3.5	106
3.77	114	3.57	108
3.84	116	3.64	110
3.84	116	3.64	110
3.84	116	3.64	110

Raw material		Yarn		
Name	Ne	Nm	tex	
CV, PES up to 60 mm	10.6	18	55.0	
	24.8	42	24.0	
	41.9	71	14.0	
	49.0	83	12.0	
	53.7	91	11.0	
	60.8	103	9.7	
PAN up to 40 mm	10.6	18	55.0	
	16.4	28	36.3	
	24.8	42	24.0	
	26.6	45	22.0	
	31.1	53	18.7	
	36.0	61	16.5	
	41.9	71	14.0	
	49.0	83	12.0	
	53.7	91	11.0	
	59.6	101	9.9	
PAN up to 60 mm	73.8	125	8.0	
	10.6	18	55.0	
	16.4	28	36.3	
	24.8	42	24.0	
	26.6	45	22.0	
	31.1	53	18.7	
	36.0	61	16.5	
	41.9	71	14.0	
	49.0	83	12.0	
	53.7	91	11.0	
	59.6	101	9.9	
	73.8	125	8.0	

Twist warp		Twist knitting	
Alpha e	Alpha m	Alpha e	Alpha m
3.64	110	3.44	104
3.7	112	3.5	106
3.77	114	3.57	108
3.84	116	3.64	110
3.84	116	3.64	110
3.84	116	3.64	110
3.7	112	3.5	106
3.77	114	3.57	108
3.77	114	3.57	108
3.84	116	3.64	110
3.84	116	3.64	110
3.84	116	3.64	110
3.84	116	3.64	110
3.9	118	3.7	112
3.9	118	3.7	112
3.9	118	3.7	112
3.9	118	3.7	112
3.57	108	3.41	103
3.64	110	3.44	104
3.64	110	3.44	104
3.7	112	3.5	106
3.7	112	3.5	106
3.7	112	3.5	106
3.7	112	3.5	106
3.77	114	3.57	108
3.77	114	3.57	108
3.77	114	3.57	108
3.77	114	3.57	108

2.2.2.6. Package weight on rotor spinning

			Cylindrical package	Conical package 1°51		Conical package 4°20	
	Lenght of H	Lenght of D/Ds	Netto package weight	Lenght of ds	Netto package weight	Lenght of ds	Netto package weight
	in mm	in mm	in g	in mm	in g	in mm	in g
Combed cotton spec. weight 0.41	150	160	1 060	149	910	134	890
		180	1 380	169	1 220	154	1 180
		200	1 750	189	1 570	174	1 510
		220	2 150	209	1 950	194	1 880
		240	2 590	229	2 370	214	2 280
		260	3 070	249	2 820	234	2 730
		280	3 590	269	3 320	254	3 210
		300	4 140	289	3 850	274	3 720
		320	4 740	309	4 420	294	4 280
		340	5 370	329	5 020	314	4 870
Carded cotton spec. weight 0.4	150	360	6 040	349	5 660	334	5 500
		160	1 030	149	890	134	870
		180	1 350	169	1 190	154	1 150
		200	1 700	189	1 530	174	1 470
		220	2 100	209	1 900	194	1 830
		240	2 530	229	2 310	214	2 230
		260	2 990	249	2 750	234	2 660
		280	3 500	269	3 240	254	3 130
		300	4 040	289	3 750	274	3 630
		320	4 620	309	4 310	294	4 170
PES/ PAN spec. weight 0.47	150	340	5 240	329	4 900	314	4 750
		360	5 900	349	5 530	334	5 360
		160	1 210	149	1 050	134	1 020
		180	1 580	169	1 400	154	1 350
		200	2 000	189	1 790	174	1 730
		220	2 460	209	2 230	194	2 150
		240	2 970	229	2 710	214	2 620
		260	3 520	249	3 240	234	3 120
		280	4 110	269	3 800	254	3 670
		300	4 750	289	4 410	274	4 270
Cellulosic viscose spec. weight 0.45	150	320	5 430	309	5 060	294	4 900
		340	6 160	329	5 760	314	5 580
		360	6 930	349	6 490	334	6 300
		160	1 160	149	1 000	134	970
		180	1 520	169	1 340	154	1 300
		200	1 920	189	1 720	174	1 660
		220	2 360	209	2 140	194	2 060
		240	2 840	229	2 600	214	2 510
		260	3 370	249	3 100	234	2 990
		280	3 940	269	3 640	254	3 520
	150	300	4 550	289	4 220	274	4 090
		320	5 200	309	4 850	294	4 690
		340	5 900	329	5 510	314	5 340
		360	6 630	349	6 220	334	6 030

			Cylindrical package	Conical package 1°51	Conical package 4°20		
	Lenght of H	Lenght of D/Ds	Netto package weight	Lenght of ds	Netto package weight	Lenght of ds	Netto package weight
	in mm	in mm	in g	in mm	in g	in mm	in g
Blends CO-MMF spec. weight 0.44	150	160	1 130	149	980	134	950
		180	1 480	169	1 310	154	1 270
		200	1 870	189	1 680	174	1 620
		220	2 310	209	2 090	194	2 020
		240	2 780	229	2 540	214	2 450
		260	3 290	249	3 030	234	2 930
		280	3 850	269	3 560	254	3 440
		300	4 450	289	4 130	274	4 000
		320	5 080	309	4 740	294	4 590
		340	5 760	329	5 390	314	5 230
		360	6 480	349	6 080	334	5 900

2.2.2.7. Production efficiency rotor

Rotor spinning	Yarn count in Ne	10	20	30	40
Theoretical prod.	kg/h	573.5	275.8	171.6	111.5
Efficiency	%	97.4	97.8	98	98.1
Draw frame					
Efficiency	%	82 – 85	82 – 85	82 – 85	82 – 85
Draw frame					
Efficiency	%	84 – 95	84 – 95	84 – 95	84 – 95
Card					
Efficiency	%	90 – 95	90 – 95	90 – 95	90 – 95
Blowroom					
Efficiency	%	90 – 95	90 – 95	90 – 95	90 – 95

2.2.3. Air-jet spinning

2.2.3.1. Machine data on air-jet spinning

Number of spinning positions	Number of robots
40	2
100 – 120	4
160	6
200	6

2.2.3.2. Raw material

Material	Air-jet		
	Staple length		Fiber fineness
	Inch	mm	
Cotton combed	$\geq 1 \frac{1}{8}"$	≥ 28.6 mm	≤ 4.3 mic
Cotton carded (max. 22% short fibers)	$\geq 1 \frac{1}{8}"$	≥ 28.6 mm	≤ 4.3 mic
Cellulosic fibers	1.57"	≤ 40.0 mm	max. 1.5 dtex
Acrylic fibers	1 $\frac{1}{2}"$	38 mm	1.3 dtex
Polyester fibers	1 $\frac{1}{2}"$	38 mm	1 – 1.3 dtex
Blends (CO/PES < 50%)	1 $\frac{1}{2}"$	PES 38 mm	max 1.3 dtex
Blends (CO/CV)	1.57"	CV ≤ 40 mm	max 1.5 dtex
Blends (CV/PES < 80%)	1 $\frac{1}{2}"$	PES 38 mm	max 1.3 dtex
	1.57"	CV ≤ 40 mm	max. 1.5 dtex
Blends (CV/PAN < 33%)	1 $\frac{1}{2}"$	PAN 38 mm	max 1.3 dtex
	1.57"	CV ≤ 40 mm	max 1.5 dtex

2.2.3.3. Yarn count range

Raw Material	Yarn count range		
	Ne	Nm	Tex
	20 – 70	33.87 – 118.54	8.4 – 29.5

2.2.3.4. Number of units per air-jet spinning machine

Raw material	Yarn count	Number of spinning units		
	in Ne			
CO combed	24 – 50	40 or 120	160	
CV < 1.0 dtex	24 – < 50	40 or 120	160	200
	50 – 70	40 or 120	160	
CV > 1.0 – 1.3 dtex	24 – 40	40 or 120	160	200
CV > 1.3 – 1.5 dtex	24 – 30	40 or 120	160	200
PES	30	40 or 120		
Blends: CV / PES	24 – 40	40 or 120	160	
Blends: CV / PAN	24 – 40	40 or 120	160	200
Blends: CO combed / MMF	20 – 40	40 or 120	160	
Blends: CO carded / PES	20 – 30	40 or 120		

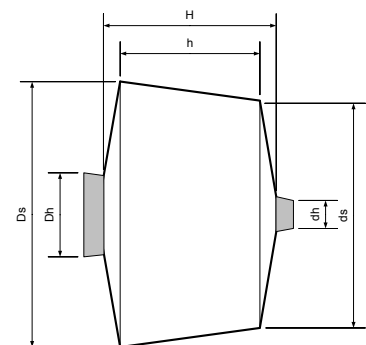
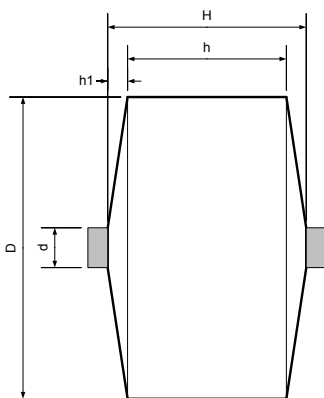
2.2.3.5. Delivery speed air-jet spinning

		Yarn count						
Raw material		20 – < 30	24 – < 30	30 – 40	> 40 – 50	> 50 – 60	> 60 – 70	Ne
		33.9 – 50.8	40.6 – 50.8	50.8 – 67.7	67.7 – 84.7	84.7 – 101.6	101.6 – 118.5	Nm
		29.5 – 19.8	24.6 – 19.8	19.7 – 14.8	14.7 – 11.8	11.7 – 9.8	9.7 – 8.4	tex
Combed cotton > 1 1/8"	Weaving			340				
	Knitting			440				
Combed cotton > 1 5/16"	Weaving			360	440			
	Knitting			320	420			
Blends combed CO/CV – PES (<50%) > 1 1/8"	Weaving	380	360					
	Knitting	420	430					
Cellulosic viscose < 1.0 Dtex	Weaving		400	400	340	300	260	
	Knitting		420	440	380	340	300	
Cellulosic viscose > 1.0 Dtex – 1.3 Dtex	Weaving		420	440				
	Knitting		400	440				
Cellulosic viscose > 1.3 Dtex – 1.5 Dtex	Weaving		420	440				
	Knitting			420				
Blends CV/ PES (<80%)	Weaving		400	400				
	Knitting		450	450	380			
Blends CV/PAN (< 33%)	Weaving							
	Knitting		450	450				
Blends carded cotton/PES (< 50%)	Weaving							
	Knitting	400						

2.2.3.6. Package weight on air-jet spinning

	Lenght of H	Lenght of D/Ds	Cylindrical package	Conical package 1°51		Conical package 4°20	
			Netto package weight	Lenght of ds	Netto package weight	Lenght of ds	Netto package weight
			in g	in mm	in g	in mm	in g
Combed cotton spec. Weight 0.41	150	160	1 060	149	910	134	890
		180	1 380	169	1 220	154	1 180
		200	1 750	189	1 570	174	1 510
		220	2 150	209	1 950	194	1 880
		240	2 590	229	2 370	214	2 280
		260	3 070	249	2 820	234	2 730
		280	3 590	269	3 320	254	3 210
Carded cotton spec. weight 0.4	150	300	4 140	289	3 850	274	3 720
		160	1 030	149	890	134	870
		180	1 350	169	1 190	154	1 150
		200	1 700	189	1 530	174	1 470
		220	2 100	209	1 900	194	1 830
		240	2 530	229	2 310	214	2 230
		260	2 990	249	2 750	234	2 660
PES/ PAN spec. weight 0.47	150	280	3 500	269	3 240	254	3 130
		300	4 040	289	3 750	274	3 630
		160	1 210	149	1 050	134	1 020
		180	1 580	169	1 400	154	1 350
		200	2 000	189	1 790	174	1 730
		220	2 460	209	2 230	194	2 150
		240	2 970	229	2 710	214	2 620
Cellulosic viscose spec. weight 0.45	150	260	3 520	249	3 240	234	3 120
		280	4 110	269	3 800	254	3 670
		300	4 750	289	4 410	274	4 270
		160	1 160	149	1 000	134	970
		180	1 520	169	1 340	154	1 300
		200	1 920	189	1 720	174	1 660
		220	2 360	209	2 140	194	2 060
	150	240	2 840	229	2 600	214	2 510
		260	3 370	249	3 100	234	2 990
		280	3 940	269	3 640	254	3 520
		300	4 550	289	4 220	274	4 090

			Cylindrical package	Conical package 1°51		Conical package 4°20	
	Lenght of H	Lenght of D/Ds	Netto package weight	Lenght of ds	Netto package weight	Lenght of ds	Netto package weight
	in mm	in mm	in g	in mm	in g	in mm	in g
Blends CO-MMF spec. weight 0.44	150	160	1 130			134	950
		180	1 480			154	1 270
		200	1 870			174	1 620
		220	2 310			194	2 020
		240	2 780			214	2 450
		260	3 290			234	2 930
		280	3 850			254	3 440
		300	4 450			274	4 000
		320				294	4 590
		340				314	5 230
		360				334	5 900



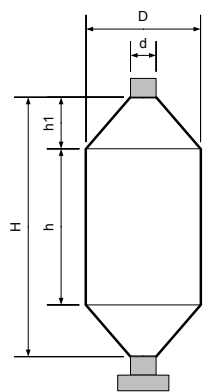
2.2.3.7. Production efficiency air-jet spinning

Yarn count	in Ne	30	40
Theoretical prod.	kg/h	472.4	354.3
Efficiency	%	92 – 96	92 – 96
Draw frame			
Efficiency	%	90 – 95	90 – 95
Card			
Efficiency	%	93 – 95	93 – 95
Blowroom			
Efficiency	%	93 – 95	93 – 95

2.2.4. Roving frame

2.2.4.1. Bobbin weight

Bobbin height	Stroke	Format of bobbin on roving frame			Weight of bobbin on roving frame according to raw material				
		D	d	H	Combed cotton	Carded cotton	PES and PAN	Cellulosic viscose	CO/MMF
inch	inch	mm	mm	mm	kg	kg	kg	kg	kg
5 (127 mm)	14	122	48	345	1 400	1 300	1 600	1 500	1 500
	16	122	48	400	1 600	1 600	1 900	1 800	1 700
	16	122	53.5	405	1 600	1 500	1 800	1 700	1 700
5.5 (140 mm)	14	135	48	345	1 700	1 600	1 900	1 900	1 800
	16	135	48	400	2 000	1 900	2 300	2 200	2 100
	16	135	53.5	405	1 900	1 800	2 200	2 100	2 000
6 (152 mm)	14	147	48	345	1 900	1 800	2 200	2 100	2 000
	16	147	48	400	2 200	2 200	2 600	2 500	2 400
	16	147	53.5	405	2 200	2 100	2 500	2 400	2 300
6.5 (165 mm)	14	160	48	345	2 100	2 100	2 400	2 400	2 300
	16	160	48	400	2 500	2 400	2 900	2 800	2 700
	16	160	53.5	405	2 500	2 400	2 800	2 700	2 600
7 (178 mm)	14	173	48	345	2 300	2 200	2 600	2 500	2 500
	16	173	48	400	2 700	2 600	3 100	3 000	2 900
	16	173	53.5	405	2 700	2 500	3 000	2 900	2 800



2.2.4.2. Roving twist

Raw material	Process/ Specification	Roving			Twist		T/m	Roving frame
		Ne	Nm	tex	alpha e	alpha m		rpm
Cotton up to 1"	carded	0.54	0.91	1100	1.32	40	38.14	1100
		0.59	1	1000	1.32	40	40	1100
		0.66	1.11	900	1.32	40	42.16	1150
		0.74	1.25	800	1.32	40	44.72	1250
		0.84	1.43	700	1.32	40	47.81	1250
		0.99	1.67	600	1.32	40	51.64	1250
Cotton up to 1 1/16"	carded and combed	0.54	0.91	1100	1.29	39	39	1200
		0.59	1	1000	1.29	39	40.1	1200
		0.66	1.11	900	1.29	39	41.37	1250
		0.74	1.25	800	1.29	39	44.23	1250
		0.84	1.43	700	1.29	39	47.77	1250
		0.99	1.67	600	1.29	39	52.33	1300
		1.18	2	500	1.26	38	56.92	1350
		1.48	2.5	400	1.26	38	65.72	1350
Cotton up to 1 1/8"	carded and combed	0.59	1	1000	1.29	39	39	1200
		0.66	1.11	900	1.26	38	40.1	1200
		0.74	1.25	800	1.23	37	41.37	1250
		0.84	1.43	700	1.23	37	44.22	1250
		0.99	1.67	600	1.23	37	47.77	1250
		1.18	2	500	1.23	37	52.33	1300
		1.48	2.5	400	1.19	36	56.92	1350
		1.79	3.33	300	1.19	36	65.73	1350
Cotton up to 1 1/2"	combed	0.74	1.25	800	1.06	32	35.78	1250
		0.84	1.43	700	1.09	33	39.44	1250
		0.99	1.67	600	1.09	33	42.6	1250
		1.18	2	500	1.09	33	46.67	1300
		1.48	2.5	400	1.12	34	53.76	1350
		1.79	3.33	300	1.12	34	62.08	1250
		2.95	5	200	1.12	34	76.03	1200
Cotton up to 1 7/8"	combed	0.84	1.43	700	0.99	30	35.86	1200
		0.99	1.67	600	0.99	30	38.73	1250
		1.18	2	500	0.99	30	42.43	1300
		1.48	2.5	400	0.99	30	47.43	1300
		1.79	3.33	300	0.99	30	54.77	1300
		2.95	5	200	0.99	30	67.08	1200
		5.91	10	100	0.99	30	94.87	1200
PES up to 40 mm	1.3 dtex	0.54	0.91	1100	0.73	22	20.98	1000
		0.59	1	1000	0.73	22	22	1000
		0.66	1.11	900	0.73	22	23.19	1100
		0.74	1.25	800	0.73	22	24.6	1100
		0.84	1.43	700	0.73	22	26.3	1200
		0.99	1.67	600	0.73	22	28.4	1250
		1.18	2	500	0.73	22	31.11	1250
		1.48	2.5	400	0.73	22	34.79	1200
		1.79	3.33	300	0.73	22	40.17	1000
		2.95	5	200	0.76	23	51.43	1000

Raw material	Process/ Specification	Yarn			Twist		T/m	Roving frame
		Ne	Nm	tex	alpha e	alpha m		rpm
PES up to 40 mm	1.7 dtex	0.54	0.91	1100	0.69	21	20.98	1000
		0.59	1	1000	0.69	21	22	1000
		0.66	1.11	900	0.69	21	23.19	1100
		0.74	1.25	800	0.69	21	24.6	1100
		0.84	1.43	700	0.69	21	26.3	1200
		0.99	1.67	600	0.69	21	28.4	1250
		1.18	2	500	0.69	21	31.11	1250
		1.48	2.5	400	0.73	22	34.79	1200
		1.79	3.33	300	0.76	23	40.17	1000
		2.95	5	200	0.8	24	51.43	1000
PES up to 60 mm	1.3 dtex 1.7 dtex 2.0 dtex	0.54	0.91	1100	0.73	22	20.98	1000
		0.59	1	1000	0.69	21	21	1000
		0.66	1.11	900	0.69	21	22.14	1100
		0.74	1.25	800	0.69	21	23.48	1100
		0.84	1.43	700	0.69	21	25.1	1200
		0.99	1.67	600	0.69	21	27.11	1250
		1.18	2	500	0.69	21	29.7	1250
		1.48	2.5	400	0.69	21	33.2	1200
		1.79	3.33	300	0.69	21	38.34	1000
		2.95	5	200	0.73	22	49.19	1100
PAN	all	0.54	0.91	1100	0.69	21	20.02	1000
		0.59	1	1000	0.69	21	21	1000
		0.66	1.11	900	0.69	21	22.14	1100
		0.74	1.25	800	0.69	21	23.48	1100
		0.84	1.43	700	0.69	21	25.1	1100
		0.99	1.67	600	0.69	21	27.11	1150
		1.18	2	500	0.69	21	29.7	1150
		1.48	2.5	400	0.69	21	33.2	1150
		1.79	3.33	300	0.69	21	38.34	1150
		2.95	5	200	0.73	22	49.19	1100
Cellulosic viscose	all	0.54	0.91	1100	0.73	22	20.98	1000
		0.59	1	1000	0.69	21	21	1000
		0.66	1.11	900	0.69	21	22.14	1000
		0.74	1.25	800	0.69	21	23.48	1100
		0.84	1.43	700	0.69	21	25.1	1100
		0.99	1.67	600	0.69	21	27.11	1150
		1.18	2	500	0.69	21	29.7	1200
		1.48	2.5	400	0.73	22	34.79	1100
		1.79	3.33	300	0.76	23	41.99	1000
		2.95	5	200	0.8	24	53.67	900
Linen	all	0.54	0.91	1100	0.63	19	18.12	900
		0.59	1	1000	0.69	21	21	1000
		0.66	1.11	900	0.69	21	22.14	1000
		0.74	1.25	800	0.69	21	23.48	1000
		0.84	1.43	700	0.69	21	25.1	1000
		0.99	1.67	600	0.69	21	27.11	1000
		1.18	2	500	0.69	21	29.7	1000
		1.48	2.5	400	0.73	22	34.79	1000
		1.79	3.33	300	0.76	23	41.99	1000
		2.95	5	200	0.8	24	53.67	1000

2.2.5. Combing preparation OMEGAlap

Raw material	Batt weight	Lap weight	Delivery	Theor. production
Cotton				
up to	g/m	kg	m/min	kg/h
1 1/16"	80	22	230	1 104
1 1/8"	80	24	230	1 104
1 3/16"	78	24	230	1048.8
1 1/4"	76	24	230	1021.1
1 3/8"	74	24	230	1021.2
1 1/2"	72	24	230	966
1 7/8"	70	24	230	966

2.2.6. Comber

Raw material	Nips	Batt weight	Noil	Theor. production
Cotton				
up to	rpm	g/m	%	kg/h
1 1/16"	525	80	16 – 22	80.1
1 1/8"	500	80	14 – 18	80.1
1 3/16"	475	78	12 – 13	65.5
1 1/4"	450	76	19 – 22	57.4
1 3/8"	425	74	14 – 22	54.2
1 1/2"	400	72	14 – 22	44.1
1 7/8"	400	70	12 – 21	44.1

Delivery speed on the draw frame in m/min

			Sliver weight in ktex and Ne						
Raw material		End spinning	1.25 – 2.0	2.1 – 2.5	2.6 – 3.3	3.4 – 3.9	4.0 – 4.4	> 4.5	ktex
			0.47 – 0.30	0.29 – 0.24	0.23 – 0.18	0.17 – 0.15	0.14 – 0.13	< 0.13	Ne
CO < 1 1/16“ regenerated fiber waste	carded	Ring			300	400	450	500	
		Rotor			400	500	550	600	
CO > 1 1/16"	carded	Ring and rotor	350	400	600	700	850	100	
CO 1 1/16" – 1 1/8"	combed	Ring and rotor	250	250	300	400	550	650	
CO 1 3/16" – 1 1/4"	combed	Ring and rotor	300	300	350	400	550	650	
CO > 1 1/4"	combed	Ring and rotor	350	350	400	450	450	550	
Polyester (PES)	carded	Ring and rotor	350	400	500	600	700	800	
Polyacrylnitril (PAN)	carded	Ring and rotor	350	400	500	600	700	700	
Cellulosic viscose (CV)	carded	Ring and rotor	350	400	500	600	750	800	
Blend CV/PAN									
Blend CO/MMF (<35%)	carded	Ring and rotor	300	350	450	650	750	800	
Blend CO/MMF (>35%)	carded	Ring and rotor	350	400	500	600	700	800	
Blend CO/MMF	CO combed	Ring and rotor	300	350	400	450	600	650	
Blend PES/CV	carded	Ring and rotor	350	400	500	600	650	700	
Blend PES/PAN									
Linen	carded	Ring and rotor				400	400	400	
CO > 1 1/16"	carded	Air-jet	300	350	450	550	700	800	
CO 1 1/16" – 1 1/8"	combed	Air-jet	250	250	250	300	350	450	
CO 1 3/16" – 1 1/4"	combed	Air-jet	250	250	300	350	400	500	
CO > 1 1/4"	combed	Air-jet	300	300	350	400	400	500	
Polyester (PES)	carded	Air-jet	300	350	400	500	550	550	
Polyacrylnitril (PAN)	carded	Air-jet	300	350	450	550	650	650	
Cellulosic viscose (CV)	carded	Air-jet	300	350	450	500	550	650	
Blend CV/PAN									
Blend CO/MMF (<35%)	carded	Air-jet	250	300	400	500	650	700	
Blend CO/MMF (>35%)	carded	Air-jet	300	350	450	500	550	600	
Blend CO/MMF	CO combed	Air-jet	250	300	350	400	450	500	
Blend PES/CV	carded	Air-jet	300	350	450	500	550	600	
Blend PES/PAN									

Raw material			Sliver count > ktex	Main yarn count									
Material	Blend ratio	Staple / Fiber finesse		- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	Ne
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	Tex
CO/CV	> 50%	> 1 1/8“ < 1 1/4“ 28.7 – 31.8mm	4.9	80	75	75	70	65	60	55			
			4.5	70	70	70	70	65	60	55			
			3.9	60	60	60	60	60	60	55			
CO/ PES – PAN	> 65%	> 1 < 1 1/16 25.5 – 27.0 mm	7	120	115	105	100						
			6.5	115	115	105	100						
			5.9	105	105	105	100						
			5.4	95	95	95	95						
			4.9	85	85	85	85	85					
			4.5	70	70	70	70	70					
			3.9	60	60	60	60	60					
			CO/ PES – PAN	> 65%	> 1 1/16“ < 1 1/8“ 27.1 – 28.6 mm	4.9	85	85	85	85	75	70	70
4.5	70	70	70			70	70	70	70				
3.9	60	60	60			60	60	60	60				
CO/ PES – PAN	> 65%	> 1 1/8“ < 1 1/4“ 28.7 – 31.8 mm	4.9	80	75	75	70	65	60	55			
			4.5	70	70	70	70	65	60	55			
			3.9	60	60	60	60	60	60	55			
PES < 40	100%	1.1 – 1.3 dtex	6.5	115	105	100	90	80	70	60			
			5.9	105	105	100	90	80	70	60			
			5.4	95	95	95	90	80	70	60			
			4.9	85	85	85	85	80	70	60			
			4.5	70	70	70	70	70	70	60			
			3.9	60	60	60	60	60	60	60			
		1.4 – 2.0 dtex	7	125	115	110	100	90	80	70			
			6.5	115	115	110	100	90	80	70			
			5.9	105	105	105	100	90	80	70			
			5.4	95	95	95	95	90	80	70			
			4.9	85	85	85	85	85	80	70			
			4.5	70	70	70	70	70	70	70			
			3.9	60	60	60	60	60	60	60			
PAN < 40	100%	1.1 – 2.0 dtex	4.9	80	80	80	80	80	75				
			4.5	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60				
CV < 40	100%	1.1 – 1.3 dtex	7	120	115	110	105	100	90	85			
			6.5	115	115	110	105	100	90	85			
			5.9	105	105	105	105	100	90	85			
			5.4	95	95	95	95	95	90	85			
			4.9	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70			
		1.4 – 2.0 dtex	3.9	60	60	60	60	60	60	60			
			7.5	130	125	120	115	110	100	90			
			7	125	125	120	115	110	100	90			
			6.5	115	115	115	115	110	100	90			
			5.9	105	105	105	105	105	100	90			
			5.4	95	95	95	95	95	95	90			
			4.9	85	85	85	85	85	85	85			
4.5	70	70	70	70	70	70	70						
3.9	60	60	60	60	60	60	60						

Raw material		Fiber finesse	Sliver count > ktex	Main yarn count									
Material	Blend ratio			- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	Ne
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	Tex
PES/CV < 40 mm	> 35%	1.1 – 1.3 mm	6.5	115	105	100	90	80	70	60			
			5.9	105	105	100	90	80	70	60			
			5.4	95	95	95	90	80	70	60			
			4.9	85	85	85	85	80	70	60			
			4.5	70	70	70	70	70	70	60			
			3.9	60	60	60	60	60	60	60			
		1.4 – 2.0 mm	7	125	115	110	100	90	80	70			
			6.5	115	115	110	100	90	80	70			
			5.9	105	105	105	100	90	80	70			
			5.4	95	95	95	95	90	80	70			
			4.9	85	85	85	85	85	80	70			
			4.5	70	70	70	70	70	70	70			
PAN/CV < 40	> 35%	1.1 – 1.3 mm	3.9	60	60	60	60	60	60				
			4.9	80	80	80	80	80	75				
			4.5	70	70	70	70	70	70				
CV/PES < 40	> 65%	1.1 – 1.3 mm	7	120	115	110	105	100	90	85			
			6.5	115	115	110	105	100	90	85			
			5.9	105	105	105	105	100	90	85			
			5.4	95	95	95	95	95	90	85			
			4.9	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70			
			3.9	60	60	60	60	60	60	60			
		1.4 – 2.0 mm	8	130	125	120	115	110	100	90			
			7	125	125	120	115	110	100	90			
			6.5	115	115	115	115	110	100	90			
			5.9	105	105	105	105	105	100	90			
			5.4	95	95	95	95	95	95	90			
			4.9	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70			
			3.9	60	60	60	60	60	60	60			
CV/PAN < 40	> 65%	1.1 – 1.3 mm	4.9	80	80	80	80	80	75				
			4.5	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60				
PES/CO	> 35%	1.1 – 1.3 mm	5.4	90	90	80	80	80	75				
			4.9	85	85	80	80	80	75				
			4.5	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60				
PES/CO	> 35%	1.4 – 2.0 mm	5.9	100	100	90	90	90					
			5.4	95	95	90	90	90					
			4.9	85	85	85	85	85					
			4.5	70	70	70	70	70					
			3.9	60	60	60	60	60					
PAN/CO	> 35%	1.1 – 1.3 mm	4.9	75	75	70	70	70	65				
			4.5	70	70	70	70	70	65				
			3.9	60	60	60	60	60	60				

2.2.8.2. Card production for rotor spinning in kg/h

Raw material			Sliver count > ktex	Main yarn count										
Material	Blend ratio	Staple		- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	- 60	Ne
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	11.7 – 9.8	Tex
CO – Noils	100%	< 1" < 25.4 mm	13.0	185	170									
			12.0	185	170									
			11.0	180	170									
			10.0	165	165									
			9.0	145	145									
			8.0	130	130									
			7.0	115	115									
			6.5	105	105									
			5.9	95	95									
CO	100%	> 1" < 1 1/16"	14.0	225	200	175								
			13.0	225	200	175								
			12.0	215	200	175								
			11.0	195	195	175	140	130						
			10.0	180	180	175	140	130						
			9.0	160	160	160	140	130						
			8.0	140	140	140	140	130						
			7.0	125	125	125	125	125						
			6.5	115	115	115	115	115						
			5.9	105	105	105	105	105						
			5.4	95	95	95	95	95						
			4.9	85	85	85	85	85						
			4.5	70	70	70	70	70						
			3.9	60	60	60	60	60						
CO	100%	> 1 1/16" < 1 1/8"	11	170	170	150								
			10	170	170	150								
			9	160	160	150	135	130	120	120				
			8	140	140	140	135	130	120	120				
			7	125	125	125	125	125	120	120				
			6.5	115	115	115	115	115	115	115				
			5.9	105	105	105	105	105	105	105				
			5.4	95	95	95	95	95	95	95				
			4.9	85	85	85	85	85	85	85				
			4.5	70	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60	60				
CO	100%	> 1 1/8" < 1 1/4"	5.4	95	95	95	95	85	80	75				
			4.9	85	85	85	85	85	80	75				
			4.5	70	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60	60				
Linen	100%		4.5	65										
			3.9	60										
Waste (CO)	> 25%		11	160	160	150	145							
			10	150	150	150	145							
			9	135	135	135	135							
			8	120	120	120	120							
			7	105	105	105	105							
			6.5	95	95	95	95							
			5.9	85	85	85	85							
			5.4	80	80	80	80							
			4.9	70	70	70	70							
			4.5	65	65	65	65							
			3.9	55	55	55	55							

[illegible]

Raw material			Sliver count > ktex	Main yarn count											
Material	Blend ratio	Staple / Fiber finesse		- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	- 60	Ne	
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	11.7 – 9.8	Tex	
PES < 40	100%	1.1 – 1.3 dtex	8	115	105	105	100	90	90	85	80				
			7	115	105	105	100	90	90	85	80				
			6.5	115	105	105	100	90	90	85	80				
			5.9	105	105	105	100	90	90	85	80				
			5.4	95	95	95	95	90	90	85	80				
			4.9	85	85	85	85	85	85	85	80				
		1.4 – 2.0 dtex	4.5	70	70	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60	60	60				
			7	125	115	115	110	100	100	95	90				
			6.5	115	115	115	110	100	100	95	90				
			5.9	105	105	105	105	100	100	95	90				
			5.4	95	95	95	95	95	95	95	90				
PAN < 40	100%	1.1 – 2.0 dtex	4.9	85	85	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70	70	70			
			3.9	60	60	60	60	60	60	60	60	60			
			5.9	105	105	95	95	95	95	95	95	95	95		
			5.4	95	95	95	95	95	95	95	95	95	95		
CV < 40	100%	1.1 – 1.3 dtex	9	130	125	120	120	115	110	110	100				
			8	130	125	120	120	115	110	110	100				
			7	125	125	120	120	115	110	110	100				
			6.5	115	115	115	115	115	110	110	100				
			5.9	105	105	105	105	105	105	105	100				
			5.4	95	95	95	95	95	95	95	95				
			4.9	85	85	85	85	85	85	85	85				
			4.5	70	70	70	70	70	70	70	70				
CV < 40	100%	1.4 – 2.0 dtex	3.9	60	60	60	60	60	60	60	60				
			9.0	145	135	130	130	125	100	100	100				
			8.0	140	135	130	130	125	100	100	100				
			7.0	125	125	125	125	125	100	100	100				
			6.5	115	115	115	115	115	100	100	100				
			5.9	105	105	105	105	105	100	100	100				
			5.4	95	95	95	95	95	95	95	95				
			4.9	85	85	85	85	85	85	85	85				
PES / CV < 40 mm	> 35%	1.1 – 1.3 dtex	4.5	70	70	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60	60	60				
			6.5	115	105	105	100	90	90	85	80				
			5.9	105	105	105	100	90	90	85	80				
			5.4	95	95	95	95	90	90	85	80				
			4.9	85	85	85	85	85	85	85	80				
		1.4 – 2.0 dtex	3.9	60	60	60	60	60	60	60	60				
			7.0	125	115	115	110	100	100	95	90				
PES / CV < 40 mm	> 35%	1.4 – 2.0 dtex	6.5	115	115	115	110	100	100	95	90				
			5.9	105	105	105	105	100	100	95	90				
			5.4	95	95	95	95	95	95	95	90				
			4.9	85	85	85	85	85	85	85	85				
			4.5	70	70	70	70	70	70	70	70				
			3.9	60	60	60	60	60	60	60	60				

Raw material			Sliver count > ktex	Main yarn count										
Material	Blend ratio	Fiber finesse		- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	- 60	Ne
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	11.7 – 9.8	Tex
PAN/CV < 40	> 35%	1.1 – 1.3 dtex	5.9	105	105	95	95	95	95	95	95	95		
			5.4	95	95	95	95	95	95	95	95	95		
			4.9	85	85	85	85	85	85	85	85	85		
			4.5	70	70	70	70	70	70	70	70	70		
			3.9	60	60	60	60	60	60	60	60	60		
CV/PES < 40 mm	> 65%	1.1 – 1.3 dtex	8.0	130	125	120	120	115	110	110	100			
			7.0	125	125	120	120	115	110	110	100			
			6.5	115	115	115	115	115	110	110	100			
			5.9	105	105	105	105	105	105	105	100			
			5.4	95	95	95	95	95	95	95	95			
			4.9	85	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70	70			
			3.9	60	60	60	60	60	60	60	60			
		1.4 – 2.0 dtex	9	145	135	130	130	125	110	110	100			
			8	140	135	130	130	125	110	110	100			
			7	125	125	125	125	125	110	110	100			
			6.5	115	115	115	115	115	110	110	100			
			5.9	105	105	105	105	105	105	105	100			
			5.4	95	95	95	95	95	95	95	95			
			4.9	85	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70	70			
CV/PAN < 40 mm	> 65	1.1 – 1.3 dtex	5.9	105	105	95	95	95	95	95	95	95		
			5.4	95	95	95	95	95	95	95	95	95		
			4.9	85	85	85	85	85	85	85	85	85		
			4.5	70	70	70	70	70	70	70	70	70		
			3.9	60	60	60	60	60	60	60	60	60		
PES/CO	> 35%	1.1 – 1.3 dtex	10	115	105	105	100	90	90	85	80			
			9	115	105	105	100	90	90	85	80			
			8	115	105	105	100	90	90	85	80			
			7	115	105	105	100	90	90	85	80			
			6.5	115	105	105	100	90	90	85	80			
			5.9	105	105	105	100	90	90	85	80			
			5.4	95	95	95	95	90	90	85	80			
			4.9	85	85	85	85	85	85	85	80			
			4.5	70	70	70	70	70	70	70	70			
PES/CO	> 35%	1.4 – 2.0 dtex	11	125	115	115	110	100	100	95	90			
			10	125	115	115	110	100	100	95	90			
			9	125	115	115	110	100	100	95	90			
			8	125	115	115	110	100	100	95	90			
			7	125	115	115	110	100	100	95	90			
			6.5	115	115	115	110	100	100	95	90			
			5.9	105	105	105	105	100	100	95	90			
			5.4	95	95	95	95	95	95	95	90			
			4.9	85	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70	70			

Raw material			Sliver count > ktex	Main yarn count										Ne
Material	Blend ratio	Fiber finesse		- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	- 60	
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	11.7 – 9.8	Tex
PAN/CO	> 35%	1.1 – 1.3 dtex	5.9	105	105	95	95	95	95	95	95	95		
			5.4	95	95	95	95	95	95	95	95	95		
			4.9	85	85	85	85	85	85	85	85	85		
			4.5	70	70	70	70	70	70	70	70	70		
			3.9	60	60	60	60	60	60	60	60	60		
CV/CO	> 50%	1.1 – 1.3 dtex	6.5	115	115	110	110	105	100	100	90			
			5.9	105	105	105	105	105	100	100	90			
			5.4	95	95	95	95	95	95	95	90			
			4.9	85	85	85	85	85	85	85	85			
			4.5	70	70	70	70	70	70	70	70			
CV/CO	> 50%	1.4 – 2.0 dtex	3.9	60	60	60	60	60	60	60	60			
			8	130	120	115	115	115						
			7	125	120	115	115	115						
			6.5	115	115	115	115	115						
			5.9	105	105	105	105	105						
			5.4	95	95	95	95	95						
			4.9	85	85	85	85	85						
			4.5	70	70	70	70	70						
			3.9	60	60	60	60	60						

2.2.8.3. Card production for air-jet spinning in kg/h

Raw material			Sliver count > ktex	Main yarn count										
Material	Blend ratio 1	Staple/Fiber finesse		- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	- 60	Ne
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	11.7 – 9.8	Tex
Combed cotton	1 1/16 < 1 1/8"	27.0 – 28.6 mm	4.9					85						
			4.5					70						
			3.9					60						
	> 1 1/8 < 1 1/4"	28.7 – 31.8 mm	4.9					75	75	70				
			4.5					70	70	70				
			3.9					60	60	60				
> 1 1/4 < 1 3/8"	31.9 – 34.8 mm	4.5					65	65	65					
		3.9					60	60	60					
		Carded cotton	100%	1/116" < 1 1/8" 27.0 – 28.6 mm	4.9			85	85	85				
4.5						70	70	70						
3.9						60	60	60						
> 1 1/82 < 1 1/4" 28.7 – 31.8 mm	4.9					75	70	60	55	50				
	4.5					70	70	60	55	50				
	3.9					60	60	60	55	50				
CO/CV	> 50%	1 1/16" < 1 1/8" 27.0 – 28.6 mm	4.5			70	70	70						
			3.9			60	60	60						
	> 50%	> 1 1/8" < 1 1/4" 28.7 – 31.8 mm	4.5			60	55	50	45	40				
			3.9			60	55	50	45	40				
CO/PES – PAN	> 65%	1 1/16" < 1 1/8" 27.0 – 28.6 mm	4.5			70	70	70						
			3.9			60	60	60						
CO/PES – PAN	> 65%	> 1 1/8" < 1 1/4" 28.7 – 31.8 mm	4.5			60	55	50	45	40				
			3.9			60	55	50	45	40				
PES < 40	100%	1.1 – 1.3 dtex	5.9		100	95	85	75	65					
			5.4		95	95	85	75	65					
			4.9		85	85	85	75	65					
			4.5		70	70	70	70	65					
			3.9		60	60	60	60	60					
		1.4 – 1.7 dtex	6.5		110	105	95	85	75					
			5.9		105	105	95	85	75					
			5.4		95	95	95	85	75					
			4.9		85	85	85	85	75					
			4.5		70	70	70	70	70					
			3.9		60	60	60	60	60					
PAN < 40	100%	1.1 – 1.7 dtex	4.9					75	75	75	75			
			4.5					70	70	70	70			
			3.9					60	60	60	60			
CV < 40	100%	1.1 – 1.3 dtex	4.9				85	85	75	70				
			4.5				70	70	70	70				
			3.9				60	60	60	60				
		1.4 – 1.7 dtex	4.9				85	85	85	80				
			4.5				70	70	70	70				
			3.9				60	60	60	60				

Raw material			Sliver count > ktex	Main yarn count										
Material	Blend ratio 1	Fiber finesse		- 10	- 16	- 20	- 24	- 30	- 33	- 36	- 40	- 50	- 60	Ne
				- 59.1	59.0 – 36.9	36.8 – 29.5	29.4 – 24.6	24.5 – 19.7	19.6 – 17.9	17.8 – 16.4	16.3 – 14.8	14.7 – 11.8	11.7 – 9.8	Tex
PES/CV < 40	> 35%	1.1 – 1.3 dtex	4.9				85	85	75	70				
			4.5				70	70	70	70				
			3.9				60	60	60	60				
		1.4 – 1.7 dtex	5.4				95	95	90	80				
			4.9				85	85	85	80				
			4.5				70	70	70	70				
	3.9				60	60	60	60						
	> 35%	1.1 – 1.3 dtex	4.9				75	75						
			4.5				70	70						
3.9						60	60							
CV/PES < 40	> 65%	1.1 – 1.3 dtex	4.9				85	85	75	70				
			4.5				70	70	70	70				
			3.9				60	60	60	60				
		1.4 – 1.7 dtex	4.9				85	85	85	80				
			4.5				70	70	70	70				
			3.9				60	60	60	60				
CV/PAN < 40	> 65%	1.1 – 1.7 dtex	4.9				75	75						
			4.5				70	70						
			3.9				60	60						
PES/CO	> 35%	1.1 – 1.3 dtex	4.9		80	75	70	60	50					
			4.5		70	70	70	60	50					
			3.9		60	60	60	60	50					
		1.4 – 1.7 dtex	5.4		90	85	75	70	60					
			4.9		85	85	75	70	60					
			4.5		70	70	70	70	60					
	3.9		60	60	60	60	60							
	> 35%	1.1 – 1.3 dtex	4.5				60	60						
			3.9				60	60						
CV/CO	> 50%	1.1 – 1.3 dtex	4.5				70	70	60	55				
			3.9				60	60	60	55				
CV/CO	> 50%	1.4 – 1.7 dtex	4.9				75	75	70	65				
			4.5				70	70	70	65				
			3.9				60	60	60	60				

2.2.9. Blowroom

Production data in the blowroom	Theoretical production in kg/h					
Machine	Cotton	Waste	Linen	MMF	Viscose	Regenerated fiber
A 21	1 100	1 100	1 100	990	1 100	1 100
Vision Shield 2	1 320	1 320	1 320			
A 79	1 100	990		800	990	990
B 60			440			
B 76	1 100	880		990	1 100	1 100
B 12	1 320	1 320	880			
A 22	1 320	1 320				
A 21			1 100			
B 25	70	70	70	70	70	70
A 12	2 200	1 760	1 760	1 540	2 200	1 760

2.2.10. Can capacity

Cans with castors

Can format		Cans with castors								
Outside		Netto weight of filling on the card in kg (without can)			Netto weight on draw frame in kg					Netto can weight on comber in kg
Diameter in mm	Height	Cotton and viscose	Polyester	PAN	Carded CV und CO	Combed cotton	PES	PAN	Blends CO/MMF	Combed cotton
400	900				10.0	13	8.5	7.5	10.0	
	1000				11.0	14.5	10	8.5	11.0	
	1070				12.0	16	10.5	9.0	12.0	
	1100				12.5	16.5	11	9.5	12.0	
	1150				13.0	17	11.5	10.0	13.0	
	1200				13.5	18	12	10.5	13.5	
450	900				12.5	16	10.5	9.0	12.0	
	1000				14.0	18	12	10.5	13.5	
	1070				15.0	19.5	13	11.0	14.5	
	1100				15.5	20	13	11.5	15.0	
	1150				16.0	21	14	12.0	15.5	
	1200				17.0	22	14.5	12.5	16.5	
470	900				13.0	17	11.5	10.0	13.0	
	1000				15.0	19	12.5	11.0	14.5	
	1070				16.0	20.5	13.5	12.0	15.5	
	1100				16.5	21.5	14	12.0	16.0	
	1150				17.0	22.5	15	13.0	17.0	
	1200				18.0	23.5	15.5	13.5	17.5	
	1270				19.0	25	16.5	14.0	18.5	
	1300				19.5	25.5	17	14.5	19.0	
500	900				15.5	19	12.5	11.0	14.5	19
	1000				17.0	21	14	12.0	16.0	21.5
	1070				18.5	23	15	13.0	17.5	23
	1100				19.0	23.5	15.5	13.5	18.0	23.5
	1150				20.0	24.5	16.5	14.0	18.5	25
	1200				21.0	26	17	15.0	19.5	26
	1270				22.5	27.5	18.5	15.5	21.0	27.5
	1300				23.0	28	18.5	16.0	21.5	28.5
600	1500				26.5	32.5	22	19.0	25.0	
	900	20	17	14	20.5	25.5	17.5	15.0	20.0	25.5
	1000	22.5	19	16	23	28.5	19.5	16.5	22.5	28.5
	1070	24	20	17	24.5	31	21	18.0	24.0	31
	1100	25	21	18	25.5	32	21.5	18.5	25.0	32
	1150	26	22	19	26.5	33.5	22.5	19.5	26.0	33.5
	1200	27.5	23	20	28	35	24	20.5	27.5	35
	1270	29	25	21	29.5	37	25.5	21.5	29.0	37
800	1300	29.5	25	21	30.5	38	26	22.0	29.5	38
	1500	34.5	29	25	35.5	44.5	30	25.5	34.5	44.5
	1000	31	26	21	32	41	27	22.0	31.0	42
	1070	33	28	23	34	44	29	24.0	33.0	46
	1100	34	29	24	35	45	29	25.0	34.0	47
	1150	36	30	25	37	48	31	26.0	36.0	49
	1200	38	31	26	39	50	32	27.0	38.0	52
	1270	40	33	28	41	53	35	29.0	40.0	55
	1300	41	34	28	43	54	35	30.0	41.0	56
	1500	48	40	33	50	63	41	35.0	48.0	66

Cans without castors

Can format		Cans without castors								
Outside		Netto weight of filling on the card in kg (without can)			Netto weight on draw frame in kg					Netto can weight on comber in kg
Diameter	Height	Cotton and viscose	Polyester	PAN	Carded CV und CO	Combed cotton	PES	PAN	Blends CO/MMF	Combed cotton
225	900				3.5	4	3	3	3.5	
	1000				4.0	4	3.5	3	4.0	
	1070				4.0	4.5	3.5	3.5	4.0	
	1100				4.5	4.5	4	3.5	4.0	
250	900				4.5	5.5	4	3.5	4.5	
	1000				5.0	6.5	4.5	4.0	5.0	
	1070				5.5	7	4.5	4.0	5.0	
	1100				5.5	7	5	4.5	5.5	
300	900				6.5	8.5	5.5	5.0	6.5	
	1000				7.0	9	6.5	5.5	7.0	
	1070				7.5	10	7	6.0	7.5	
	1100				8.0	10	7	6.0	7.5	
350	900				8.5	11	7.5	6.5	8.5	
	1000				9.5	12.5	8.5	7.5	9.5	
	1070				10.5	13.5	9	8	10.0	
	1100				10.5	13.5	9.5	8	10.5	
400	900	10.5	9	8	11.0	14.5	9.5	8.5	11.0	
	1000	12	10	9	12.5	16	10.5	9.5	12.0	
	1070	12.5	11	9.5	13.0	17	11.5	10.0	13.0	
	1100	13	11	9.5	13.5	17.5	12	10.5	13.0	
	1150	13.5	11.5	10	14.0	18.5	12.5	11.0	14.0	
	1200	14	12	10.5	14.5	19	13	11.0	14.5	
450	900	13.5	11.5	10	13.5	18	11.5	10.0	13.5	
	1000	15	12.5	11	15.0	20	13	11.5	15.0	
	1070	16	13.5	11.5	16.0	21.5	14	12.0	16.0	
	1100	16	14	12	16.5	22	14.5	12.5	16.0	
	1150	17	14.5	12.5	17.5	23	15	13.0	17.0	
	1200	17.5	15	13	18.0	24	15.5	13.5	17.5	
470	900	14	12	10.5	900	19	12.5	11.0	14.0	
	1000	16	13.5	11.5	1000	21	14	12.0	16.0	
	1070	17	14.5	12.5	1070	22.5	15	13.0	17.0	
	1100	17.5	15	13	1100	23	15.5	13.0	17.5	
	1150	18	15.5	13.5	1150	24	16	14.0	18.0	
	1200	19	16	14	1200	25	16.5	14.5	19.0	
	1270	20	17	15	1270	26.5	17.5	15.5	20.0	
	1300	20.5	17.5	15	1300	27.5	18	15.5	20.5	
500	900	14.5	12	10.5	17.0	21	14	12.0	16.0	21
	1000	16	13.5	12	19.0	23	15.5	13.5	17.5	23.5
	1070	17.5	15	12.5	20.0	25	16.5	14.0	19.0	25
	1100	18	15	13	20.5	25.5	17	14.5	19.5	25.5
	1150	18.5	16	13.5	21.5	26.5	17.5	15.5	20.0	27
	1200	19.5	16.5	14.5	22.5	28	18.5	16.0	21.0	28
	1270	21	18	15	24.0	29.5	19.5	17.0	22.5	29.5
	1300	21.5	18	15.5	24.5	30	20	17.5	23.0	30.5
	1500				28.5	34.5	23	20.0	26.5	

Can format		Cans without castors								
Outside		Netto weight of filling on the card in kg (without can)			Netto weight on draw frame in kg					Netto can weight on comber in kg
Diameter	Height	Cotton and viscose	Polyester	PAN	Carded CV und CO	Combed cotton	PES	PAN	Blends CO/MMF	Combed cotton
600	900				22.5	28	19	16.5	22.0	28
	1000				25	31.5	21.5	18.0	24.5	31.5
	1070				27	33.5	23	19.5	26.0	33.5
	1100				27.5	34.5	23.5	20.0	27.0	34.5
	1150				29	36	24.5	21.0	28.0	36
	1200				30	37.5	25.5	22.0	29.5	37.5
	1270				32	40	27	23.0	31.0	40
	1300				32.5	41	27.5	23.5	32.0	41
	1500				37.5	47	32	27.5	36.5	47

2.2.11. Total waste per machine in %

Machines						1.7 and 1.3 dtex			
	Cotton					Linen	Viscose	PES	PAN
	up to 1"	up to 1 1/16"	up to 1 1/8"	up to 1 1/2"	up to 1 7/8"		up to 40 mm	up to 40 mm	up to 50 – 60 mm
Air-jet spinning machine	8.2	8.2	5.2	5.2	5.2		1.2	1.2	1.2
Winder	1	1	1	1	1		1	1	1
Ring spinning machine	1.5	2.2	2.7	3	3.2		2.2	2.2	2.2
Rotor spinning machine	1	0.9	0.8			1.1	0.7	0.7	0.7
Roving frame	1	1	1	1	1		1	1	1
2. draw frame	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Comber	15	17	17	17	17				
UNIlap	0.5	0.5	0.5	0.5	0.5				
1. draw frame (before UNIlap)	0.6	0.6	0.6	0.6	0.6		0.6	0.6	0.6
Card	6.4	5	4.6	4.3	4.1	3.8	1.5	0.6	0.6
Blowroom total	3	2.5	2.5	1.5	1.5	8	0.3	0.3	0.3

2.3. Expenditure

2.3.1. Ring spinning personnel and energy

Operators per shift (8 h)													
Ne	10	20	30	40	50	60	70	80	90	100	120	140	160
Total production in kg/h	131.5	66.9	42.8	29.7	21.8	16.5	12.7	9.8	8.3	6.9	5.1	3.6	2.5
Ring spinning machine (1 824 spindles per machine)	0.9	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.3
Roving frame (144 positions per machine)	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2
Draw frame	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Comber	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Combing preparation	0.2	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Draw frame for combing	0.1	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Card	0.1												
Blowroom total	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2

Installed power per machine in kW	
Ring spinning machine (1824 spindles)	109.4
SERVOrail	2.8
Roving frame (144 positions)	38.7
Draw frame	11.2
Comber	13.9
SERVOLap	2.3
Combing preparation	14
Draw frame for combing	8.5
Card	27.8
Blowroom total	122.7

Machines per operator	
Ring spinning machine (1 824 spindles)	1 to 3
Roving frame (144 positions)	2 to 3
Draw frame	up to 10
Comber	up to 10
Combing preparation	up to 5
Draw frame	up to 10
Card	20 – 40
Blowroom total	1 to 5

Spindles per operator													
Yarn count classification	Coarse			Medium			Fine			Very fine			
Ne	10	20	30	40	50	60	70	80	90	100	120	140	160
Ring spinning machine (1 824)	2 026	3 039	3 039	3 648	3 648	4 560	4 560	4 560	6 080	6 080	6 080	6 080	6 080
Roving frame (144)	480	480	480	480	480	480	480	480	720	720	720	720	720

Energy consumption in kW/kg yarn													
Yarn count classification	Coarse				Medium				Fine				Very fine
Ne	10	20	30	40	50	60	70	80	90	100	120	140	160
Total production in kg/h	131.5	66.9	42.8	29.7	21.8	16.5	12.7	9.8	8.3	6.9	5.1	3.6	2.5
Ring spinning machine (1 824 spindles)	0.832	1.635	2.556	3.684	5.018	6.63	8.614	11.163	13.181	15.855	21.451	30.389	43.76
SERVOTrail	0.021	0.042	0.065	0.094	0.128	1.17	0.22	0.286	0.337	0.406	0.549	0.778	1.12
Roving frame (144 positions)	0.294	0.578	0.904	1.303	1.775	2.345	3.047	3.949	4.663	5.609	7.588	10.75	15.48
Draw frame	0.085	0.167	0.262	0.377	0.514	0.679	0.882	1.143	1.349	1.623	2.196	3.111	4.48
Comber	0.021	0.208	0.325	1.303	1.775	0.842	1.094	1.418	1.675	2.014	2.725	3.861	5.56
SERVOLap	0.017	0.034	0.054	0.077	0.106	0.139	0.181	0.235	0.277	0.333	0.451	0.639	0.92
Combing preparation	0.106	0.209	0.327	0.471	0.642	0.848	1.102	1.429	1.687	2.029	2.745	3.889	5.6
Draw frame for combing	0.065	0.127	0.199	0.286	0.39	0.515	0.669	0.867	1.024	1.232	1.667	2.361	3.4
Card	0.211	0.416	0.65	0.936	1.275	1.685	2.189	2.837	3.349	4.029	5.451	7.722	11.12
Blowroom total	0.933	1.834	2.867	4.131	5.628	7.436	9.661	12.52	14.783	17.783	24.059	34.083	49.08

2.3.2. Compact spinning personnel and energy

Operators per shift (8 h)													
Ne	10	20	30	40	50	60	70	80	90	100	120	140	160
Total production kg/h	132.2	68.9	44.1	31.1	22.6	17.1	13.3	10.2	8.3	7.1	4.9	3.4	2.3
Compact spinning (1 824 spindles)	1	0.7	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.3
Roving frame (144 positions)	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2
Draw frame	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Comber	(4) 0.2	(2) 0.1	(2) 0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Combing preparation	0.2	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Draw frame for combing	0.1	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Card	(5) 0.1	(3) 0.1	0.1										
Blowroom total	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2

Installed power per machine in kW	
Compact spinning machine (1 824 spindles)	115.9
Servotrail	2.8
Roving frame (144 positions)	38.7
Draw frame	11.2
Comber	7
E 26	2.3
Combing preparation	14
Draw frame for combing	8.5
Card	27.8
Blowroom total	122.7

Machines per operator	
Compact spinning (1 824 spindles)	1 to 3
Roving frame (144 positions)	2 to 3
Draw frame	up to 10
Comber	up to 10
Combing preparation	up to 10
Draw frame	up to 10
Card	20 – 40
Blowroom total	1 to 5

Spindles per operator													
Yarn count classification	Coarse	Medium			Fine				Very fine				
Ne	10	20	30	40	50	60	70	80	90	100	120	140	160
Compact spinning machine (1 824 spindles)	1 824	2 605	3 040	3 648	3 648	4 560	4 560	4 560	6 080	6 080	6 080	6 080	6 080
Roving frame (144 positions)	480	480	480	480	480	480	480	480	720	720	720	720	720

Energy consumption in kW/kg yarn													
Yarn count classification	Coarse	Medium			Fine				Very fine				
Ne	10	20	30	40	50	60	70	80	90	100	120	140	160
Total production in kg/h	132.2	68.9	44.1	31.1	22.6	17.1	13.3	10.2	8.3	7.1	4.9	3.4	2.3
Compact spinning machine (1 824 spindles per machine)	0.877	1.682	2.628	3.727	5.128	6.778	8.714	11.363	13.963	16.324	23.653	34.088	50.391
SERVOfail	0.021	0.041	0.063	0.09	0.124	0.163	0.211	0.275	0.337	0.394	0.571	0.824	1.217
Roving frame (144 positions per machine)	0.293	0.562	0.878	1.244	1.712	2.263	2.91	3.794	4.662	5.451	7.898	11.382	16.826
Draw frame	0.085	0.163	0.193	0.36	0.5	0.655	0.842	1.098	1.349	1.577	2.286	3.294	4.87
Comber	0.053	0.102	0.159	0.225	0.31	0.409	0.526	0.686	0.843	0.986	1.429	2.059	3.043
SERVOfap	0.017	0.033	0.052	0.074	0.102	0.17	0.173	0.225	0.277	0.324	0.469	0.675	1
Combing preparation	0.106	0.203	0.317	0.45	0.619	0.819	1.053	1.373	1.687	1.972	2.857	4.118	6.087
Draw frame for combing	0.064	0.123	0.193	0.273	0.376	0.497	0.639	0.833	1.024	1.197	1.735	2.5	3.7
Card	0.21	0.403	0.63	0.834	1.23	1.626	2.09	2.725	3.349	3.945	5.673	8.176	12.087
Blowroom total	0.928	1.781	2.782	3.945	5.429	7.175	9.226	12.029	14.783	17.28	25.041	36.088	53.348

2.3.3. Semi-automated rotor spinning personnel and energy

Operators per shift (8 h)				
Yarn count in Ne	10	20	30	40
Total production in kg/h	314.7	157.8	84.7	55.2
Semi rotor spinning machine (600 rotors)	1.8	1.3	1.1	1
Draw frame	0.1	0.1	0.1	0.1
Draw frame	0.1	0.1	0.1	0.1
Card	0.1	0.1	0.1	0.1
Blowroom	0.4	0.2	0.2	0.2

Installed power per machine in kW		Machines per operator	
Semi rotor spinning machine (600 rotors)	167.2	Semi rotor spinning machine (600 rotors)	1
Draw frame 2. passage	11.2	Draw frame 2. passage	up to 10
Draw frame 1. passage	8.5	Draw frame 1. passage	up to 10
Card	27.8	Card	20 to 40
Blowroom	122.7	Blowroom	2 to 5

Rotors per operator				
Yarn count classification	Coarse	Medium		
Yarn Count in Ne	10	20	30	40
Total production in kg/h	314.7	157.8	84.7	55.2
Semi rotor spinning machine (600 rotors)	333	462	545	600

Energy consumption in kW/kg yarn				
Yarn count classification	Coarse	Medium		
Ne	10	20	30	40
Total production in kg/h	314.7	157.8	84.7	55.2
Semi rotor machine	0.532	1.06	1.974	3.029
Draw frame 2. passage	0.036	0.071	0.132	0.203
Draw frame 1. passage	0.027	0.054	0.1	0.154
Card	0.088	0.176	0.328	0.504
Blowroom total	0.39	0.778	1.449	2.223

2.3.4. Fully automated rotor spinning personnel and energy

Operators per shift (8h)				
Ne	10	20	30	40
Total production in kg/h	335.8	162.1	100.9	65.6
Auto rotor spinning machine (600 rotors)	0.9	0.5	0.4	0.3
Draw frame	0.1	0.1	0.1	0.1
Draw frame	0.2	0.1	0.1	0.1
Card	0.1	0.1	0.1	0.1
Blowroom total	0.5	0.2	0.2	0.2

Installed power per machine in kW	
Auto rotor spinning machine (600 rotors)	300.4
Draw frame 2. passage	11.2
Draw frame 1. passage	8.5
Card	27.8
Blowroom total	122.7

Machines per operator	
Auto rotor spinning machine (600 rotors)	1 to 3
Draw frame 2. passage	up to 10
Draw frame 1. passage	up to 10
Card	20 to 40
Blowroom total	2 to 5

Rotors per operator				
Yarn count classification	Coarse	Medium		
Ne	10	20	30	40
Total production in kg/h	335.8	162.1	100.9	65.6
Auto rotor spinning machine (600 rotors)	667	1 200	1 500	2 000

Energy consumption in kW/kg yarn				
Yarn count classification	Coarse	Medium		
Ne	10	20	30	40
Total production in kg/h	335.8	162.1	100.9	65.6
Auto rotor spinning machine (600 rotors)	0.896	1.853	2.977	4.579
Draw frame 2. passage	0.033	0.069	0.111	0.171
Draw frame 1. passage	0.025	0.052	0.084	0.13
Card	0.083	0.171	0.276	0.424
Blowroom total	0.365	0.757	1.216	1.87

2.3.5. Air-jet spinning personnel and energy

Operators per shift (8h)							
Ne	20	30	40	50	60	70	80
Total Production in kg/h	139.4	96.9	72.5	50.3	37.8	30	22.9
Air-jet spinning machine (200 positions)	0.7	0.6	0.6	0.5	0.5	0.5	0.5
Draw frame	0.1	0.1					
Draw frame	0.1	0.1	0.1	0.1	0.1		
Draw frame	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Card	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Blowroom total	0.2	0.2	0.2	0.2	0.2	0.2	0.2

Installed power per machine in kW	
Air-jet spinning machine (200 positions) (Including compressed air)	90.5
Draw frame	11.2
Draw frame	8.5
Draw frame	8.5
Card	27.8
Blowroom total	68.5

Machines per operator	
Air-jet spinning machine (200 positions)	1 To 2
Draw frame	up to 10
Card	20 – 40
Blowroom total	up to 5

Units per operator							
Ne	20	30	40	50	60	70	80
Total production in kg/h	139.4	96.9	72.5	50.3	37.8	30	22.9
Air-jet spinning machine (200 positions)	300	360	360	400	400	400	400

Energy consumption in kW/kg yarn							
Ne	20	30	40	50	60	70	80
Total production in kg/h	139.4	96.9	72.5	50.3	37.8	30	22.9
Air-jet spinning machine (200 positions)	0.649	0.934	1.248	1.799	2.394	3.017	3.952
Draw frame	0.08	0.116	0.154	0.229	0.296	0.373	0.489
Draw frame	0.061	0.88	0.117	0.169	0.225	0.283	0.371
Draw frame	0.061	0.88	0.117	0.169	0.225	0.283	0.371
Card	0.199	0.288	0.383	0.553	0.735	0.927	1.214
Blowroom total	0.491	0.707	0.945	1.362	1.812	2.283	2.991

2.3.6. Floor space

2.3.6.1. Floor space winder

Units per machine	Floor space in m²
8	26
10	28
12	29
14	31
16	33
18	35
20	36
22	38
24	40
26	41
28	43
30	45
32	46
34	48
36	50
38	51
40	53
42	55
44	56
46	58
48	60
50	61
52	63
54	65
56	66
58	68
60	70
62	71
64	73

Winder, width of machine: 1 090 mm

2.3.6.2. Floor space ring spinning machine

G 36, G 32 (Link)

	Machine gauge in mm	
	70	75
	Width of machine in mm	
	1090	1090
Spindles per machine	Floor space per machine in m ²	
480	59	63
528	64	67
576	68	72
624	72	77
672	77	81
720	81	86
768	86	91
816	90	95
864	94	100
912	99	104
960	103	109
1008	107	114
1056	112	118
1104	116	123
1152	120	128
1200	125	132
1248	129	137
1296	133	142
1344	138	146
1392	142	151
1440	148	157
1488	152	162
1536	157	167
1584	161	171
1632	165	176
1680	170	181
1728	174	185
1776	178	190
1824	183	195

K 46, K 42

	Machine gauge in mm	
	70	75
	Width of machine in mm	
	1 090	1 090
Spindles per machine	Floor space in m ²	
480	59	63
528	64	67
576	68	72
624	72	77
672	77	81
720	81	86
768	86	91
816	90	95
864	94	100
912	99	104
960	103	109
1 008	107	114
1 056	112	118
1 104	116	123
1 152	120	128
1 200	125	132
1 248	129	137
1 296	133	142
1 344	138	146
1 392	142	151
1 440	152	161
1 488	156	166
1 536	160	170
1 584	165	175
1 632	169	180
1 680	174	184
1 728	178	189
1 776	182	194
1 824	187	198

2.3.6.4. Floor space semi-automated rotor spinning machine

R 36 with round cans

	Can diameter in mm		
	400	450	500
	Width of Machine inclusive Cans in mm		
	1 760	2 140	2 700
Rotors per machine	Floor space per machine in m ²		
120	100	111	126
140	110	122	138
160	119	132	150
180	129	143	162
200	139	153	174
220	148	164	187
240	158	175	199
260	167	185	211
280	177	196	223
300	187	207	235
320	196	217	247
340	206	228	259
360	216	239	271
380	225	249	283
400	235	260	295
420	244	271	308
440	254	281	320
460	264	292	332
480	273	303	344
500	283	313	356
520	293	324	368
540	302	335	380
560	312	345	392
580	321	356	404
600	331	366	416

R 66 with round cans

Rotors per machine	Round can diameter in mm			
	400	450	470	500
	Width of machine inklusive can in mm			
	2 230	2 330	2 350	2 700
	Floor space per machine in m²			
120	125	130	131	142
140	137	142	143	154
160	148	153	155	167
180	160	165	167	180
200	171	177	179	193
220	183	189	191	206
240	194	201	202	219
260	205	212	214	232
280	217	224	226	245
300	228	236	238	258
320	240	248	250	270
340	251	260	262	283
360	262	272	274	296
380	274	283	286	309
400	285	295	298	322
420	297	307	310	335
440	308	319	322	348
460	320	331	334	361
480	331	342	346	374
500	342	354	358	386
520	354	366	369	399
540	365	378	381	412
560	377	390	393	425
580	388	402	405	438
600	400	413	417	451
500	352	365	368	398
520	364	376	380	411
540	375	388	392	423
560	387	400	404	436
580	398	412	416	449
600	410	424	428	462
620	421	436	440	475
640	432	447	451	488
660	444	459	463	501
680	455	471	475	514
700	467	483	487	527

2.3.6.6. Floor space air-jet spinning machine

J 26 with round cans

Units per machine	Can diameter in mm	
	470	500
	Width of machine inclusive robots in mm	
	3 060	3 060
Floor space per machine in m ²		
20	86	87
40	101	103
60	117	118
80	133	134
100	149	150
120	165	166
140	180	182
160	196	198
180	212	214
200	228	230

F 38 with SERV0trail

			Can diameter in mm	
			500	600
Gauge	Units per machine	Lenght of machine in mm	Floor space in m²	
110 mm	32	5 775	86	99
	48	7 535	102	116
	64	9 295	117	134
	80	11 055	133	152
	96	12 815	149	170
	112	14 575	164	187
	128	16 335	180	205
	144	18 095	195	223
	160	19 855	211	241
	176	21 615	226	258
	192	23 375	242	276
	208	25 135	257	294
	224	26 895	273	312
130 mm	36	6 885	91	104
	48	8 445	104	119
	60	10 005	116	134
	72	11 565	129	149
	84	13 125	142	164
	96	14 685	155	179
	108	16 245	168	194
	120	17 805	181	209
	132	19 365	194	224
	144	20 925	207	239
	156	22 485	220	254
	168	24 045	233	269

2.3.6.8. Floor space draw frame

(R)SB-D 50

		Can diameter in mm (delivery)			
		400 – 500	600	800	1 000
Doubling	Can diameter in mm infeed	Floor space per machine in m ²			
6	600	98	98	105	105
	800	97	97	104	104
	1 000	97	97	104	104
	1 200	101	101	108	108
8	600	102	102	109	109
	800	106	106	113	113
	1 000	106	106	113	113
	1 200	111	111	119	81

2.3.6.9. Floor space comber

E 86 with SERVOlap

Can diameter in mm	
600	1 000
Width of machine in mm	
7 634	8 152
Length of machine in mm	
2 077	2 077
Floor space per machine in m ²	
37	43

2.3.6.10. Floor space combing preparation

OMEGAlap with SERVOlap

Can diameter in mm	
600	1 000
Width of machine (cans inclusive) in mm (über Kannen)	
5 038	6 890
Length of machine in mm	
7 835	7 963
Floor space per machine m ²	
129	171

2.3.6.11. Floor space card

C 70 with CBA coiler

Can diameter in mm		
	600	800
Floor space in m ²	39	41

3. Spinning Mill Calculation Formulas

3.1. Yarn count

The numbering systems

As with wires in the metal industry, all linear textile structures (e.g. yarns, slivers, etc.) must also have a quite specific thickness, which varies according to their range of application. However, since these structures are not uniform and can be compressed to differing extents, their degree of fineness can only rarely be determined by measuring their diameter. Fineness (count) has to be calculated indirectly. This is done by relating the measurable mass and the measurable length.

Numbering by mass

Numbering by mass indicates a product's units of mass per defined unit of length. i.e.

$$\text{count} = \frac{\text{mass}}{\text{length}}$$

The international tex count system

The tex system is a decimal system which can be used for all linear structures from the fiber to the yarn:

$$T_t = \frac{m}{l} \left[\frac{\text{grams}}{1\,000\text{ m}} = \frac{\text{g}}{\text{km}} \right]$$

The count expressed in tex indicates how many grams 1 000 metres of a product weigh, «tex» is a physical unit and is placed behind the figure. Here too, as with the other decimal measuring systems, the usual multiples and fractions of tex are used. i.e. mtex, dtex (fibers), ktex (slivers, cables, etc.).

$$\left[\text{tex} = \frac{\text{g}}{\text{km}} ; \text{mtex} = \frac{\text{mg}}{\text{km}} ; \text{ktex} = \frac{\text{kg}}{\text{km}} = \frac{\text{g}}{\text{m}} \right]$$

The denier count

The denier count was used for continuous fibers such as silk and man-made fibers. It goes back to old French units of weight and length. The denier count indicates how many denier (0.05 g) a thread 450 metres long weighs, or the weight in grams of 9 000 metres of yarn:

$$T_d = \frac{0.05\text{ g}}{450\text{ m}} = \frac{\text{g}}{9\,000\text{ m}}$$

60 den indicates that 9 000 metres of a thread weigh 60 grams. Calculation formula:

$$T_d = \frac{(y)\text{g} \cdot 9\,000}{\text{m}} = (x)\text{ den}$$

Numbering by length

Numbering by length indicates the fineness of the linear product as the ratio of length to mass. i.e.

$$\text{count} = \frac{\text{length}}{\text{mass}}$$

The metric count

The metric count is based on the metric quantities of the unit system:

$$N_m = \frac{l}{m} \left[\frac{\text{m}}{\text{g}} = \frac{\text{mm}}{\text{mg}} = \frac{\text{km}}{\text{kg}} \right]$$

Nm 1 indicates that 1 metre of material weighs 1 gram; Nm 10 that 10 metres of material weigh 1 gram.

The English count

English yarn numbering is based on the English unit of weight (pound; lb) and the English unit of textile length (hank).

However, different hank lengths are used as follows:

840 yards for cotton yarns

560 yards for worsted yarns

300 yards for bast fiber yarns

256 yards for woollen yarns.

There are thus different English yarn count figures. e.g.

$$N_{\text{cotton}} = \frac{\text{hank (840 yards)}}{\text{lb}} = \left[\frac{768\text{ m}}{453.6\text{ g}} \right]$$

$$N_{\text{worsted}} = \frac{\text{hank (560 yards)}}{\text{lb}} = \left[\frac{512\text{ m}}{453.6\text{ g}} \right]$$

Conversion tables

The formulas and tables shown are used to convert yarn, sliver and fiber counts from one numbering or designation system to another and to the international tex system (see ISO 1144-1973: Tex system for defining the counts of textile fibers. intermediate products yarns and related products).

The tables on the following pages (conversion formula) show the conversion formulae to and from each system for the most commonly used of these designation systems.

Designation systems for yarn and fiber counts according to international standard ISO 1144-1973.

Unit	Abbreviation	Conversion
millitex	mtex	1 mtex = 1mg/km = 1µg/m
decitex	dtex	1 dtex = 1 dg/km = 0.1 mg/m
tex	tex	1 tex = 1 g/km = 1 mg/m
kilotex	ktex	1 ktex = 1 kg/km = 1 g/m

Count system	Abbreviation	Unit of mass	Unit of length	Unit of yarn count	Multiplier for the yarn count for conversion to the tex system
Tex	Tt	1 gram	1 kilometre	g/km	–
Denier	Td	1 gram	9 000 metres	g/9 000 m	0.1111
Linen dry spun Hemp Jute	Tj	1 pound	14 400 yards (spindle unit)	lb/14 400 yd	34.45
Número en cuartos de onza	To	1/4 onza	500 cañas cuartos de onza	0.25 onza/ 500 años	10.71
Woollen (Aberdeen)	Ta	1 pound	14 400 yards	lb/14 400 yd	34.45
Woollen (Catalonian)	Tc _w	1 gram	504 metres	g/504 m	1.984

Example:

840 denier corresponds in tex to

$840 \times 0.1111 = 93.32 \text{ tex}$ or $933.2 \text{ dtex} = \sim 933 \text{ dtex}$

Count system	Abbreviation	Unit of length	Unit of mass	Unit of yarn count	Multiplier for the yarn count for conversion to the tex system
Asbestos (American)	Na _A	100 yards (cut)	1 pound	100 yd/lb	4 961
Asbestos (English)	Ne _A	50 yards	1 pound	50 yd/lb	9 921
Cotton bump yarn	N _B	1 yard	1 ounce	yd/oz	31 000
Cotton (English)	Ne _C	840 yards (hank)	1 pound	840 yd/lb	590.5
Cotton (Catalonian)	Nc _C	500 cañas	1.1 libra catalana	500 cañas/1.1 lb cat.	565.9
Glass (U.S.A. and U.K.)	N _G	100 yards	1 pound	100 yd/lb	4 961
Linen (wet or dry spun)	Ne _L	300 yards (lea)	1 pound	300 yd/lb	1 654
Metric	Nm	1 metre	1 gram	m/g	1 000
Numero en puntos	Np	1 320 metres	1 libra de Alcoy	1 320 m/lb de Alc.	358.7
Spun silk	N _S	840 yards	1 pound	840 yd/lb	590.5
Type	Nt	1 000 yards	1 pound	1 000 yd/lb	496.1
Wollen (Alloa)	Nal	11 520 yards (spindle)	24 pounds	11 520 yd/24 lb	1 033
Woollen (American cut)	Nac	300 yards	1 pound	300 yd/lb	1 654
Woollen (American run)	Nar	100 yards	1 ounce	100 yd/oz	310
Woollen (Cardado Covilhã)	Np _W	1 metre	5 gram	m/5 g	5 000
Woollen (Dewsbury)	Nd	1 yard	1 ounce	yd/oz	31 000
Woollen (Galashiels)	Ng	300 yards (cut)	24 ounces	300 yd/24 oz	2 480
Wollen (Hawick)	Nh	300 yards (cut)	26 ounces	300 yd/26 oz	2 687
Woollen (Irish)	Ni _W	1 yard	0.25 ounce	yd/0.25 oz	7 751
Woollen (West of England)	Nwe	320 yards (snap)	1 pound	320 yd/lb	1 550
Woollen (Yorkshire)	Ny	256 yards (skein)	1 pound	256 yd/lb	1 938
Woollen (Yorkshire)	Ny	1 yard	1 dram	yd/dram	1 938
Worsted	New _W	560 yards (hank)	1 pound	560 yd/lb	885.8

Conversion formula

	1 Nm	2 Ne _C	3 Ne _L	4 Ne _W
Metric count	Nm	Ne _C = 0.590541 Nm	Ne _L = 1.65352 Nm	Ne _W = 0.885812 Nm
Engl. cotton count	Nm = 1.69336 Ne _C	Ne _C	Ne _L = 2.80000 Ne _C	Ne _W = 1.50000 Ne _C
Engl. bast fiber yarn count	Nm = 0.604772 Ne _L	Ne _C = 0.357143 Ne _L	Ne _L	Ne _W = 0.535714 Ne _L
Engl. worsted count	Nm = 1.12891 Ne _K	Ne _C = 0.666667 Ne _K	Ne _L = 1.86667 Ne _K	Ne _W
English woollen count	Nm = 0.516072 Ne _W	Ne _C = 0.304762 Ne _W	Ne _L = 0.853333 Ne _W	Ne _W = 0.457143 Ne _W
International count	$Nm = \frac{9\,000}{Td}$	$Ne_C = \frac{5\,314.87}{Ts}$	$Ne_L = \frac{14\,881.6}{Ts}$	$Ne_W = \frac{7\,972.31}{Ts}$
Scottish count	$Nm = \frac{29.0291}{Ts}$	$Ne_C = \frac{17.1429}{Ts}$	$Ne_L = \frac{48.0000}{Ts}$	$Ne_W = \frac{25.7143}{Ts}$
Count in the tex system	$Nm = \frac{1\,000}{Tt\,(\text{tex})}$	$Ne_C = \frac{590.541}{Tt\,(\text{tex})}$	$Ne_L = \frac{1\,653.52}{Tt\,(\text{tex})}$	$Ne_W = \frac{885.812}{Tt\,(\text{tex})}$

	5 Ny	6 Td	7 Ts	8 Tt
Metric count	Ny = 1.93771 Nm	$Td = \frac{9\,000}{Nm}$	$Ts = \frac{29.0291}{Nm}$	$Tt\,(\text{tex}) = \frac{1\,000}{Nm}$
Engl. cotton count	Ny = 3.28125 Ne _C	$Td = \frac{5\,314.87}{Ne_C}$	$Ts = \frac{17.1429}{Ne_C}$	$Tt\,(\text{tex}) = \frac{590.541}{Ne_C}$
Engl. bast fiber yarn count	Ny = 1.17188 Ne _L	$Td = \frac{14\,881.6}{Ne_L}$	$Ts = \frac{48.0000}{Ne_L}$	$Tt\,(\text{tex}) = \frac{1\,653.52}{Ne_L}$
Engl. worsted count	Ny = 2.18750 Ne _K	$Td = \frac{7\,972.31}{Ne_K}$	$Ts = \frac{25.7143}{Ne_K}$	$Tt\,(\text{tex}) = \frac{885.812}{Ne_K}$
English woollen count	Ny	$Td = \frac{17\,439.4}{Ne_W}$	$Ts = \frac{56.2500}{Ne_W}$	$Tt\,(\text{tex}) = \frac{1\,937.71}{Ne_W}$
International count	$Ny = \frac{17\,439.4}{Td}$	Td	Ts = 0.00322545 Td	Tt (tex) = 0.111111 Td
Scottish count	$Ny = \frac{56.2500}{Ts}$	Td = 310.034 Ts	Ts	Tt (tex) = 34.4482 Ts
Count in the tex system	$Ny = \frac{1\,937.71}{Tt\,(\text{tex})}$	Td = 9 Tt (tex)	Ts = 0.0290291 Tt (tex)	Tt (tex)

Preferred values

(rounded to preferred round values in accordance with international standard ISO 1144-1973)

If at all possible, the introduction of the tex system should be combined with a rational limitation of the total number of counts manufactured, i.e. by replacing the values of closely neighbouring counts with a single rounded value. The ISO recommends the preferred rounded values given in column 2 of the following table for this purpose.

The range covered by each rounded value is shown in column 1.

The relevant rounded value is defined by ascertaining the range (column 1) in which the precise value determined according to section 1 is to be found.

The figures in the table apply to all units of the tex system. The scope of the table can be extended by multiplication or division of the listed figures by 10 or 100.

Ranges and preferred rounded values in the tex system					
1		2	1		2
Range		Preferred rounded value	Range		Preferred rounded value
Over	Up to		Over	Up to	
9.0	9.4	9.2	31	33	32
9.4	9.8	9.6	33	35	34
9.8	10.25	10	35	37	36
10.25	10.75	10.5	37	39	38
10.75	11.25	11	39	41	40
11.25	11.75	11.5	41	43	42
11.75	12.25	12	43	45	44
12.25	12.75	12.5	45	47	46
12.75	13.5	13	47	49	48
13.5	14.5	14	49	51	50
14.5	15.5	15	51	54	52
15.5	16.5	16	54	58	56
16.5	17.5	17	58	62	60
17.5	18.5	18	62	66	64
18.5	19.5	19	66	70	68
19.5	20.5	20	70	74	72
20.5	21.5	21	74	78	76
21.5	22.5	22	78	82	80
22.5	23.5	23	82	86	84
23.5	24.5	24	86	90	88
24.5	25.5	25	90	94	92
25.5	27	26	94	98	96
27	29	28	98	102.5	100
29	31	30	102.5	107.5	105

3.2. Performance

Peripheral speed

General:

The peripheral speed of cylinders, rolls, discs, etc., is calculated by means of the following formula:

$$v_p = d_c \cdot \pi \cdot n_c$$

v_p	=	[m/min]	peripheral speed
d_c	=	[m]	diameter of the cylinder
n_c	=	[rpm]	speed of the cylinder

Example

Peripheral speed of the ring traveler

$$v_t = d_r \cdot \pi \cdot n_{spi}$$

v_t	=	[m/min]	peripheral speed of the ring traveler
d_r	=	[m]	diameter of the ring
n_{spi}	=	[rpm]	spindle speed

Delivery speed

The delivery speed is identical to the peripheral speed of the delivery roller or cylinder, and is calculated using the following equation:

$$v_d = d_{dr} \cdot \pi \cdot n_{dr}$$

v_d	=	[m/min]	delivery speed
d_{dr}	=	[m]	diameter of the delivery roller
n_{dr}	=	[rpm]	speed of the delivery roller

In ring spinning (and roving) the delivery speed can also be calculated using the following equation:

$$v_d = \frac{n_{spi}}{T}$$

v_d	=	[m/min]	delivery speed
n_{spi}	=	[rpm]	spindle speed
T	=	[T/m]	yarn twist (roving twist)

Production

General

$$P = \frac{f \cdot v_d \cdot 60}{1\,000} \cdot \eta$$

P	=	[kg/h]	production
f	=	[ktex]	fineness / count
v_d	=	[m/min]	delivery speed
η	=	[%]	machine efficiency

In ring spinning (and roving) the delivery speed can also be calculated using the following equation:

$$P = \frac{n_{spi} \cdot f \cdot 60}{T \cdot 1\,000} \cdot \eta$$

P	=	[kg/h]	production
n_{spi}	=	[rpm]	spindle speed
f	=	[ktex]	fineness / count
T	=	[T/m]	yarn twist (roving twist)
η	=	[%]	machine efficiency

For combing the following equation has to be used:

$$P_{combing} = \frac{f \cdot d_{fr} \cdot n_c \cdot q \cdot d \cdot \pi}{1\,000} \cdot \eta$$

$P_{combing}$	=	[kg/h]	production
d_{fr}	=	[m/nip]	delivery rate feed rollers
n_c	=	[nips/min]	nipping rate
f	=	[g/m]	fineness / count
q	=		degree of exploitation
d	=		number of combing heads
η	=	[%]	machine efficiency

3.3. Draft

Draft from the drive

General	
---------	--

D	=	draft
v_d	=	[m/min] peripheral speed of the delivery roller
v_f	=	[m/min] peripheral speed of the feed roller

$$D_{\text{tot}} = v_1 \cdot v_2 \cdot v_3 \cdot \dots v_n$$

D_{tot}	=		total draft resulting from individual drafts
v_1	=	[m/min]	peripheral speed of the 1 st roller
v_2	=	[m/min]	peripheral speed of the 2 nd roller
v_3	=	[m/min]	peripheral speed of the 3 rd roller
v_n	=	[m/min]	peripheral speed of further roller

Without having to calculate the two speeds separately, a calculation can be made direct from the drive:

$$D = \frac{d_d}{d_f} \cdot i$$

D	=		draft
d_d	=	[mm]	diameter of the delivery roller
d_f	=	[mm]	diameter of the feed roller
i	=		gear ratio from delivery to feed roller

Draft from product counts

$$D = \frac{L_d}{L_f} = \frac{f_d}{f_f}$$

D	=		draft
L _d	=	[m]	length delivered
L _f	=	[m]	length fed
f _d	=	[Nm]*	fineness / count of delivered material
f _f	=	[Nm]*	fineness / count of fed material

If the yarn is doubled in the feed the following formula applies:

D	=	draft
f_d	=	[Nm]* fineness / count of delivered material
f_f	=	[Nm]* fineness / count of fed material
d	=	doubling

* Every unit for fineness may be used (Nm, Ne, tex,..) as long as f_f and f_d are in the same unit.

Attenuation

Elimination is not an intentional reduction of the number of fibers but arises as an unavoidable side effect of the necessity for cleaning; it occurs in the blowroom, in carding, and in combing. (Fiber loss is intentional in combing, as the aim is to remove short fibers.) However, since drafting takes place simultaneously here, the term “attenuation” is used. This is defined by:

$$A = D \cdot \frac{100}{100-p}$$

A	=	attenuation
D	=	draft
p	=	[%] waste in percent

The reduction of the number of fibers in the cross-section logically leads to a reduction in diameter of the strand. In terms of fineness, the following relationship is obtained:

$$\frac{d_d}{d_f} = \frac{\sqrt{f_d}}{\sqrt{f_d}}$$

d_d	=	[mm]	diameter of the delivery roller
d_f	=	[mm]	diameter of the feed roller
f_d	=	[Nm]*	fineness / count of delivered material
f_f	=	[Nm]*	fineness / count of fed material

* Every unit for fineness may be used (Nm, Ne, tex,...) as long as f_r and f_d are in the same unit.

3.4. Twist

Twist factor, coefficient of twist

The twist factor or coefficient of twist α is used as a directly comparable measure of twist in yarns of different counts.

The coefficient of twist cannot be calculated, it is an empirical value. Since no international standard exists, different values are used:

The ranges are roughly as follows:

- using the English system, between 1.5 and 6
- using the metric system, between 45 and 180
- using the tex system, between 1 500 and 6 000

The twist equation according Köchlin

Twist per inch

$$T / \text{inch} = \alpha_e \cdot \sqrt{N_{e_c}}$$

T	=	twist
α_e	=	twist coefficient in accordance with the English count system
N_{e_c}	=	English count for cotton

Twist per meter

$$T / \text{meter} = \alpha_m \cdot \sqrt{N_m}$$

T	=	twist
α_m	=	twist coefficient in accordance with the metric count system
N_m	=	metric count

Important conversions

$$\alpha_e = \alpha_m \cdot 0.033 = \frac{\alpha_{\text{tex}}}{958}$$

$$\alpha_{\text{tex}} = \alpha_m \cdot 31.6 = \alpha_e \cdot 958$$

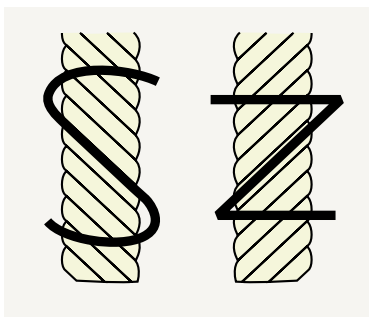
$$\alpha_m = \alpha_e \cdot 30.3$$

$$T / \text{inch} = T / \text{meter} \cdot 0.0254$$

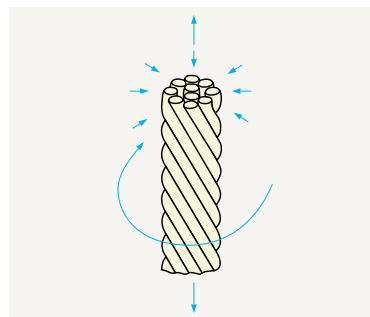
Yarn twist from delivery and spindle speed

$$T / \text{meter} = \frac{n_{\text{spi}}}{v_d}$$

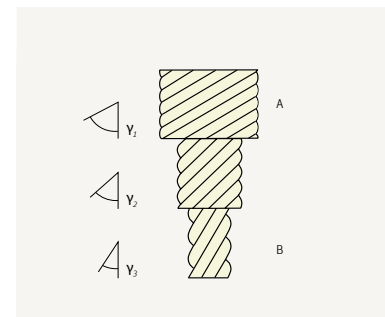
T	=	twist
n_{spi}	=	[rpm] spindle speed
v_d	=	[m/min] delivery speed



Yarn twist in S or Z direction



Yarn twist increases yarn strength



Alpha of twist

4. Technological Spinning Mill Data

4.1. Moisture content

Moisture content of yarns at different relative humidities (in% of weight in wet state)

Relative humidity (%)	Cotton	Flax	Jute	Cupro	Acetate	Wool	Silk	Polyamide 6.6
30	5.0	6.5	6.6	5.6	2.0	8.8	6.2	2.3
40	5.5	7.1	7.8	6.9	2.5	9.7	7.5	2.9
50	6.0	7.5	8.9	9.1	3.3	10.4	8.2	3.4
60	6.5	8.2	9.8	10.2	4.0	11.0	9.3	4.0
65	6.8	8.6	10.4	11.2	4.5	11.4	10.0	4.4
70	7.1	8.8	11.0	12.5	4.9	11.6	10.5	4.7
80	8.2	10.0	13.0	15.0	6.0	12.5	12.0	5.7
90	10.0	12.0	15.8	18.2	7.7	14.7	14.2	6.9
100	14.2	18.0	25.4	26.2	10.7	23.4	19.3	10.0

Moisture additions

Commercial additions in% of dry weight

Cotton

Raw cotton, cotton yarn and folded yarn	8 1/2%
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Cellulose filament

Viscose and cuprammonium rayon	13%
Acetate rayon	9%

Bast fibers

Hemp, linen	12%
Jute, raw	3 3/4%
Jute yarn up to NeL 6	18%
Jute yarn over NeL 6	16%

Cellulose staple fibers

Viscose and cuprammonium rayon staple fibers	14%
Acetate staple fibers	9%

Man-made staple fibers

Natural silk	11%	Polyamide 6.6	4 1/2%
		Polyamide 6	3 1/2%
Wool		Polyester	1/2%
Pure washed wool, reclaimed wool	17%	Polyacryl	1 1/2%
Combed sliver in oil	19%	Polyvinyl chloride	1/4%
Combed sliver without oil	18 1/4%	Polypropylene	1%
Worsted yarn	18 1/4%		
Woollen yarn	17%		

4.2. Cotton quality characteristics guide

Upper half mean length (UHML) – HVI mode

less than 0.99"	less than 25.1 mm	short
0.99 – 1.10"	25.1 – 27.9 mm	medium
1.11 – 1.26"	28.2 – 32.0 mm	long
above 1.27"	above 32.0 mm	extra long

Fiber strength (1/8" in g/tex) – HVI mode (USDA)

23 and below	weak
24 – 25	intermediate
26 – 28	average
29 – 30	strong
31 and above	very strong

Fiber strength (Pressley (0) in 1 000 lb/inch²)

70 and less	very weak
70 – 74	weak
85 – 80	average
81 – 86	strong
87 – 92	very strong
93 and more	outstanding

Fiber strength (Pressley (1/8") in g/tex)

21 and less	very weak
21 – 23	weak
24 – 26	average
27 – 29	strong
30 and more	very strong

Fiber strength (Stelometer (1/8") in g/tex) – ICC mode

17 and less	very weak
17 – 19	weak
20 – 22	average
23 – 26	strong
26 and more	very strong

Fiber elongation (%)

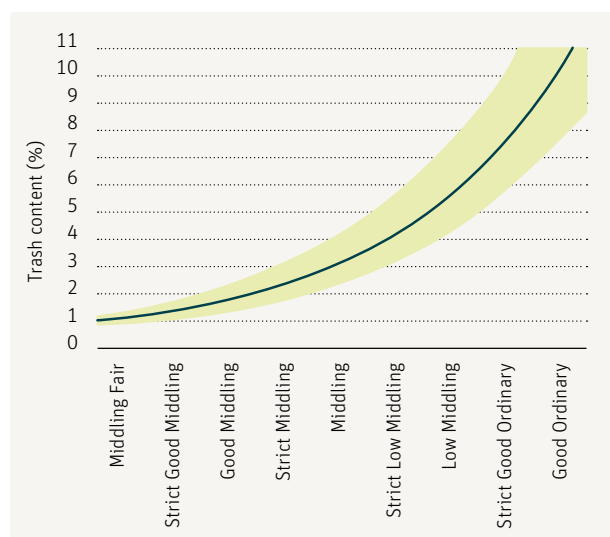
below 5.0	very low
5.0 – 5.8	low
5.9 – 6.7	average
6.8 – 7.6	high
above 7.7	very high

Micronaire (approx. conversion $\text{Mic} = \frac{\text{mtex}}{39.4}$)

below 3.0	very fine
3.0 – 3.9	fine
4.0 – 4.9	average
5.0 – 5.9	coarse
6.0 and above	very coarse

Non-fibrous content of raw cotton (trash) (Shirley)

Under 1.2%	very clean
1.21 – 2.0%	clean
2.01 – 4.0%	average
4.01 – 7.0%	dirty
7.01 and above	very dirty



4.3. Staple conversion chart

Category	Inches	32 ^{nds}	Decimals	USDA UPLAND HVI length (inches)	USDA Am. Pima HVI length (inches)	mm	mm gost
Short	13/16"	26	0.8125	0.80 – 0.85		20.64	
	(27/32")	27	0.84375			21.43	
	7/8"	28	0.875	0.86 – 0.89		22.23	24/25
	29/32"	29	0.90625	0.90 – 0.92		23.02	25/26
	15/16"	30	0.9375	0.93 – 0.95		23.81	26/27
Medium	31/32"	31	0.96875	0.96 – 0.98		24.61	27/28
	1"	32	1.0	0.99 – 1.01		25.40	28/29
	1 1/32"	33	1.03125	1.02 – 1.04		26.19	30/31
	1 1/16"	34	1.0625	1.05 – 1.07		26.99	31/32
	1 3/32"	35	1.09375	1.08 – 1.10		27.78	32/33
Medium to long	1 1/8"	36	1.125	1.11 – 1.13		28.58	33/34
	1 5/32"	37	1.15625	1.14 – 1.17		29.37	34/35
	1 3/16"	38	1.1875	1.18 – 1.20		30.16	35/36
	1 7/32"	39	1.21875	1.21 – 1.23		30.96	36/37
	1 1/4"	40	1.25	1.24 – 1.26	- 1.20	31.75	36/37
Long	1 9/32"	41	1.28125	1.27 – 1.29		32.54	37/38
	1 5/16"	42	1.3125	1.30 – 1.32	1.21 – 1.25	33.34	38/39
	(1.11/32")	43	1.34375	1.33 – 1.35		34.13	
	1 3/8"	44	1.375	1.36 +	1.26 – 1.31	34.93	39/40
	(1 13/32")	45	1.40625			35.72	
Extra long	1 7/16"	46	1.4375		1.32 – 1.36	36.51	40/41
	(1 15/32")	47	1.46875			37.31	
	1 1/2"	48	1.5		1.37 – 1.42	38.10	41/42
	(1 17/32")	49	1.53125			38.89	
	1 9/16"	50	1.5625		1.43 – 1.47	39.69	
	(1 19/32")	51	1.59375			40.48	
	1 5/8"	52	1.625		1.48 +	41.28	
	(1 21/32")	53	1.65625			42.07	
	1 11/16"	54	1.6875			42.86	
	(1 23/32")	55	1.71875			43.66	
	1 3/4"	56	1.75			44.45	

Millimeters according to GOST-standard are expressed as so-called „modal length“. They do not correlate with mathematically converted millimeters. The staple length of Central Asian cottons is also given as „TYP“. The minimum length is guaranteed as follows:

„Typ“	Ia	Ib	I	II	III	IV	V	VI	VII
mm min.	40.2	39.2	38.2	37.2	35.2	33.2	31.2	30.2	29.2

4.4. Universal cotton standards for color and leaf grade of American upland cotton

4.4.1. Color grades

	White	Light spotted	Spotted	Tinged	Yellow stained
Good middling	11	12	13		
Strict middling	21	22	23	24	25
Middling	31	32	33	34	35
Strict low middling	41	42	43	44	
Low middling	51	52	53	54	
Strict good ordinary	61	62	63		
Good middling	71				
Below grade	81	82	83	84	85

4.4.2. Leaf grades

Leaf grade		Equals HVI trash measurement (% area)
1	Good middling	0.08
2	Strict middling	0.12
3	Middling	0.18
4	Strict low middling	0.34
5	Low middling	0.55
6	Strict good ordinary	0.86
7	Good ordinary	1.56



Cotton field

4.5. Fiber properties of major cotton grades *

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
Angola	Fokion	50		1 3/32 – 1. 1/8	3.6 – 4.9	30
	Azahar	40		1 3/32 – 1. 1/8	4.3 – 4.8	32
	Others	10		1 3/32 – 1. 1/8		
Argentina	DP 402 BG/RR		28		4	31
	DP 604 BG		27		4.5	29
	Guazuncho 3 INTA		29		4.2	32
	Guazuncho 2000 R		29		3.8	30
	NuOpal BR		27		4	30
	DP 404 BG		28		4.5	29
	DP 447 BG		28		4.5	30
Australia	Sicot 74 BRF	70		1.2	4.5	31
	Sicot 71 BRF	13		1.18	4.4	31
	Sicot 71 RRF	5		1.17	4.5	30
Azerbaijan	Agh Giil, Flash, Karizma	62.5		1.1/8 – 1 5/32	4.4 – 5.0	31.3 – 34.0
	Aghdash-3	10		1 3/32	4.9 – 5.2	27.5
	Aznixi-195	17.5		1 3/32	5.0 – 5.3	29.4
	30 – 38	10		1 3/32	5.0 – 5.2	25.5 – 26.0
Bangladesh	Medium staple					
	CB – 9	<1	28		4.6	30.1
	CB – 12	28	29		4	30.5
	CB – 13	1	29.5		3.8	29
	CB – 14	3	30.5		3.9	30.4
	Rupali- 1 (Hybrid)	44	30.8		4.2	29.5
	Shuvro (Hybrid)	8	29.9		3.9	30.4
	DM-2 (Hybrid)	16	32.3		3.9	30.3
	Short-staple					
	Hill cotton -1	90	16		6.5	77
	Hill cotton -2	10	18		6.3	77
Barbados	West Indian Sea Island cotton	100	39		3.28	36.8
Benin	H 279-1	100		1 3/32 – 1 5/32	3.5 – 4.4	28 – 31
Bolivia	DELTAOPAL	100	28		3.5 – 4.0	28
Brazil						
	FM	73				
	FM 913GL		29.6		4.1	29.7
	FM 940GLT		29.1		4.2	31.9
	FM 944GL		29.5		4	30.9
	FM 975WS		29.4		4.2	29.6
	FM 980GLT		29.4		4.3	30.4
	FM 982GL		29.9		4.1	30.2
	DP	4				
	DP 1240B2F		28.4		4.1	29.1
	DP 1227B2F		28.9		4	29.4

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
	TMG	22				
	TMG 42 WS		30		4	30.5
	TMG 46 BSF		28.7		4.6	29.3
	TMG 81 WS		29		4.6	30.4
	TMG 82 WS		30		3.94	30.1
Bulgaria	IPTP VENO	60	29 – 31		4.2	29.1
	Chirpan – 5394	20	29 – 31		4.2	29.1
	Helius	10	29 – 31		4	28.9
	Darmi	10	30 – 32		4.1	28.9
Burkina Faso	FK 37		28.8		3.9	30.5
	FK 95 BG II		27.5		4.1	28.1
	STAM 59A		28.2		4.1	28.7
	FK64		29.4		4.1	29.2
Cameroon	IRMA L457	67	28.6		3.8	29.4
	IRMA L484	33	30		4.2	30.3
Chad	A 51	49		1 316 – 1 7/32	3.7 – 4.9	28 – 30
	STAM F	51		1 18 – 1 316	3.7 – 4.2	39 – 43
China	Upland	96.9	28 – 29		3.7 – 4.9	28 – 30
	Long staple	3	38 – 40		3.7 – 4.2	39 – 43
Colombia	Interior region					
	ST 5288 B 2F	27.13	28.64		4.52	27.14
	FM 1740	1.81	28.47		4.49	28.32
	Coast region					
	DP 1048 Bollgard 2 RR Flex	14.5	28.6		4.42	29.02
	DP 90 Nacional	4.64	28.97		4.4	30.6
	FM 1740 B 2F	11.65	28.64		45.23	30.02
	FM 9162 B 2F	20.85	29.22		4.44	31.59
	ST 5288 B 2F	9.41	29.29		4.55	30.3
Egypt	Extra long staple					
	GIZA 45	<1	34.7		3.06	40.1
	GIZA 87	<1	34.9		3.02	42.3
	GIZA 88	5	32.6		3.9	40.7
	GIZA 93	<1	35.8		2.87	42.2
	GIZA 92	2	32.4		3.58	46.6
	Long staple					
	GIZA 86	85	85.3		4.55	40.9
	GIZA 80	<1	84.4		4.53	35.5
	GIZA 90	7.0 – 8	83.2		4.27	34.5
Ethopia	Upland cotton					
	ACALA-SJ2		28.6		3.2 – 4.1	25.9
	DP-90		29.6		4.2	24.7 – 29.3
	SAM 59A		30.9		4.4	32.5
	CANDIA		29.9		4.4	31.5

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
	R 36		27.9		4.3	24
	ABA		29.7		4.2	25.7
	Local races		22 – 24			
Ghana	FK			1.13	4.1	31
	STAM			1.16	4.3	30
Greece	ST 402	10.3	28.5 – 29.0		4.4 – 4.7	28 – 30
	CELIA	5.9	28.0 – 29.0		4.0 – 4.3	29 – 30
	DP 332	5.3	28.0 – 29.0		4.3 – 4.5	29 – 30
	HERSI	5	26.5 – 30.0		4.1 – 4.2	30 – 31
	ST 474	4.2	27.0 – 28.0		4.7 – 4.9	29 – 30
	FLORA	4.1	28.6 – 29.5		4.1 – 4.2	29 – 30
	ELSA	3.6	28.8 – 29.7		4.1 – 4.4	29 – 30
	DP 419	3.5				
	ST 318	3.2	28.0 – 29.0		4.3 – 4.5	28.5 – 30.0
	LIDER	3.2	27.7 – 29.4		4.3 – 4.5	28 – 29.5
	BABYLON	3.1	28.6 – 29.6		4.1 – 4.4	29 – 31
	DP 396	3	28.0 – 29.0		4.5 – 4.7	28 – 29
	CARMEN	2.9	28.0 – 28.5		4.2 – 4.3	28 – 30
	FLASH	2.4	29.0 – 30.0		4.6 – 4.8	33 – 35
	IDEAL	2.2	28.0 – 28.7		4.4 – 4.7	30 – 31
	DP 401	2.1	28.5 – 29.0		4.2 – 4.3	28.7 – 31.0
	FANTOM	2	27.6 – 28.1		3.8 – 4.1	28 – 29
	ST 405	1.8	28.0 – 28.6		4.6 – 4.8	28.4 – 29.3
	SANDRA	1.5	28.0 – 28.5		4.4 – 4.8	28.5 – 29.5
	CAMPO	1.5	28.0 – 28.5		4.3 – 4.5	27 – 28
India	Punjab	4.4				
	ICS-202 J-34 SG		25 – 28		4.0 – 4.5	32 – 27
	Bengal Deshi SG		15 – 19		6.0 – 6.7	14 – 17
	Haryana	4.8				
	ICS-202 J-34 SG		25 – 27		4.0 – 4.6	24 – 26
	Bengal Deshi SG		15 – 18		6.5 – 7.0	14 – 17
	Rajasthan	3.6				
	ICS-202 J-34 SG		25 – 26		4.0 – 4.6	24 – 26
	Bengal Deshi SG		15 – 18		6.5 – 7.0	14 – 17
	Gujarat	23.3				
	ICS-105 Shankar-6		27 – 30		3.5 – 4.4	26 – 29
	ICS-102 V-797		22 – 24		4.6 – 5.5	20 – 23
	Madhya Pradesh	5.4				
	ICS-105 H-4/ MECH-1		27 – 29		3.5 – 4.9	26 – 28
	ICS-105 LRA-5166		25 – 27		3.5 – 4.9	24 – 27
	ICS-104-Y-1		23 – 25		4.0 – 5.5	22 – 24
	Maharashtra	33.5				
	ICS-105 H-4/ MECH-1		27 – 29		3.5 – 4.4	26 – 28
	ICS-105 LRA-5166		26 – 27		3.5 – 4.4	24 – 26

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
	ICS-103 NHH-44		24 – 26		3.5 – 4.4	22 – 25
	ICS-104 Y-1		22 – 24		4.0 – 5.5	20 – 23
	Karnataka	5				
	ICS-107 DCH-32		33 – 36		2.8 – 3.3	32 – 34
	ICS-105 LRA-5166		26 – 28		3.5 – 4.2	24 – 27
	ICS-105 NHH-44		22 – 24		4.0 – 5.5	20 – 22
	ICS-105 Bunny/ Brahma		30 – 31		3.5 – 4.2	28 – 30
	Andhra Pradesh	18.5				
	ICS-105 H-4/ MECH-1		27 – 29		3.5 – 4.2	26 – 28
	ICS-106 MCU -5		31 – 33		3.5 – 4.4	30 – 32
	ICS-105 Brahma/ Brahma		30 – 31		3.5 – 4.2	29 – 30
	ICS-105 LRA-5166		25 – 27		3.6 – 4 – 2	24 – 26
	Tamil Nadu	1				
	ICS-107 DCH-32		34 – 36		2.7 – 3.4	32 – 35
	ICS-106 MCU -5		31 – 33		3.4 – 4.0	30 – 32
	ICS-105 LRA-5166		25 – 27		3.8 – 4.5	24 – 26
	Others	1.5				
Iran	Varanim	76	30		4.1	27.5
	Bakhtegan	12	29.8		4.1	28
	Sahel	10	30		4.1	28
	Golestan	1	29.5		4.3	27
	Sepid	0.5	30.8		4.2	27.2
	Amaghan	0.5	29.4		4.3	26.5
Israel	Pima	85	36 – 38		3.7 – 4.5	40 – 42
	Acalpi	15	34 – 35		3.4 – 4.1	34 – 37
Ivory Coast	R 405-5			1 1/16 – 1 1/8	3.6 – 4.0	29
	R 405-96			1 1/16 – 1 1/8	3.6 – 4.0	29
Kazakhstan	PA-3044	11	32.6		4.7	25 – 29.5
	M-4005	18	33		4.7	25 – 29.5
	M-4007	29	33.1		4.7	25 – 29.5
	M-4011	31	33.2		4.7	25 – 29.5
	Bereke-07	9	33		4.7	25 – 29.5
	Myrzashol	1	33.3		4.6	25 – 29.5
Kenya	Coastal Central / Eastern	70				
	HART 89 M			1 1/16 – 1 3/32	3.8 – 4.1	19 – 25
	Western	30				
	KSA 81 M			1 1/16 – 1 3/32	3.8 – 4.1	19 – 22
Kyrgyzstan	Kirgiskij-3		32 – 33		4.5 – 4.8	25.5 – 26.5
	Ferghana-3		30 – 31		4.6 – 4.8	24.5 – 25.5
	Others		30 – 31		4.6 – 4.7	24.5 – 25.5
Madagascar	Various		28 – 30		4.0 – 4.8	30 – 34
Malawi	Makoka-2000		28.5		4.4	density 198
	Rasam-17		28.4		4.3	density 199

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
	IRM-81		29.6		4.1	density 182
	Chureza		28.7		4.6	density 190
	SZ-9314		29.4		4.2	density 184
Mali	NTA 90-5		30.7		4.1	29
	NTA 93-15		29.5		3.8	26.9
	STAM 59 A		29.1		4.1	28.2
	STAM 279 A		28.5		4.3	26.6
Mexico	DP 451 BRR			1 1/32 – 1 3/32	3.6 – 4.9	25 – 28
	Nucotn-35 B			1 1/16 – 1 1/8	3.8 – 4.9	25 – 28
	DP 458 BRR			1 3/32 – 1 1/8	3.8 – 4.9	25 – 28
	Sure-Grow-125BR			1 3/32 – 1 1/8	3.9 – 4.7	26 – 29
	Fibermax			1 3/32 – 1 1/8	3.9 – 4.7	25 – 28
Mozambique	CA-324	75		1 3/32 – 1 1/8	4.0 – 4.4	
	ALBAR SZ 9314	25		1 3/32 – 1 1/8	4.4 – 4.6	
Myanmar	Ngwe Chi-6	68.32	28 – 30		4.8 – 5.0	41.8 – 42.9
	Shwe-Daung-8	4.58	28 – 31		4.8 – 5.1	402.44.5
	Ngwe Chi-9	0.012	29 – 30		4.8 – 5.3	41.3 – 42.9
	Mahlaing	9.82	19 – 24		5.8 – 7.8	39.6 – 48.24
	Wagyi	12.21	21 – 24		5.9 – 7.2	41.8 – 45
	Others	5.06				
Nevis	Asian	100		1.47	3.2	37.2
Nigeria	Samcott varieties			1 1/16 – 1 5/32	3.6 – 4.6	
Pakistan	SLH-41		26.4		4.4	95.8 Pressley
	CIM-70		29		4.2	92.5 Pressley
	CIM-109		27.2		4.4	92 Pressley
	CIM-240		27.5		4.7	93.7 Pressley
	CRIS-9		26.5			97 Pressley
	BH-36		28		4.3	100.3 Pressley
	CIM-1100		29		3.9	94 Pressley
	CIM-448		28.5		4.5	93.8 Pressley
	CIM-443		27.6		4.9	96 Pressley
	CIM-446		27		4.7	97.4 Pressley
	CIM-482		28.5		4.5	98 Pressley
	BH-118		27.6		4.6	96.2 Pressley
	Marvi CRIS-5A		26.8			97.4 Pressley
	CIM-473		29.6		4.3	98 Pressley
	CIM-499		29.6		4.4	96.2 Pressley
	CIM-707		32.2		4.2	97.3 Pressley
	CIM-506		28.7		4.5	97.5 Pressley
	CIM-496		29.7		4.6	98.9 Pressley
	CIM-134		27.5			93.5 Pressley
	CIM-467		29		4.6	97.5 Pressley
	CIM-534		27.5		4.5	97.2 Pressley
	CIM-121		27.5		4.9	97.2 Pressley

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
	CIM-554		29		4.7	98.5 Pressley
	CRIS-342		27.5		4.3	95.5 Pressley
	CIM-573		28.5		4.6	93.8 Pressley
	CIM-573		28.4		4.6	90.2 Pressley
	Bt CIM-598		31.6		4.3	94.8 Pressley
	BH-167		29		4.7	92.7 Pressley
	SLH-317		29.1		4.4	96.7 Pressley
	Bt CIM-595		29		4.7	97.5 Pressley
	Bt CIM-599		28.9		4.6	95 Pressley
	Bt CIM-602		29.1		4.2	94.8 Pressley
	CIM-608		28.5		4.6	93.9 Pressley
	CRIS-129		28.5			98.5 Pressley
	B. CIM-600		29		4.7	94.8 Pressley
	Bt. Cyto-177		29		4.3	99.9 Pressley
	Cyto-124		30.3		4.4	92.4 Pressley
Paraguay	Various			1 3/32	3.5 – 4.9	28
Peru	Pima	15.4		1 1/2 – 1 5/8	3.4 – 3.8	35 – 39
	Tanguis	64.6		1 1/8 – 1 3/16	4.6 – 5.4	34 – 37
	Del Cerro	1.5		1 5/16 – 1 1/16	3.3 – 3.8	
	Aspero	18.5		1 1/32 – 1 3/32	6.1 – 6.4	28 – 31
Philippines	Visayas					
	UPL-C 2	38	29.8		5.1	19.44
	NSIC-CT 11	62	28.72		4.74	18.63
	Mindanao Lower Valley					
	CRDI-1	5	28.4		4.31	19.45
	UPL-C 2	10	29.8		5.1	19.44
	NSIC-Ct11	36	28.72		4.74	18.83
	Bioseed 6527	49	27.91		4.12	20.41
	Mindanao Upper Valley					
	NSIC-Ct11	59	28.72		4.74	18.63
	Bioseed 6527	41	27.91		4.12	20.41
Senegal	STAM 129	100	27.3 – 29.3		3.99	22.8
South Africa	DP 210 BRF	32	29.2 – 30		3.5 – 4.9	28
	DP 12 BRF	40	28.4 – 29.2		3.5 – 4.9	28
	Candia BRF	20	29.0 – 30.0		3.5 – 4.9	28
	PM 3225 B 2RF	8	26.9 – 27.6		3.5 – 4.9	27 – 28
Spain	Upland	97	27 – 29		3.7 – 4.9	29+
	Extra long staple	3	33+		3.5 – 4.9	35+
Sudan	Extra long staple	31	33 – 37		3.6 – 4.2	30 – 40
	Medium to long staple	10	29 – 34		4.0 – 4.4	28 – 34
	Medium staple	59	25 – 29		4.4 – 5.1	24 – 31
Syria	Various			1.06 – 1.17	4.3 – 4.7	21.1 – 26.2
Tajikistan	Medium staple		32.1 – 34.2		4.5 – 4.7	25 – 28.5

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
	Long staple		39.2 – 41			32.6 – 38.2
Tanzania	UK 91	99		1.1 – 1.14	3.9 – 4.7	26.8 – 28.2
	ALAI 90	1		1.11 – 1.15	3.9 – 4.2	27.5 – 29.3
Thailand	Tak Fa 2	30		1.19	3.9	21.7
	Tak Fa 84-4	70		1.23	3.9	24.6
	Tak Fa 86-5	14		1.25	2.6	21.5
	Tak Fa 3	8		0.78	5.8	20.8
	Local varieties	5		<1	5	15 – 17
Togo	STAM 129 A	100	28.9		3.8 – 4.9	29.8 – 31.1
Turkey	Aegean Region					
	Lydia	11	29 – 32		4.0 – 4.5	34.38
	lash	12	29 – 31		4.6 – 4.8	33.35
	Carisma	13	28 – 30		4.4 – 4.9	30.32
	Pg 2018	5	28 – 30		4.6 – 4.8	31.33
	Gloria	47	30 – 32		4.1 – 4.4	33.36
	Cloudia	7	30 – 32		4.1 – 4.5	32.34
	Others	5				
	Cukurova Region					
	BA 119	9	29 – 31		4.4 – 4.6	31 – 33
	Flash	13	29 – 31		4.6 – 4.8	33 – 35
	Carisma	21	28 – 30		4.4 – 4.9	30 – 32
	BA 525	10	28.5 – 30.0		4.4 – 4.8	28 – 30
	loria	16	29 – 31		4.2 – 4.8	32 – 36
	DP 125	9	28 – 30		4.2 – 5.0	29 – 32
	STV 488	8	28.5 – 30.0		4.4 – 4.8	32 – 35
	DP 332	7	28 – 29		4.5 – 5.0	32 – 34
	Others	7				
	Southeast Anatolia					
	BA 119	18	28 – 31		4.4 – 4.6	31 – 33
	BA 440	12	28 – 30		4.6 – 4.8	31 – 33
	PG 2018	3	28 – 30		4.6 – 4.8	31 – 33
	STV 468	39	28 – 30		4.4 – 4.8	32 – 35
	Candia	19	29 – 31		4.1 – 4.5	30 – 32
	Others	9				
Turkmenistan	Medium staple	95				
	Yoloten-7		33 – 34		3.8 – 4.8	90 Pressley
	S-2606		33 – 34		3.8 – 4.8	90 Pressley
	S-4727		31 – 32		3.5 – 4.5	90 Pressley
	133		33 – 34		3.8 – 4.8	90 Pressley
	Serdar		33 – 34		3.8 – 4.8	90 Pressley
	Ash-36 C		33 – 34		3.8 – 4.8	90 Pressley
	Dashoguz-120		33 – 34		3.8 – 4.8	90 Pressley
	149-F		33 – 34		3.8 – 4.8	90 Pressley
	Long staple	5				
	Yoloten-14		40 – 41		3.3 – 4.0	98 – 110 Pressley

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
Uganda	BPA	100		1 1/8 – 1 3/16	3.7-4.3	27 – 33
USA	Georgia	50.3		1 1/8	4.7	29
	Alabama	14.2		1 1/8	4.7	30
	Florida	3.8		1 1/8	4.8	29
	North Carolina	17.3		1 3/32	4.9	28
	South Carolina	10.6		1 3/32	4.8	27.7
	Virginia	3.8		1 3/32	4.8	28.5
	Arkansas	32.4		1 5/32	4.7	33
	Louisiana	11.2		1 3/32 – 1 1/8	5	32
	Mississippi	32.7		1 1/8	4.9	32
	Missouri	18.9		1 5/32 – 1 3/16	4.3	32.1
	Tennessee	15.8		1 5/32	4.6	31.4
	Texas	95.5		1 3/32 – 1 1/8	4.1	30.7
	Oklahoma	4.2		1 1/8	4.1	31.4
	Kansas	0.3				
	Arizona	50.9		1 1/8 – 1 5/32	4.6	31
	California	28.1		1 5/32 – 1 3/16	4.4	34
	New Mexico	21		1 1/8 – 1 5/32	4.1	30.7
American Pima	Arizona	11.4		1 15/32 – 1 1/2	3.7	42.5
	California	72.8		1 1/2	3.9	43.8
	Texas	10.8		1 15/32	4	41.9
	New Mexico	5				
Uzbekistan	S 6524		35.7		4.6	32.9
	Bukhara 102		36		4.5	21.8
	Bukhara 6		36.3		4.4	32.7
	Bukhara 8		36.4		4.4	32.7
	An-Bayaut 2		35.3		4.5	30.8
	Namangan 77		35.7		4.6	31.8
	Andijan 35		35.3		4.7	31.7
	Khorezm 127		35.7		4.7	31.2
	Omad		36.1		4.5	30.6
	S 4727		35.7		4.7	31
	Mehnat		35.7		4.6	31.2
	Anijan 36		35.9		4.6	30.5
	S 6541		35.3		4.6	31.3
	Sulton		35.6		4.6	31.4
	Ibrat		35.7		4.6	31.3
	Andijan 37		35.8		4.6	31.6
	Beshkakhramon		36		4.6	31.8
	Porloq		37 – 39		3.7 – 4.6	29 – 34
Vietnam	VN 01-2		30		4.1	27

Country	Variety	Area in%	Staple length in mm	Staple length in inch	Micronaire	Strength in g/tex (HVI Cal.)
	VN35KS		29		4.3	27
	VN 04-3		31		4.2	28
	VN 04-4		32		4.3	28
Zambia	Chureza	45	29 – 30		3.9 – 4.0	30 – 32
	CDT II	35	29 – 30		3.9 – 4.5	30 – 32
	CDT V	20	29 – 30		3.9 – 4.2	30 – 32
Zimbabwe	Albar	100		13/32 – 15/32	3.5 – 4.9	28 – 31

* Data of Cotton Varieties by Origins, Bremen Cotton Exchange, January 2018 partially adjusted for HVI cal. Level (excert)

5. General Data

5.1. Units of measurement

Basic SI units

Basic quantity	Basic unit	Abbreviation	DIN symbol
Length	Metre	m	l
Mass	Kilogram	kg	m
Time	Second	s	t
Electric current	Ampere	A	I
Thermodynamic temperature	Kelvin	K	T
Luminous intensity	Candela	cd	I
Amount of substance	Mol	mol	

Derived SI units

Quantity	DIN symbol	Name	Symbol	Ratio
Frequency	f	Hertz	Hz	$1 \text{ Hz} = 1 \text{ s}^{-1} = 1/\text{s}$
Force	F	Newton	N	$1 \text{ N} = 1 \text{ kg} \cdot \text{m}/\text{s}^2$
Pressure and mechanical stress	p	Pascal	Pa	$1 \text{ Pa} = 1 \text{ N}/\text{m}^2$
Work, energy, heat	W, E, Q	Joule	J	$1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s}$
Power, energy flow, heat flow	P	Watt	W	$1 \text{ W} = 1 \text{ N} \cdot \text{m}/\text{s} = 1 \text{ J}/\text{s}$
Electric charge	Q	Coulomb	C	$1 \text{ C} = 1 \text{ A} \cdot \text{s}$
Electric voltage, potential difference	U	Volt	V	$1 \text{ V} = 1 \text{ W}/\text{A}$
Electric capacitance	C	Farad	F	$1 \text{ F} = 1 \text{ C}/\text{V} = 1 \text{ A} \cdot \text{s}/\text{V}$
Electric resistance	R	Ohm	Ω	$1 = 1 \text{ V}/\text{A}$
Luminous flux	Φ	Lumen	lm	$1 \text{ lm} = 1 \text{ cd} \cdot \text{sr}$
Illumination	E	lux	lx	$1 \text{ lx} = 1 \text{ lm}/\text{m}^2$
Electric conductance	G	Siemens	S	$1 \text{ S} = 1 \Omega^{-1} = 1/\Omega = 1 \text{ A}/\text{V}$
Self inductance	L	Henry	H	$1 \text{ H} = 1 \text{ Wb}/\text{A} = 1 \text{ V} \cdot \text{s}/\text{A}$

Units which are not included in the SI system but are permissible

Physical quantity	Designation	Unit symbol	Relation to SI system
Area	Are	a	$1 \text{ a} = 10^2 \text{ m}^2$
	Hectare	ha	$1 \text{ ha} = 10^4 \text{ m}^2$
Volume	Litre	l	$1 \text{ l} = 1 \text{ dm}^3 = 10^{-3} \text{ m}^3$
Angle	Degree	°	$1^\circ = 60' = 3600''$
			$1^\circ = 0.017453 \text{ rad} = \frac{\pi}{180} \text{ rad}$
	Minute	'	$1' = 0.0002908 \text{ rad} = \frac{\pi}{10800} \text{ rad}$
	Second	''	$1'' = 0.000004848 \text{ rad} = \frac{\pi}{648000} \text{ rad}$
Mass	Tonne	t	$1 \text{ t} = 10^3 \text{ kg}$
Time	Minute	min	$1 \text{ min} = 60 \text{ s}$
	Hour	h	$1 \text{ h} = 60 \text{ min} = 3600 \text{ s}$
	Day	d	$1 \text{ d} = 24 \text{ h} = 1440 \text{ min} = 86400 \text{ s}$
Pressure	Bar	bar	$1 \text{ bar} = 10^5 \text{ Nm}^{-2} = 10^5 \text{ Pa}$
Temperature	Degree celsius	°C	$1^\circ\text{C} = 1 \text{ K}$
			$0^\circ\text{C} = 273.15 \text{ K}$

Conversion factors

Quantity	Required	Given	X conversion factor
Force	N =	kp	· 9.81
	kp =	N	· 0.102
Power (1 W = 1 J/s = 1 Nm/s)	Nm/s =	kpm/s	· 9.81
	kpm/s =	Nm/s	· 0.102
	kW =	PS	· 0.7355
	PS =	kW	· 1.36
Work and energy (1 J = 1 Nm = 1 Ws)	Nm =	kpm	· 9.81
	kpm =	Nm	· 0.102
	kWh =	PSh	· 0.7355
	PSh =	kWh	· 1.36
Heat	J =	cal	· 4.187
	cal =	J	· 0.2388
Pressure (N/m² = Pa) (1 bar = 100 000 Pa)	Pa =	mm WS	· 9.81
	mm WS =	Pa	· 0.102
	bar =	at	· 0.981
	at =	bar	· 1.02
	Pa =	mm Hg	· 133.3
	mm Hg =	Pa	· 0.0075
Temperature			
	K = °C + 273.15		
	°C = K - 273.15		

5.2. Conversion factors for German and Anglo-American units of measurement and weight

Basic SI units

Multiply	by	to obtain
Acres	0.404687	Hectares [ha]
Acres	0.004047	Square kilometers [km ²]
Ares/ar	1076.39	Square feet [ft. 2]
Atmospheres	1.033	[kg/cm ²]
Atmospheres	14.696	lbs. per sq. in.
Atmospheres	760	mm mercury-column
Board feet	0.0833	Cubic feet [ft. 3]
British Thermal Units: B.T.U.	0.252	[kcal]
British Thermal Units: B.T.U.	0.293	Watt-hour [wh]
B.T.U./cub. ft.	8.9	[kcal/m ³]
B.T.U./lbs.	0.5556	[kcal/kg]
Centimeters [cm]	0.3937	Inches [in.]
Cubic centimeters [ccm]	0.06102	Cubic inches [cuin.]
Cubic feet	0.028316	Cubic meter [cbm] oder [m ³]
Cubic feet	6.22905	British Imp. Gallons
Cubic feet	28.3170	Liters [l]
Cubic inches	16.38716	Cubic centimeters [ccm] oder [cm ³]
Cubic meter [cbm]	35.3145	Cubic feet
Cubic meter [cbm]	1.30794	Cubic yards
Cubic meters	219.97	Gallons. British
Cubic meters	264.2	Gallons. American
Cubic meters/kg	16.01	[cub. ft./lbs.]
Cubic yards	0.764559	Cubic meters [cbm] oder [m ³]
Circular mils	0.000507	Square millimeters [sq. mm]
Degrees centigrade	°C x 9 : 5 + 32	Degrees Fahrenheit [°F]
Degrees centigrade	°C x 4 : 5	Degrees Reaumur [°R]
Degrees Fahrenheit	(°F – 32) x 5 : 9	Degrees centigrade [°C]
Degrees Fahrenheit	(°F – 32) x 4 : 9	Degrees Reaumur
Degrees Reaumur	°R x 5 : 4	Degrees centigrade
Degrees Reaumur	(°R x 9 : 4) + 32	Degrees Fahrenheit
Fathoms	1.8288	Meters [m]
Foot pounds	0.13826	Kilogram meters [kgm]
Feet	30.4801	Centimeters [cm]
Feet	0.304801	Meters [m]
Feet	304.801	Millimeters [mm]
ft.-lbs.	1.356	Joules [J]
Gallons. British Imp.	0.160538	Cubic feet
Gallons. British Imp.	1.20091	Gallons U.S.
Gallons. British Imp.	4.54596	Liters/Liter [l]
Gallons U.S.	0.832702	Gallons. British Imp.
Gallons U.S.	3.78543	Liters [l]

Multiply	by	to obtain
Grams/Gram	0.03527	Ounces [Oz.]
Grams/Gram	15.43	Grains [gr]
Grains	0.0648	Grams/Gram [g]
Gram	0.0022	[Lbs.]
Gram/cm³	0.0361	[Lbs./cub. in.]
Gram/cm³	62.4	[Lbs./cub. ft.]
Hectares [ha]	2.47104	Acres [ar]
Hectares	11 960	Sq. yards
Hectolitres	3.531	Cub. ft.
Hectolitres	2.838	Bushels U.S.
Horse-power. metric [PS]	0.98632	Horse-power U.S. or British
Horse-power U.S. or British	1.01387	Horse-power. metric [PS]
Horse-power U.S. or British	0.707	[B.T.U./s]
Horse-power U.S. or British	0.178	[kcal/s]
Horse-power U.S. or British	0.746	[Kilowatts]
Inches	2.54001	Centimeters [cm]
Inches	25.4001	Millimeters [mm]
Inches mercury column	0.0333	[atm.]
Inches mercury column	13.6	Inches of water-column
Inches mercury column	0.49	[Lbs./sq.in.]
Inches of water-column	0.0735	Inches mercury column
Inches of water-column	0.036	[Lbs./sq.in.]
Joules	0.738	[ft.-lbs.]
Joules	0.000239	[kcal]
[kcal]	426.9	[kg/m]
[kcal]	3.97	[B.T.U.]
[kcal]	4 184	Joules [J]
[kcal/m³]	0.1124	[B.T.U./cub.ft.]
[kcal/kg]	1.8	[B.T.U./lbs.]
[kg/m³]	0.062	[Lbs./cub.ft.]
[kg/PS]	2.235	[Lbs./HP]
Kilograms [kg]	2.20462	Pounds [lb.]
Kilogram meters [kgm]	7.233	Foot pounds
Kilograms per meter [kg/m]	0.671972	Pounds per foot [lbs./ft.]
Kilograms per sq. centimeter	14.2234	Pounds per sq. inch
Kilograms per sq. meter	0.204817	Pounds per sq.ft. [lbs./ft. 2]
Kilograms per sq. millimeter	1422.34	Pounds per sq. inch
Kilograms per sq. millimeter	0.634973	Long tons per sq. inch
Kilometers	0.62137	Miles statute U.S.
Kilometers	0.53959	Miles nautical U.S.
Kilometers	3 280.8693	Feet [ft.]
Kilometers per hour	0.54	Knots
Kilowatts	0.948	[B.T.U./s]
Kilowatts	0.293	[kcal/s]

Multiply	by	to obtain
Kilowatts	738	[ft.-lbs./s]
Kilowatts	1.341	Horse-power U.S. or British
Kilowatts	1.36	[PS]
Kilowatt-hour [kWh]	860	[kcal]
Knots	1.151	U.S. Miles per hour
Liters [l]	0.219975	Gallons. British Imp.
Liters [l]	0.26417	Gallons. U.S.
Litres	0.0353	Cubic feet
Litres	61.022	Cubic inch
Litres	1.7598	Pints Engl.
Meters [m]	3.28083	Feet [ft.]
Meters [m]	39.37	Inches [in.]
Meters [m]	1.09361	Yards
Meters [m]	0.5468	Fathoms
Miles. statute U.S.	1.60935	Kilometers [km]
Miles. statute U.S.	0.8684	U.S. Miles nautical
Miles. nautical U.S.	6080.204	Feet
Miles. nautical U.S.	1.85325	Kilometers [km]
Miles. nautical U.S.	1.1516	U.S. Miles statute
U.S. Miles per hour	1.467	[ft./s]
U.S. Miles per hour	0.869	Knots
Millimeters [mm]	0.003937	Inches
Ounces	28.3495	Grams/Gramm [g]
Pints Engl.	0.568	Litres [l]
Pounds. avoirdupois	453.593	Grams/Gramm [g]
Pounds. avoirdupois	0.453592	Kilograms [kg]
Pounds p. foot [lb.p.ft.]	1.48816	Kilograms per meter [kg/m]
Pounds p.sq.ft. [lb./ft. 2]	4.88241	Kilograms per sq. meter [kg/m ²]
Pounds p.sq.inch [lbs./in. 2]	0.0703	Kilograms per sq. centimeters [kg/cm ²]
Pounds p. cubic foot	16.0184	Kilograms per cubic meter [kg/m ³]
Quarts Engl.	1.136	Litres
Square millimeters [mm]	0.0015	Square in
Square millimeters [mm ²]	1973.5	Circular miles
Square centimeters [cm ²]	0.1550	Square inches [in. 2]
Square feet [ft. 2]	0.0929034	Square meter [m ²]
Square inches	6.45163	Square centimetres [cm ²]
Square inches	645.163	Square millimeters [mm ²]
Square kilometers [km ²]	247.104	Acres [ar]
Square kilometers [km ²]	0.3861	Square miles U.S.
Square meters [m ²]	10.7639	Square feet
Square meters [m ²]	1.19599	Square yards
Square miles U.S.	259.0	Hectares [ha]
Square miles U.S.	2.59	Square kilometers [km ²]
Square yards	0.83613	Square meters [m ²]

Multiply	by	to obtain
Therm (100 000 B.T.U.)	2 500	[kcal]
Tons. long	1016.05	Kilograms [kg]
Tons. long	2 240	[Lbs.]
Tons. long	1.01605	Tons. metric
Tons. long	1.120	Tons. short
Tons. long p.sq. inch	1.57494	Kilograms per sq. millimeter [kg/mm²]
Tons. metric	2 204.62	Lbs.
Tons. metric	0.98421	Tons. long
Tons. metric	1.10231	Tons. short
Tons. short	907.185	Kilograms [kg]
Tons. short	0.892857	Tons. long
Tons. short	0.907185	Tons. metric
Tons. British Shipping	42.00	Cubic feet
Tons. British Shipping	0.9532381	Tons. U.S. Shipping
Tons. U.S. Shipping	40.00	Cubic feet
Tons. U.S. Shipping	1.050	Tons. British Shipping
[Ton/sq.in.]	157.5	[kg/cm²]
[Ton/sq.in.]	152.4	[atm.]
Watt	0.00134	Horse-power British or U.S.
Watt	0.00136	[PS]
Watt	0.7373	[ft.-lbs./s]
Watt-hour	3.415	B.T.U.
Yards	0.914402	Meters

5.3. Specific gravity (g/cm³)

Certain textile fiber materials	
Wool	1.32
Cotton	1.52
Hemp. flax	1.48
Natural silk	1.30
Viscose	1.52
Acetate	1.13
Cuprammonium rayon	1.52
Polyamide 6.6	1.14
Polyamide 6	1.14
Polyvinylchloride	1.35
Polyester	1.38
Polyacryl	1.16
Glass fibers	2.5
Polyvinylides	1.7
Polypropylene	0.91

By comparison: solids	
Aluminium	2.60 – 2.70
Asbestos	2.15 – 2.80
Fats	0.92 – 0.94
Cast iron	7.25
Steel	7.85
By comparison: liquids	
Petroleum at 15°C	0.69
Benzene at 0°C	0.90
Water at 4°C	1.00
By comparison: gases at 0°C in kg/m	
Carbon dioxide	1.98
Oxygen	1.43
Water vapour	0.77

6. Spinning Documentation Abbreviations

Abbreviation	Meaning
A 12	UNIfloc bale opener
A 21	Condenser for fiber transport in blowroom
A 22	Condenser for fiber transport in blowroom
A 79	UNIstore feeding machine; opening, cleaning
A 81	UNIBlend fiber mixing in blowroom
AT	Waste transport in blowroom
B 12	UNILclean fiber cleaning
B 25	Waste opener
B 72	UNImix fiber mixing in blowroom
B 76	UNImix fiber mixing in blowroom
C 70	Card
E 26	Lap transport system SERVOLap from combing preparation to comber
E 36	Combing preparation
E 86R	Comber with fully automatic lap change and batt piecing system ROBOLap
F 38 - 110	Fully automated roving frame with gauge 110 mm
G 32	Ring spinning machine
G 36	Ring spinning machine
J 26	Air-jet spinning machine
K 42	Compact spinning machine
K 46	Compact spinning machine
Link	Connection from ring or compact spinning machine to the winder
MT	Pneumatic fiber transport in blowroom
OMEGAlap	Combing preparation machine
R 36	Semi-automated rotor spinning machine
R 66	Fully automated rotor spinning machine
R 66	Rotor spinning machine
RSB-D 50	Autoleveler draw frame
SB-D 22	Double-head draw frame without autoleveling
SB-D 50	Draw frame without autoleveling
SERVOLap	Lap transport system E 26 from combing preparation to comber
SERVOTrail-F-A	Bobbin transport system from roving frame to ring or compact spinning machine
SL	Bobbin lift system on rotor spinning machine
SPIDERweb	Mill monitoring system
UNILclean	Fiber cleaning
UNIcontrol	Control of blowroom
UNIfloc	Bale opener
UNImix	Fiber mixing in blowroom
UNIstore	Fiber feeding machine; opening, cleaning
Vision shield	Detects and removes foreign material in blowroom



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